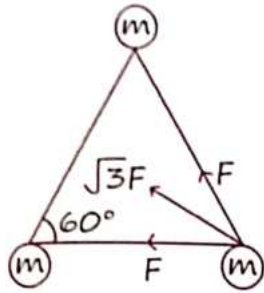


Q.2. In Given figure find net force on mass m placed at corner.

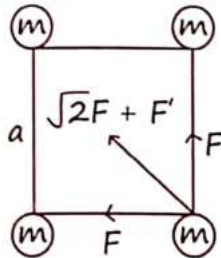


Sol.

$$F_{\text{net}} = \sqrt{F^2 + F^2 + 2FF\cos 60^\circ}$$

$$F_{\text{net}} = \sqrt{3}F = \frac{\sqrt{3}Gm^2}{a^2}$$

Q.3. In Given figure find net force on mass m placed at corner.



Sol. Each side of the square is of length a .

The net force on one mass (F_{net}) due to the other two masses is

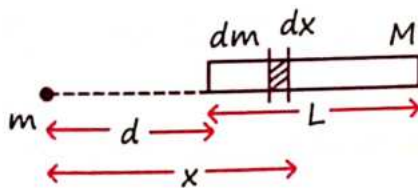
$$F_{\text{net}} = \frac{Gm^2}{a^2} \left(\frac{2\sqrt{2}+1}{2} \right)$$

+ A particle with mass m is located a distance d from one end of a rod with mass M and length L .

The gravitational force exerted by the rod on the particle is

$$F_{\text{net}} = \frac{GMm}{d(d+L)}$$

due to rod on particle



4 Gravitational Field

A gravitational field is a region around a mass where other masses experience a force of attraction.

Gravitational Field intensity

This is how strong the gravitational field is at a certain point.

It is represented by E_g or I .

$$\vec{I} = \frac{\vec{F}}{m} \text{ Unit} = \text{N/Kg},$$

Intensity due to a Point Mass:-

At a distance r from the center of a body with mass M , the intensity of the gravitational field is given by: $I = \frac{GM}{r^2}$

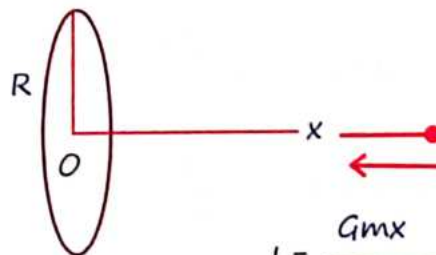
Here, G is the gravitational constant, M is the mass of the body (like Earth), and r is the distance from the center of the mass.

Note:-

The intensity of the gravitational field is a vector quantity, meaning it has both magnitude and direction.

+ The direction of the gravitational field intensity is always towards the center of the mass creating the field (e.g., towards the center of the Earth).

Gravitational field intensity due to ring:-



$$I = \frac{Gmx}{(R^2 + x^2)^{3/2}}$$

Variation of "g" due to rotation of earth about its own axis:-

$g_{\text{eff}} = g_0 - R\omega^2 \cos^2 \theta$ Equator pr acclⁿ due to gravity 0.34% kam hoti hai pole se!

- At pole, $\theta = 90^\circ$, $g_{\text{eff}} = g_0$ (maximum)
- At equator, $\theta = 0^\circ$, $g_{\text{eff}} = g_0 - R\omega^2$ (minimum)

Where,

g_0 = standard gravitational acceleration (9.8 m/s^2).

R is the radius of the Earth.

ω , is the angular velocity of Earth's rotation.

θ is the latitude (measured from the equator).

Special Points:-

Rotational speed and weightlessness

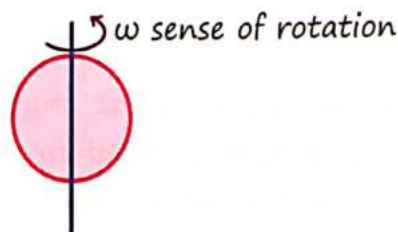
$T = 84.6 \text{ min}$ This indicates that if earth rotates 17 times its present rotational speed so body at equator feels weightless.

Increase in Angular Velocity (ω)

- With increased angular velocity, the centrifugal force acting on objects on Earth's surface increases.
- This would lead to a decrease in the effective gravity (g) experienced by objects at all places except the poles.
- At the poles, the centrifugal force is zero because the poles are on the axis of rotation, so gravity remains unaffected there.
- If $\omega \uparrow$, then g = decrease at all place except pole

Direction of Earth's Rotation and Rocket Launch

- The Earth rotates from west to east, which means if a rocket is launched in the same direction ($W \rightarrow E$), it benefits from the Earth's rotational speed.
- Launching a rocket in the direction of the Earth's rotation (west to east) provides it with additional velocity due to the Earth's rotational speed.
- This can be beneficial for launching rockets into orbit as it reduces the amount of fuel needed to reach the required orbital speed.



Q.4. The depth 'd' at which the value of acceleration due to gravity becomes $\frac{1}{n}$ times the value at the earth's surface is (R radius of earth)

Sol. Acceleration due to gravity at depth d under the surface $g' = g_s \left(1 - \frac{d}{R}\right)$

$$\text{Given : } g' = \frac{g_s}{n}$$

$$\therefore \frac{g_s}{n} = g_s \left(1 - \frac{d}{R}\right)$$

$$\text{or } \frac{1}{n} = \left(1 - \frac{d}{R}\right)$$

$$\Rightarrow d = R \left(\frac{n-1}{n}\right)$$

Q.5. The height from earth's surface at which acceleration due to gravity becomes is $\frac{g}{4}$ (where g is acceleration due to gravity on the surface of earth and R is radius of earth).

Sol. $g_h = g_0 \frac{R^2}{(R+h)^2}$

$$\frac{g_0}{4} = g_0 \frac{R^2}{(R+h)^2}$$

$$\frac{R}{R+h} = \sqrt{\frac{1}{4}} = \frac{1}{2}$$

$$2R = R+h$$

$$h = R$$

6 Gravitational Potential

- Gravitational potential at any point in gravitational field is equal to the work done per unit mass in bringing a very light body from infinity to that point.

$$\text{Gravitational potential, } V_g = \frac{W}{m}$$

- SI unit is J/kg and it is a scalar quantity.
- Dimensional formula is $[L^2 T^{-2}]$.

7 Gravitational PE

Gravitational potential energy is the energy possessed by an object due to its position in a gravitational field. It is the work done to move an object from one point to another in the field.

$$W_{CF} = -\Delta U$$

$$U_B = -[W_{A \rightarrow B}]_{GF}$$

- Here, W_{CF} represents the work done by the conservative force.
- ΔU is the change in gravitational potential energy.

8 Gravitational potential due to point mass at a distn " r "

$$U = -\frac{GMm_0}{r} \text{ \& } V = \frac{U}{m_0} = -\frac{GM}{r}$$

- Since, work W is negative, the gravitational potential is always negative.

Note:-

Gravitational potential energy per unit mass:

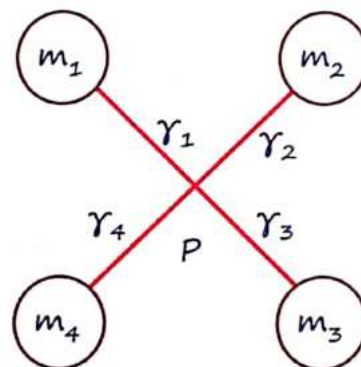
$$V = \frac{U}{M} \quad \vec{I} = \frac{\vec{F}}{M} \quad \vec{I} = -\frac{dv}{dr} \quad \vec{F}_{CF} = -\frac{dU}{dr}$$

$$\vec{I} = -\left[\frac{\delta v}{\delta x} \hat{i} + \frac{\delta v}{\delta y} \hat{j} + \frac{\delta v}{\delta z} \hat{k} \right]$$

$$\Delta V = -\int \vec{I} \cdot d\vec{r}$$

$$\Delta U = -\int \vec{F} \cdot d\vec{r}$$

9 Gravitational potential due to combination of point mass

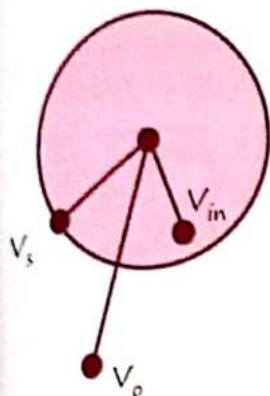


$$V_p = -\frac{Gm_1}{r_1} - \frac{Gm_2}{r_2} - \frac{Gm_3}{r_3} - \frac{Gm_4}{r_4}$$

$$U_p = -\frac{GMm}{r} \quad \text{Total no. of pairs} = \frac{N(N-1)}{2}$$

N is the total no. of masses.

10 Gravitational potential due to

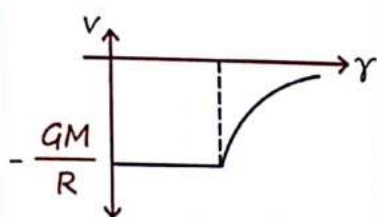


a> Uniform thin shell

$$V_{out} = - \frac{GM}{r}$$

$$V_{surface} = - \frac{GM}{R}$$

$$V_{in} = - \frac{GM}{R}$$



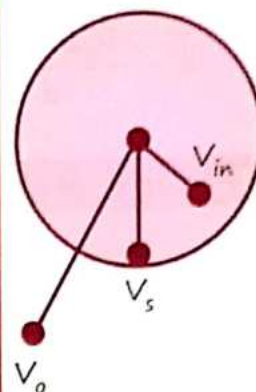
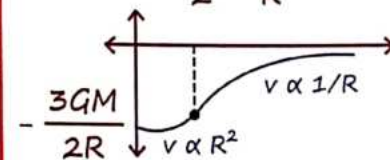
b> Uniform Solid Sphere

$$V_{out} = - \frac{GM}{r}$$

$$V_{surface} = - \frac{GM}{R}$$

$$V_{in} = - \frac{GM}{2R^3} [3R^2 - r^2]$$

$$V_{centre} = - \frac{3}{2} \frac{GM}{R}$$



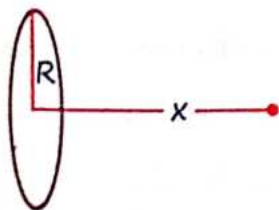
11 Gravitational potential due to uniform ring

At a axial point:-

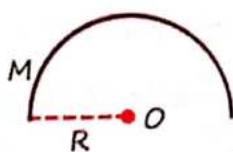
$$V_p = - \frac{GM}{\sqrt{x^2 + R^2}}$$

At center ($x = 0$)

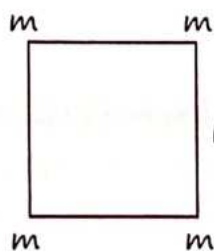
$$V = - \frac{GM}{R}$$



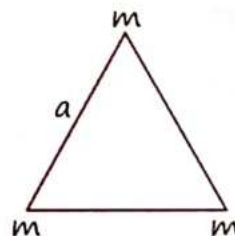
At centre of half ring $V_o = - \frac{GM}{R}$



Ex.: find potential energy of the given system



Ex:

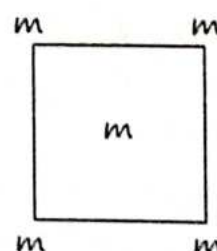
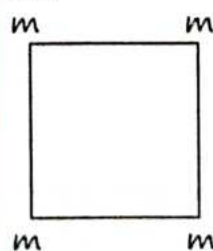


$$U = - \frac{4Gm^2}{l} - \frac{2Gm^2}{\sqrt{2}l}$$

$$U = - \frac{3Gm^2}{a}$$

Q.6. Four point mass m placed at corner of square of side a , find work done in bringing 5th mass m from infinity to centre of square.

Sol.



$$U_i = U_o \text{ (Let)}$$

$$W = \Delta U = U_f - U_i \quad U_f = U_o - \frac{4Gm^2}{a\sqrt{2}}$$

$$= -\frac{4Gm^2\sqrt{2}}{a}$$

12 G. p.E. of earth-mass system

From surface to height "h": -

$$\Delta U = \frac{mgh}{(1 + \frac{h}{R})} = \frac{GMmh}{R^2 \left[1 + \frac{h}{R}\right]}$$

Q.7. Object of mass m raised to height $h = 2R$ from earth surface change in its potential energy.

$$\text{Sol. } \Delta U = \frac{mgh}{1 + \frac{h}{R}} = \frac{mg2R}{1 + \frac{2R}{R}} = \frac{2}{3} mgR$$

13 Escape Velocity

Escape velocity is the minimum velocity an object must have to escape the gravitational pull of a celestial body without any further propulsion.

$$V_e = \sqrt{\frac{2GM}{R}} \quad V_e = \sqrt{2gR}, \quad V_e = \sqrt{\frac{8}{3} G\rho\pi R^2}$$

→ Escape velocity interns of density.

$$\rho = \text{const}^n \quad V_e \propto R$$

$$M = \text{const}^n \quad V_e \propto 1/\sqrt{R}$$

Note:-

Escape velocity does not depend upon angle of projection & mass of object.

14 velocity of object at ∞ when projected with $v_g > V_{ea}$

$$V_{\infty} = \sqrt{V_g^2 - V_e^2}$$

Q.8. Object is projected with double velocity of escape then its velocity in space

$$\text{Sol. } V_{\text{space}} = \sqrt{V_g^2 - V_e^2} = \sqrt{4V_e^2 - V_e^2} = \sqrt{3} V_e$$

If $V_g < V_e$ THEN $h_{\text{max}} = ?$

$$h_{\text{max}} = \frac{R}{\left[\frac{V_e^2}{V_g^2} - 1\right]}$$

Q.9. Object is projected with one-fourth of escape velocity then maximum height attained?

$$\text{Sol. } h = \frac{R}{\left(\frac{V_e}{V_g}\right)^2 - 1} = \frac{R}{16 - 1} = \frac{R}{15}$$

15 Escape velocity from height "h" from surface

$$V = \sqrt{\frac{2GM}{R+h}}$$

For earth to become a black hole

$$R = 1.48 \text{ mm}$$

If object is projected with speed V_g then,

$V_g > V_e \rightarrow \text{ME} = +V_e$, Hyperbolic

$V_g = V_e \rightarrow \text{parabolic}$, ME = 0

$V_o < V_g < V_e \rightarrow \text{Elliptical close}$, ME = -ve

$V = V_o \rightarrow \text{Circular}$ TE = -ve

$V < V_o \rightarrow \text{Elliptical}$ TE = -ve

$V \ll V_o \rightarrow \text{Projectile open path}$

16 Orbital velocity

$$V_o = \sqrt{\frac{GM}{R+h}}$$

$$V_e = \sqrt{2} V_o$$

Satellite near surface of earth:-

$$V_o = \sqrt{gR} = \sqrt{\frac{4}{3} \pi G \rho R^2}$$

Note:-

$V_{\text{satellite}}$ inc. by $\sqrt{2}$ times or 41.4 %
then it'll escape to ∞

Q.10. The radii of the circular orbits of two satellites A and B of the earth are $4R$ and R , respectively. If the speed of satellite A is $3v$, then the speed of satellite B will be

Sol. $V \propto \frac{1}{\sqrt{R}}$

$$\frac{3v}{V} = \sqrt{\frac{R}{4R}}$$

$$V = 3v \times 2 = 6v$$

17 Energies of satellite

Potential energy, $U = -\frac{GMm}{R+h}$

Kinetic energy $K = \frac{GMm}{2(R+h)}$

Total energy $TE = -\frac{GMm}{2(R+h)}$

Binding energy $BE = \frac{GMm}{2r}$

$$U : K : TE : BE :: -1 : \frac{+1}{2} : \frac{-1}{2} : \frac{+1}{2}$$



* B.E. of a particle placed on earth:-

$$BE = \frac{GMm}{R}$$

18 Time period of satellite

$$F \propto r^n \quad T.P. \propto r^{(1-n)/2}$$

$$T = \frac{2\pi r}{V_o} = 2\pi \sqrt{\frac{(R_e + h)^3}{GM_e}}$$

independent of mass of satellite

$$T^2 \propto r^3$$

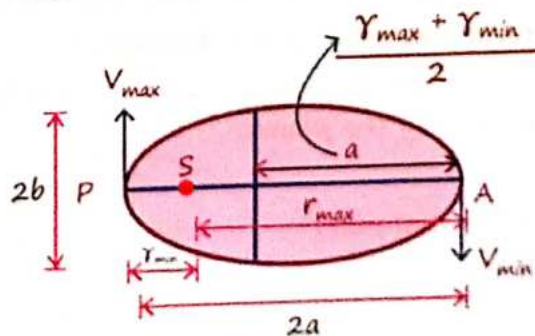
$$K = \frac{4\pi^2}{GM} \quad T = \sqrt{K} r^{3/2}$$

19 Kepler's law

Law of Orbits:

* Planets move in elliptical orbits with the Sun at one focus.

$$\frac{V_{\max}}{V_{\min}} = \frac{r_{\max}}{r_{\min}} = \frac{a(1+e)}{a(1-e)}$$



$$r_{\max} + r_{\min} = 2a$$

Law of Areas:

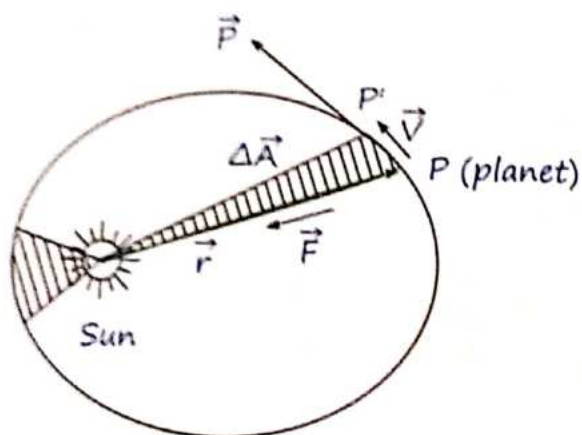
- A line segment joining a planet and the Sun sweeps out equal areas during equal intervals of time. This can be written as:

$$\frac{dA}{dt} = \text{const}^n$$

$$\gamma V = \text{const}^n$$

$$\frac{L}{2m} = \text{const}^n$$

$$\gamma_{\text{mean}} = \frac{\gamma_{\text{max}} + \gamma_{\text{min}}}{2} = a$$



Law of Periods:

- The square of the orbital period (T) of a planet is directly proportional to the cube of the semi-major axis (a) of its orbit.

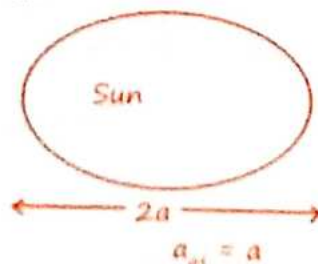
$$T^2 \propto a^3 \propto \left[\frac{\gamma_{\text{max}} + \gamma_{\text{min}}}{2} \right]^3$$

Semi major axis

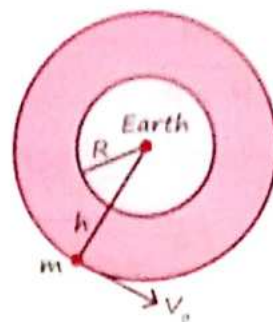
Law of Periods: The square of the time period of revolution of a planet is proportional to the cube of the semi-major axis of the ellipse traced out by the planet.

$$T^2 \propto (\text{Avg rate})^3$$

$$T^2 \propto R^3$$



20 Satellite on circular path



$$|\gamma| = \text{const}^n$$

$$\text{All energies} = \text{const}^n$$

$$L = \text{const}^n \quad \tau = 0$$

$$\vec{P} = \text{variable}$$

$$\vec{V} = \text{variable (due to dir}^n)$$

$$\omega = 0 \quad V = \text{const}^n$$

$$\tau = \frac{dL}{dt} F = \frac{dP}{dt}$$

Note:-

$$+ \text{ Total energy} = - \frac{GMm}{(\gamma_{\text{max}} + \gamma_{\text{min}})}$$

escape energy = +

$$\frac{GMm}{R_e}$$

$$29> V_g = KV_e$$

$$h_{\text{max}} \text{ from Surface of Earth} = \frac{RK^2}{1-K^2}$$

$$\text{If } F \propto \frac{1}{\gamma^n}$$

$$\therefore V_o \propto \frac{1}{\gamma^{n-1/2}}$$

$$\therefore T.P. \propto \gamma^{n+1/2}$$

Q.11. A satellite of mass m moving around planet of mass M in circular path radius r_1 then energy required to shift this path to radius r_2 .

Sol. $E_{\text{given}} = E_{\text{total final}} - E_{\text{total initial}}$

$$= -\frac{GMm}{2r_2} - \left(-\frac{GMm}{2r_1} \right)$$

$$= \frac{GMm}{2} \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$

Q.12. The time period of a satellite in a circular orbit of radius R is T . The period of another satellite in a circular orbit of radius $4R$ is

- (i) $4T$ (ii) $\frac{T}{2}$
- (iii) $8T$ (iv) $\frac{T}{8}$

Sol. (iii) $T^2 \propto R^3$

$$= \left(\frac{T}{T^1} \right)^2 = \left(\frac{R}{4R} \right)^3$$

$$= \left(\frac{T}{T^1} \right)^2 = \frac{1}{64}$$

$$= \frac{T}{T^1} = \frac{T}{\sqrt{64}}$$

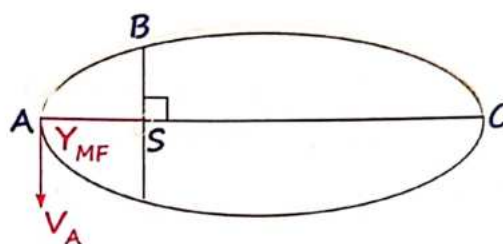
$$= \boxed{8T = T^1}$$

Q.13. The kinetic energies of a planet in an elliptical orbit about the Sun, at positions A, B and C are K_A , K_B and K_C respectively. AC is the

major axis and SB is perpendicular to AC at the position of the sun S as shown in the figure. Then

- (i) $K_A < K_B < K_C$ (ii) $K_A < K_B < K_C$
- (iii) $K_B < K_A < K_C$ (iv) $K_B < K_A < K_C$

Sol. (ii)

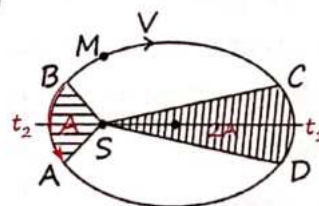


disⁿ ↓
Speed ↓ = $A > V_B > C$

Q.14. The figure shows elliptical orbit of a planet m about the sun S . The shaded area SCD is twice the shaded area SAB . If t_1 is the time for the planet to move from C to D and t_2 is the time to move from A to B then

- (i) $t_1 = 4t_2$ (ii) $t_1 = 2t_2$
- (iii) $t_1 = t_2$ (iv) $t_1 > 4t_2$

sol. (ii)



$\boxed{2t_2 = t_1}$

MR*

Question krte time Galat option ko dhoondo, sahi apne aap mil jayenge.

1 Deforming Force

A deforming force is any force that changes the shape or configuration of an object when applied.

For example: - bending a rod or compressing a spring involves applying a deforming force.

Elasticity:

- + Elasticity is the property of a material to resist deformation. This implies that materials which are more elastic can withstand deformation better without undergoing a permanent change in shape.

Example: - Steel is more elastic than rubber. This means steel returns to its original shape more efficiently after the deforming force is removed compared to rubber.

Elastic Limit

- + The elastic limit is the maximum extent to which a material can be deformed and still return to its original shape once the force is removed. Beyond this limit, the material will be permanently deformed. For example, stretching a spring too far may cause it to lose its ability to return to its original shape.

Perfectly Elastic Bodies

- + These are ideal materials that can completely return to their original shape immediately after the deforming force is removed. Examples include quartz and phosphor bronze.

- + Ex. Quartz is near approach to perfect elastic body.

$$\gamma = \infty \quad e = 0.$$

Perfectly Plastic Bodies:

- + These materials do not regain their original shape at all after the deforming force is removed. They remain deformed. Examples include putty, paraffin, and wax.

2 Stress

$$\frac{F_{\text{restoring}}}{\text{Area}} = \text{stress}$$

Stress is not a vector quantity unlike a force, the stress cannot be assigned a specific direction. Force acting on the portion of a body on a specified side of a section has definite direction.

Types of Stress**Normal Stress**

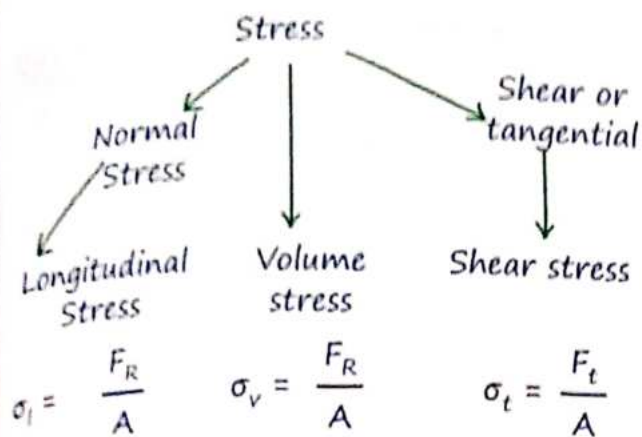
- + Deals with forces applied perpendicular to the surface, causing changes in length.

Volumetric Stress

- + Deals with forces applied uniformly over the entire surface, causing changes in volume.

Tangential Stress

- + Deals with forces applied parallel to the surface, causing changes in shape.



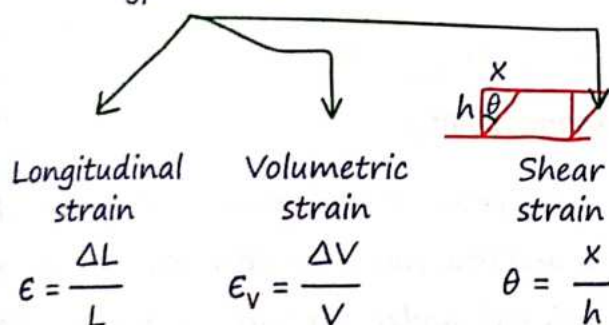
* Volume Stress = ΔP

3 Strain

- It's the fractional change in configuration due to an applied force.

$$\text{Strain} = \frac{\text{Change in size the body}}{\text{Original size of the body}}$$

type of strain



Strain is unitless and dimensionless

- Longitudinal Strain occurs when there is a change in length.
- Volumetric Strain occurs when there is a change in volume.
- Shearing Strain occurs when there is a change in shape due to layers sliding over each other.

4 Hooke's Laws

Within the elastic limit, the stress applied to a material is directly proportional to the strain it produces.

i.e.,

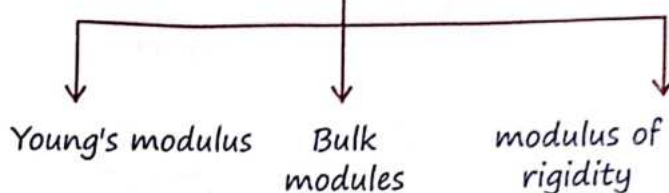
Stress $\propto$ Strain

$$\sigma \propto \epsilon \quad \sigma = E\epsilon \quad \boxed{E = \frac{\sigma}{\epsilon}}$$

Where, E = modulus of elasticity, depends on the nature of the material of the body and is independent of its dimensions (i.e. length, volume etc.)

Material Prop.

Modulus of Elasticity (E) = $\frac{\text{Stress}}{\text{Strain}}$



5 Young's Modulus

Young's modulus is the ratio of normal stress to the longitudinal strain.

$$Y = \frac{\text{Longitudinal Stress}}{\text{Longitudinal Strain}}$$

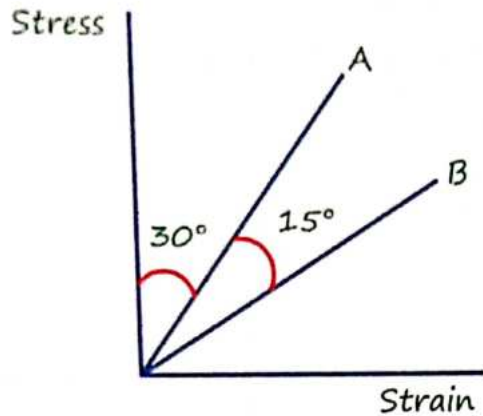
$$\boxed{Y = \frac{FL}{A\Delta L}}$$

- Hooke's law is valid only in the linear part of stress-strain curve.
- The Young's modulus and shear modulus are relevant only for solids since only solids have lengths and shapes.
- Metals have larger values of Young's modulus than alloys and elastomers. A material with large value of Young's modulus requires a large force to produce small changes in its length.

- * Slope of stress strain graph is Young's modulus.

Q.1. Find ratio of young's modulus

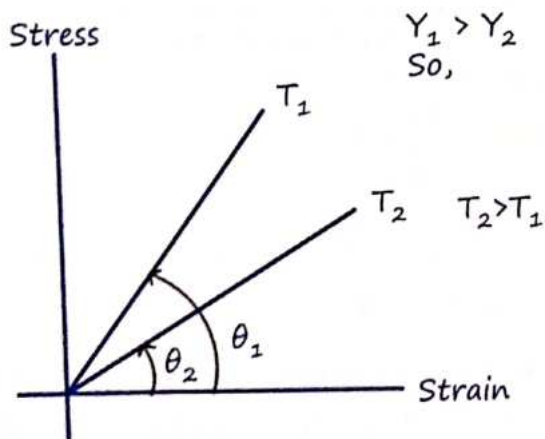
Sol. Slope of stress-strain graph is young's modulus.



$$Y = \frac{Y_A}{Y_B} = \frac{(\text{Slope})_A}{(\text{Slope})_B} = \frac{\tan 60^\circ}{\tan 45^\circ} = \sqrt{3} : 1$$

- * Young Modulus $\propto \frac{1}{\text{Temperature}}$

e.g.



6 Bulk's Modulus

- * The bulk modulus is the ratio of volumetric stress to the volumetric strain.

$$\beta = \frac{\text{volume Stress}}{\text{volume strain}}$$

- * Bulk modulus is relevant for solids, liquid and gases. It refers to the change in volume when every part of the body is under the uniform stress so that the shape of the body remains unchanged.

$$\beta = -V \left[\frac{\Delta P}{\Delta V} \right]$$

Isothermal
($T = \text{const}$)
 $\beta = P$

Adiabatic
($Q = \text{const}$)
 $\beta = P\gamma$

Isobaric
($P = \text{const}$)
 $\beta = 0$

$\gamma = \frac{C_p}{C_v}$ Isochoric
($V = \text{const}$)
 $\beta = \infty$

- * $B_{\text{solid}} < B_{\text{liquid}} > B_{\text{gas}}$

Compressibility

- * Compressibility is a measure of how much a material can be compressed (reduced in volume) under pressure. It indicates the relative volume change of a material in response to a pressure change.

○ Compressibility (C) of a material is the reciprocal of its bulk modulus of elasticity (K)

- Mathematically, it is expressed as

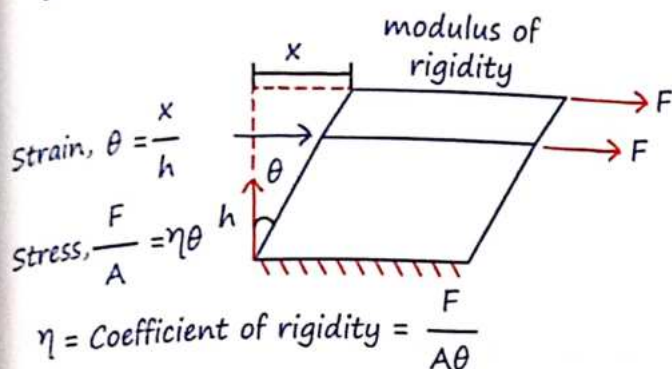
$$C = \frac{1}{\beta} = \frac{1}{-V} \frac{\Delta V}{\Delta P}$$

- Density of Compressed Liq. $\uparrow$

$$\rho' = \rho \left[1 + \frac{\Delta P}{K} \right] = \rho [1 + C\Delta P]$$

7 Shear Modulus

- Shear modulus or modulus of rigidity is the ratio of tangential stress to the shearing strain.



Potential Energy Stored in Wire:-



$x = \text{elongation}$
work done in elongation x
 $w = Fx$

Energy stored $= \frac{1}{2} Fx = U$

$U = \frac{1}{2} \frac{A\gamma x^2}{L}$

using Hooke's law

Potential Energy Stored in wire unit volume:-

$$\frac{U}{V} = \frac{1}{2} \gamma (\text{Strain})^2 = \frac{1}{2} \frac{(\text{Stress})^2}{\gamma}$$

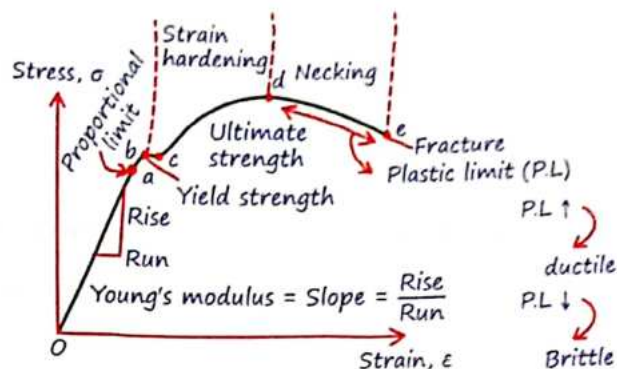
$$= \frac{1}{2} \text{Stress} \times \text{Strain.}$$

Breaking Stress (P) :- Breaking stress does not depend upon area.
Material Property.

$P = \frac{\text{Breaking Force}}{\text{Area}}$

* For a Particular Material: - $B. \text{Force} \propto A$

Stress-strain curve



Elastic region, O-a:-

- Hooke's law Valid. (Stress $\propto$ Strain)

F_{elastic} Conservative.

Completely regain its shape.

O-b:-

- Conservative limit.

Stress $\neq$ Strain

Almost Regain its shape.

F_{elastic} Conservative.

Plastic region b-c:-

- F_{elastic} is not Conservative

Non-linear region where permanent deformation occurs.

Fracture Point (D)

- The point where the material breaks.

Q.2. If the longitudinal strain of a iron rod is 0.01 and its poisson's ratio of 0.2, then the lateral strain will be

Sol. Since,

Poisson's ratio $= \frac{\text{Lateral Strain}}{\text{Longitudinal Strain}}$

$\Rightarrow \text{Lateral strain} = (\text{Poisson's ratio}) \times (\text{Longitudinal strain})$

$\Rightarrow \text{Lateral strain} = (0.2) (0.01) = 0.002$

Q.3. A wire elongates by L mm when a load W is hanged from it. If the wire goes over a pulley and two weights W each is hung at the two ends, the elongation of the wire will be (in mm).

Sol. Due to the arrangement of the pulley, the length of wire is $L/2$ on each side and so the elongation will be $L/2$. For both sides, elongation = L

Poisson's Ratio

$$\sigma = \frac{\text{Lateral Strain}}{\text{Longitudinal Strain}}$$

$$\sigma = - \frac{\Delta D/D}{\Delta L/L}$$

Volume strain = $2 \times$ lateral strain

* longitudinal strain

$$\sigma = 0.5 \quad V = \text{constant.}$$

Q.4. If the elastic potential energy density store in a material is $3 \times 10^4 \text{ J/m}^3$ due to the application of longitudinal stress of $1 \times 10^{11} \text{ N/m}^2$ then, the strain developed in it would be

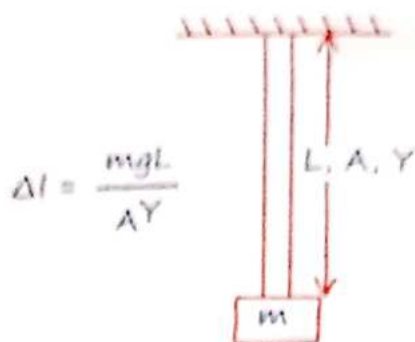
$$\text{Sol. } U_e = \frac{1}{2} \text{ stress} \times \text{strain}$$

$$\text{strain} = \frac{2U_e}{\text{stress}}$$

$$\text{strain} = \frac{2 \times 3 \times 10^4}{10^{11}} = 6 \times 10^{-7}$$

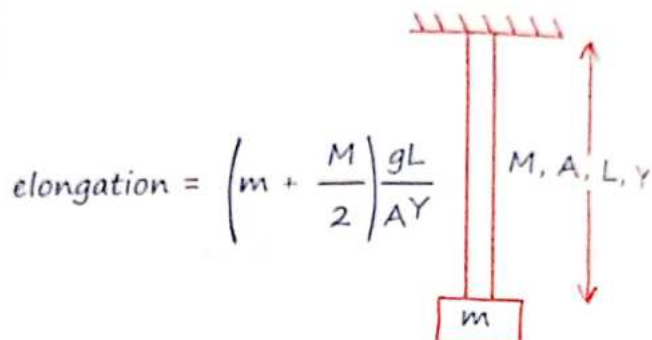
Special Case:-

* Elongation in massless rod due to attached block of mass m :



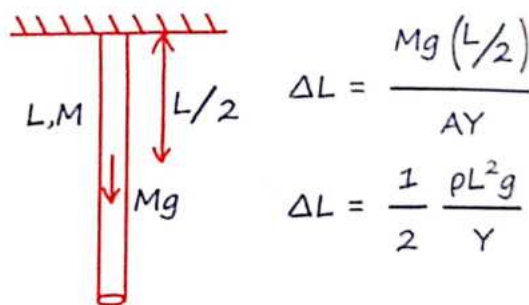
$$\Delta l = \frac{mgL}{AY}$$

* Rod have mass M :



$$\text{elongation} = \left(m + \frac{M}{2} \right) \frac{gL}{AY}$$

* Elongation due to own weight:-



$$\Delta L = \frac{Mg(L/2)}{AY}$$

$$\Delta L = \frac{1}{2} \frac{\rho L^2 g}{Y}$$

* $l_{\max}$ of a wire which can hang under his own weight:-

$$l_{\max} = \frac{P}{\rho g}$$

* Linear Strain = $\alpha \Delta T$

$$\text{Linear Stress} = Y \alpha \Delta T$$

Q.5. Two wires are made of the same material and have the same volume. However, wire 1 has a cross-sectional area A and wire 2 has cross-sectional area $3A$. If the length of the wire increases by Δx on applying force F , how much force is needed to stretch wire 2 by the same amount?

Sol. For the same material, Young's modulus is the same and it is given that the volume is the same and the area of the cross-section for the wire L_1 is and that of L_2 is $3A$

$$V = V_1 = V_2$$

$$V = A \times L_1 = 3A \times L_2 \rightarrow L_2 = L_1/3$$

$$Y = (F/A)/(\Delta L/L)$$

$$F_1 = YA(\Delta L_1/L_1)$$

$$F_2 = Y3A(\Delta L_2/L_2)$$

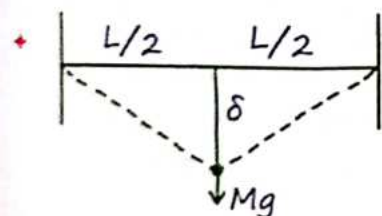
Given $\Delta L_1 = \Delta L_2 = \Delta x$ (for the same extension)

$$F_2 = Y3A(\Delta x/(L_1/3)) = 9 \cdot (YA\Delta x/L_1) =$$

$$9F_1 = 9F$$

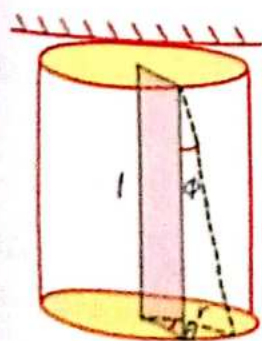
* Fractional Change in Radius of Sphere:-

$$\beta = \text{Bulk Modulus.} \quad \frac{dR}{R} = \frac{F}{3A\beta} = \frac{Mg}{3A\beta}$$



$$\delta = \frac{L}{2} \left[\frac{Mg}{AY} \right]^{2/3}$$

* Relation between shear angle and angle of twist.



θ = Twist Angle
 ϕ = Shear Angle

$$PE = \frac{1}{2} C\theta^2 = \frac{1}{2} Ke^2$$

$$r\theta = l\phi$$

* Relation among Y , K , η & σ :-

$$Y = \frac{9\eta K}{3K + \eta}$$

$$Y = 2\eta [1 + \sigma]$$

$$Y = 3K [1 - 2\sigma]$$

$$Y = \frac{3K - 2\eta}{6K + 2\eta}$$

* If length of wire l_1 at tension T_1 and l_2 at tension T_2 , then find original length $T_1 \rightarrow l_1 \Rightarrow T_2 \rightarrow l_2$

$$L = \frac{T_1 l_2 - T_2 l_1}{T_1 - T_2} = \frac{T_2 l_1 - T_1 l_2}{T_2 - T_1}$$

Q.6. When a block of mass M is suspended by a long wire of length L , the length of the wire becomes $(L+l)$. The elastic potential energy stored in the extended wire is:

Sol. Using Formula

$$U = \frac{1}{2} \times \text{stress} \times \text{strain} \times \text{volume}$$

$$\text{stress} = \frac{\text{Force}}{\text{Area}} = \frac{Mg}{A}$$

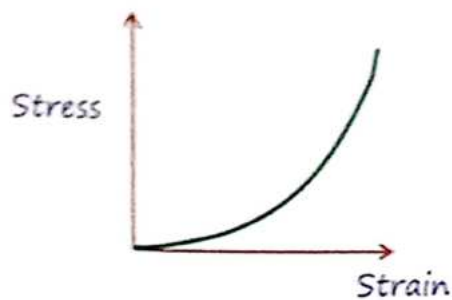
$$\text{strain} = \Delta L/L = l/L$$

$$\text{volume} = \text{area} \times \text{length} = A \times L$$

$$\text{Hence, } U = \frac{1}{2} \times \frac{Mg}{A} \times \frac{l}{L} \times A \times L$$

$$U = \frac{1}{2} Mgl$$

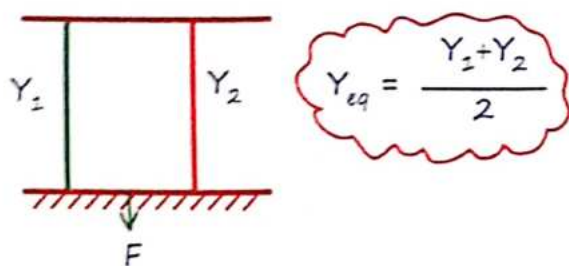
Stress-Strain graph for Elastomer:- e.g.,
Tissue of Aorta,



* Parallel & Series Combination of Young's Modulus:-

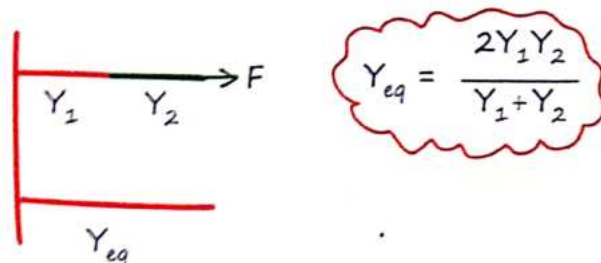
1> Parallel Combination:-

In both wire have same elongation but different stress



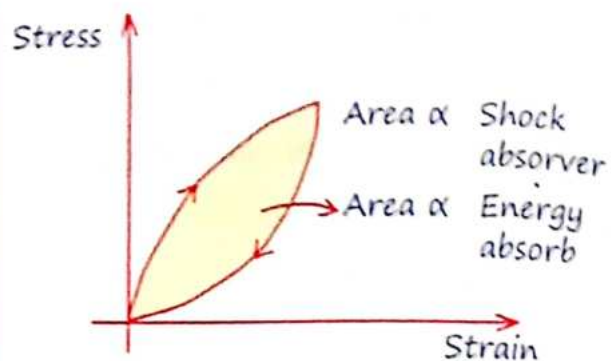
2> Series Combination:-

In both wire have same stress but different elongation.

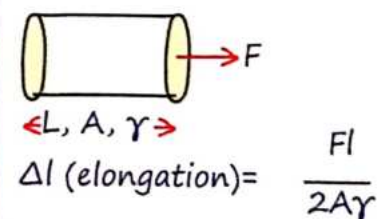
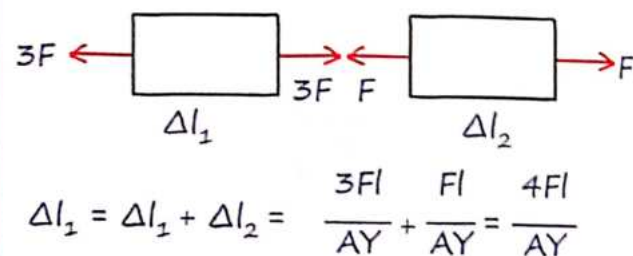
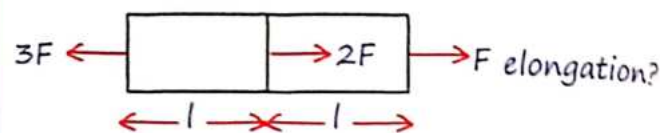
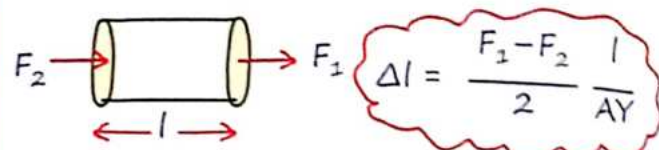
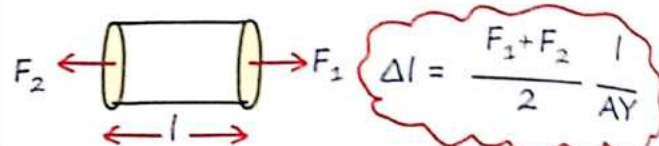


NOTE:-

* On removing deforming Force Solid regain its shape by this graph.



Special Type Questions:-



MR*

“Push Yourself, because No One Else is going to do it for you.”

Mass - quantity of matter contains in a body.
 → Scalar, Unit - kg
 1 kg = 10^3 gm

15

Fluid Mechanics

Volume - 3-D space occupied by object is called volume
 1 L = 10^{-3} m^3

Substances that can flow when an external force is applied. This includes both liquids and gases.

Types of Fluids

Liquids:

○ Have a definite volume but take the shape of their container.

○ Examples: Water, oil, and milk.

Gases:

○ Do not have a definite volume or shape and expand to fill their container.

○ Examples: Air, helium, and carbon dioxide.

+ Relative density (R.D.): - Specific gravity

○ Relative density is the ratio of the density of a substance (object) to the density of water.

$$\text{R.D.} = \frac{\rho_{\text{obj.}}}{\rho_{\text{water}}} = \text{Unitless.}$$

$$\rho_w = 1 \text{ gm/cm}^3 \text{ (Water)}$$

$$\rho_{\text{oil}} = 0.8 \text{ gm/cm}^3 \text{ (Oil)}$$

$$\rho_{\text{Hg}} = 13.6 \text{ gm/cm}^3 \text{ (Mercury)}$$

$$\rho_{\text{milk}} = 1.04 \text{ gm/cm}^3$$

Example 1: Oil $\rho_{\text{oil}} = 0.9 \text{ gm/cm}^3$

$$\text{R.D.} = \frac{\rho_{\text{oil}}}{\rho_w} = \frac{0.8 \text{ g/cm}^3}{1 \text{ g/cm}^3} = 0.8$$

The relative density of oil is 0.8, meaning oil is less dense than water.

Example 2: Mercury

$$\text{R.D.} = \frac{\rho_{\text{Hg}}}{\rho_w} = \frac{13.6 \text{ g/cm}^3}{1 \text{ g/cm}^3} = 13.6$$

The relative density of mercury is 13.6, meaning mercury is much denser than water.

Note:-

+ Objects with a relative density less than 1 will float in water.

+ Objects with a relative density greater than 1 will sink in water.

2 Density

linear mass density $\lambda = \frac{dm}{dl}$
 areal mass density $\sigma = \frac{dm}{dA}$
 volume mass density $\rho = \frac{dm}{dV}$

Density of an object is the ratio of its mass to its volume. i.e.,

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

+ it is a scalar quantity with dimensional formula

$$[ML^{-3}]$$

+ Density of water = 1 g/cm^3 or 10^3 kg/m^3

Density of mixture:-

$$\rho_{\text{mix}} = \frac{M_{\text{mix}}}{V_{\text{Total}}}$$

$$\rho_{\text{mix}} = \frac{m_1 + m_2}{V_1 + V_2}$$

Same Mass, Different Density

m = same ρ = different

$$\rho_{\text{mix}} = \frac{\frac{2m}{\rho_1} + \frac{2m}{\rho_2}}{\frac{2m}{\rho_1} + \frac{2m}{\rho_2}} = \frac{2\rho_1\rho_2}{\rho_1 + \rho_2}$$

$$\rho_{\text{mix}} < \rho_2$$

Same volume, different density

$V = \text{same}$ $\rho = \text{different}$

$$P_{\text{mix}} = \frac{\rho_1 V + \rho_2 V}{2V} = \frac{\rho_1 + \rho_2}{2}$$

3 Hydrostatics or static fluid

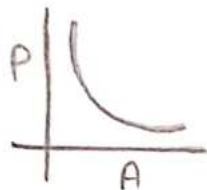
- + This branch deals with fluids at rest.
- + It studies how fluids exert pressure and the effects of this pressure.
- + Example: Water in a tank or a swimming pool.

Thrust:-

- + Thrust is the total normal force exerted by a liquid at rest on a given surface.
- + SI Unit: Newton (N).

Pressure (P):-

$$P = \frac{F}{A}$$



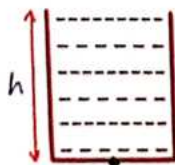
- + It is a scalar quantity
- + Unit: N/m^2
- + Dimension: $[\text{ML}^{-1}\text{T}^{-2}]$
- + $1 \text{ atm} = 1.01 \times 10^5 \text{ N/m}^2$

Variation of "P" with depth:-

$$\Delta P = \rho gh$$

ρ = density of liquid.

h = depth.



Note:-

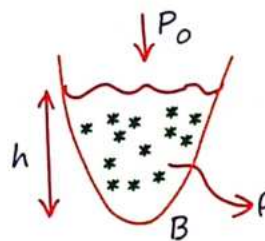
- + ΔP does not depend upon amount, liquid, shape of container.
- + The mean pressure is given by:

$$\frac{\rho gh}{2}$$

This means that on average, the pressure exerted by the liquid on the walls of the container is half of the pressure at the bottom of the container.

4 Pascal's Law

Static liquid, "P" on horizontal level must be same.



$$*P_B = P_0 + \rho gh.*$$

Relation between Absolute & Gauge Pressure:-

$$P_{\text{absolute}} = P_{\text{net}}$$

$$P_{\text{gauge}} = P_{\text{due to liq. only.}}$$

$$P_{\text{absolute}} = P_0 + P_{\text{gauge}}$$

Moving Container "P" Calculation:-

(Be "g_{eff}" lelana hai!)

1. Lift is moving up:- (with "a")

$$P_{\text{net}} = \rho(g + a)H$$

2. Lift is moving down:- (with "a")

$$P_{\text{net}} = \rho(g - a)H$$

3. Lift is in free fall:- ($a = g$)

$$P_{\text{net}} = 0$$

MR*

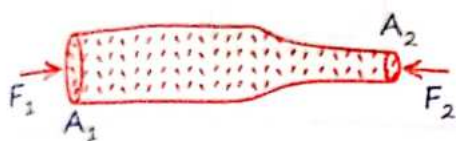


Pascal चर्चा :-

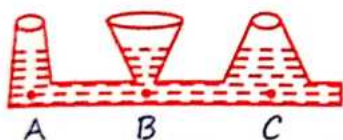
Static fluid me pressure balance karte hai

$$\frac{F_1}{\pi r_1^2} = \frac{F_2}{\pi r_2^2}$$

- pressure is the same at any two points at the same level in the fluid. The shape of the container does not matter



$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

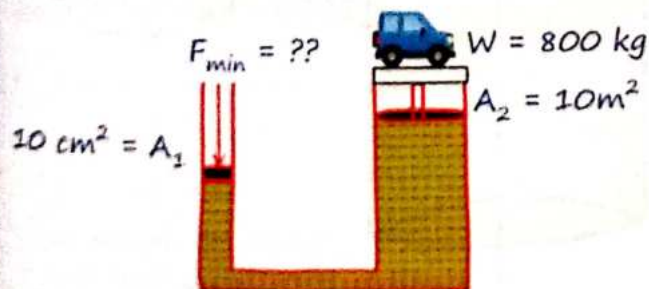


$$P_A = P_B = P_C$$

Q.1. In a hydraulic jack as shown, mass of the car $W = 800 \text{ kg}$, $A_1 = 10 \text{ cm}^2$, $A_2 = 10 \text{ m}^2$. The minimum force F required to lift the car is

- (a) 1N (b) 0.8 N
(c) 8 N (d) 16 N

Sol.



$$\frac{F_{\min}}{10 \text{ cm}^2} = \frac{800 \times 10}{10 \times 10^4 \text{ cm}^2}$$

$$F_{\min} = 0.8 \text{ N}$$

MR*

- When air bubble move from bottom to surface of lake then $p_1 v_1 = p_2 v_2$

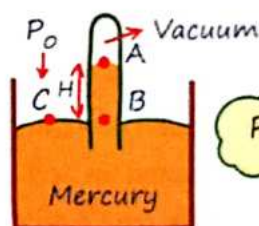
2> MR**

Agar Koi Cheez linearly vary
Krti hai toh uska Avg:-

$$X_{\text{avg}} = \frac{X_i + X_f}{2}$$

5 Measurement of atmospheric Pressure

1. Barometer:- "Torricilli"



$$P_B = P_0 = \rho_{\text{Hg}} g H$$

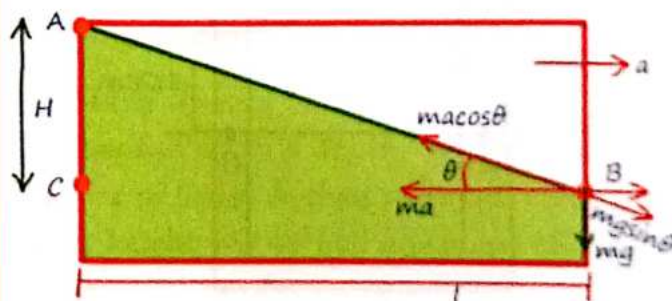
$$H = 76 \text{ cm of Hg} = 1 \text{ atm}$$

The Garib Ramlal Exp:-

Height of water column in Barometer:-

$$H = 10.1 \text{ m}$$

6 Horizontal Accelerating Lift

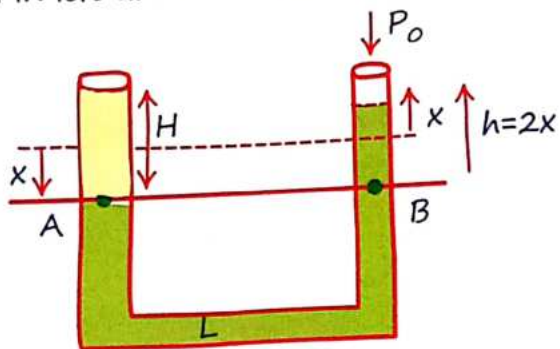


$$\tan \theta = \frac{a}{g} = \frac{H}{l}$$

$$\sin \theta = \frac{a}{\sqrt{a^2 + g^2}}$$

7 U-Tube

Initially one liquid is on same horizontal level, other liquid is put in left arm of container then 1st liquid moves x down from initial level in left arm of U-tube.



$$P_A = P_B$$

$$\rho_w g H = \rho_L g (2x)$$

ek aise line select Karo jiske about "P" same hona chahiye.

Q.2. A liquid X of density 3.36 g cm^{-3} is poured in a U-tube, which contains Hg. Another liquid Y is poured in left arm with height 8 cm, upper levels of X and Y are same. What is density of Y? (given - height of $x = 10 \text{ cm}$)

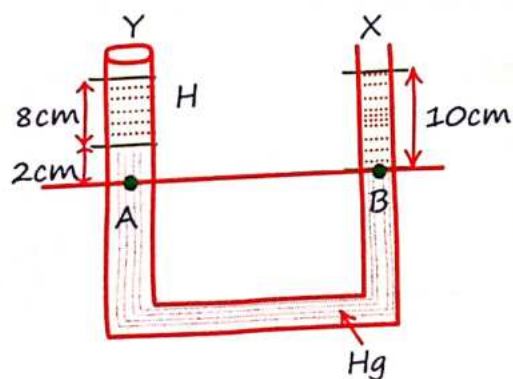
(a) 0.8 gcc^{-1}

(b) 1.2 gcc^{-1}

(c) 1.4 gcc^{-1}

(d) 1.6 gcc^{-1}

Sol.



$$P_A = P_B$$

$$P_0 + \rho_Y g 8 \text{ cm} + \rho_{\text{Hg}} g 2 \text{ cm} = P_0 + \rho_X g 10 \text{ cm}$$

$$\rho_Y 8 + 13.6 \times 2 = 3.36 \times 10$$

$$8\rho_Y + 27.2 = 33.6$$

$$8\rho_Y = 6.4$$

$$\rho_Y = \frac{6.4}{8}$$

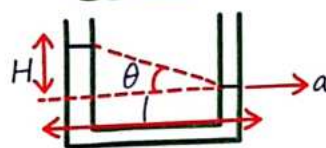
Important Case

1> When U-tube is given horizontal acc

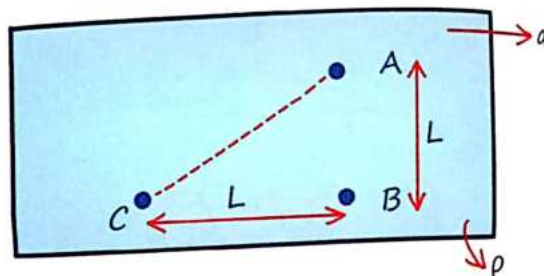
(a) rise is liquid column:-

$$\frac{a}{g} = \frac{H}{l} = \tan \theta$$

$$H = al/g$$



2>

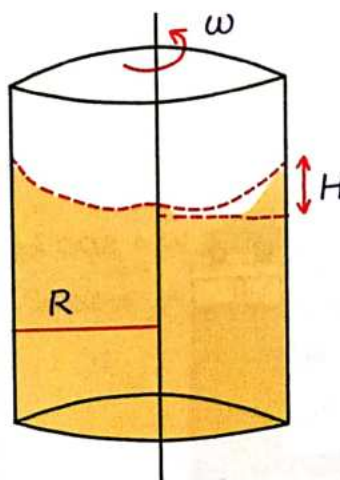


$$P_C - P_A = \rho L (a + g)$$

$$P_B - P_A = \rho g L$$

$$P_C - P_B = \rho a L$$

3> A vessel is rotated about vertical axis. Find rise in water "H":-



$$H = \frac{R^2 \omega^2}{2g}$$

8 Open-Tube Manometer

Gas Container:

- Contains the gas whose pressure we want to measure.

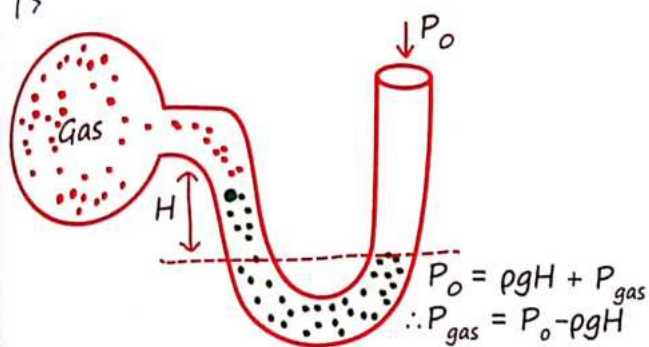
U-Shaped Tube:

- One end is connected to the gas container, and the other end is open to the atmosphere.

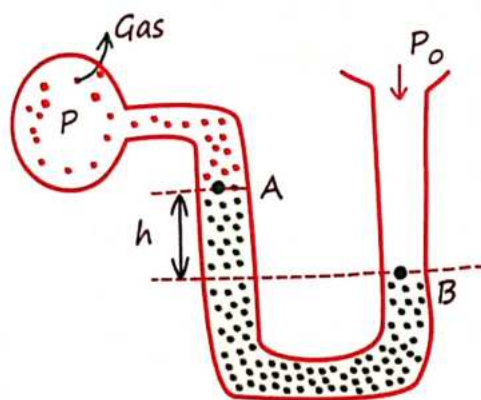
Liquid Column:

- The liquid inside the U-shaped tube moves in response to the pressure differences.

I >



II >



9 Archimedes Principle

F_B (Buoyant Force) = Weight of displaced liquid.

- depends on density of liquid, volume of solid submerged in liquid.

* does not depend on density, location of object inside liquid.

Volume of displaced liq. = Volume of Solid in liq.
 Volume of solid inside liquid

$$F_B = mg = \rho V_{\text{in}} g$$

Reason is "P" difference
 liquid

* Apparent Weight:-

σ = object

ρ = Liquid

$\sigma > \rho$	$\sigma = \rho$	$\sigma < \rho$
Sink!	Submerge & Float.	Float on Surface
$N = mg_{\text{eff}} \left[1 - \frac{\rho}{\sigma} \right]$	$N = 0$	$N = 0$
$N = mg_{\text{eff}} \left[1 - \frac{\text{Chota}}{\text{bada}} \right]$	Object Remains where it's Placed!	$\frac{V_{\text{in}}}{V_T} = \frac{\sigma}{\rho}$
$N = mg_{\text{eff}} \left[1 - \frac{1}{RD} \right]$		$\frac{V_{\text{in}}}{V_T} = \frac{\text{Chota}}{\text{bada}}$

* Object of density " σ " is released then find acclⁿ of object inside liquid:-
 ($\sigma > \rho$)

$$a = g \left[1 - \frac{\rho}{\sigma} \right]$$

* A ball of density "D" is immersed in liquid of density "d" to a depth "h" below the surface of liquid & then released. Upto what height will the ball jump out of the liquid:-

Condition = $d > D$

$$H = h \left[\frac{d}{D} - 1 \right]$$

*A container is at rest then inside volume is V_{in} and when accelⁿ up with a_0 the inside volume is V_{in}' of an object then:-

$$V_{in} = V_{in}'$$

Container ke accelⁿ se up/down volume emerged in liquid depend hi nai Krta.

Buoyant Force with Cavity:-

$$V_{matter} = V_T - V_C$$

$$*N = \sigma[V_T - V_C]g - \rho V_T g$$

To Find V_{cavity} :-

$$1> N = mg - F_B$$

$$F_B = \rho_L V_T g$$

$$V_{matter} = \frac{m}{\sigma}$$

$$V_C = V_T - V_M$$

Problem Solving Strategy

* Rise / Fall of liq:-

When ICE placed on liquid will Melt.

$$\rho_L > \rho_0 = \text{Rise}$$

$$\rho_L < \rho_0 = \text{Fall}$$

$$\rho_L = \rho_w = \text{Same.}$$

$$\rho_0 = \text{density of water.}$$

$$\rho_L = \text{Surr. Liquid.}$$

10 Hydrodynamics or fluid dynamics

+ This branch deals with fluids in motion.

Example: Water flowing in a river or air flowing around an airplane wing.

Ideal Fluid:-

+ Non-viscous

+ Incompressible

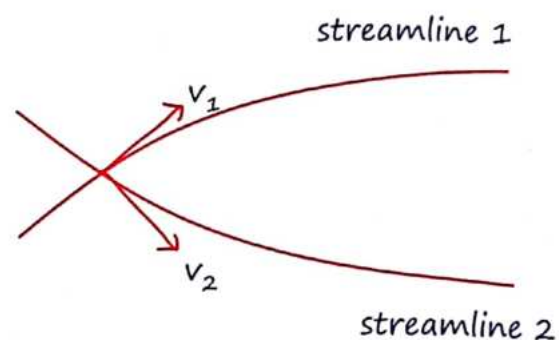
$$+ \rho = \text{const}^n$$

+ Streamline & IRROTATIONAL Flow.

11 Properties of Streamlines

+ The tangent at any point of a streamline gives the direction of flow of fluid particle at that point.

+ In a steady flow, no two streamlines can cross each other, for if they do so, two tangents can be drawn at the point of intersection. It means the oncoming fluid particle can go either one way or the other. Thus, the flow would not be steady



+ Fluid velocity remains constant at any point of a streamline, but it may be different at different points of the same streamline.

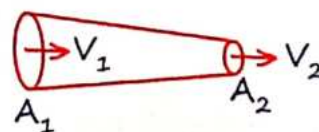
+ Fluid velocity is greater at the regions where streamlines are closely spaced. This can be proved from the equation of continuity that we discuss in the upcoming section.

Equation of Continuity:-

(Conservⁿ of Mass)

$$* \text{Area} \times \text{Velocity} = \text{Const}^n$$

Rate of Volume Flow remains Constⁿ.

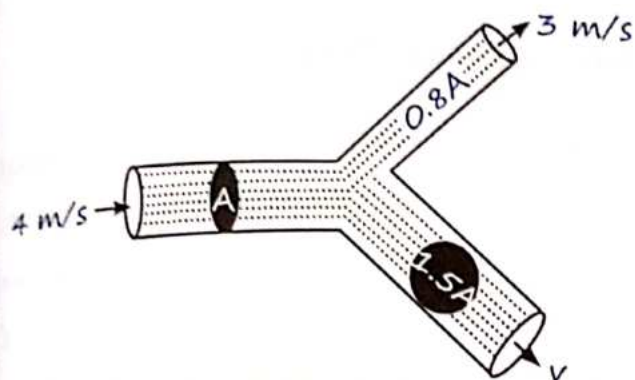


$$A_1 V_1 = A_2 V_2$$

water flowing from the tap.

Q3. An incompressible liquid is flowing through a horizontal pipe as show in figure. Find the value of v .

Sol. $(AV)_{\text{incoming}} = (AV)_{\text{outgoing}}$



$$4A = 3(0.8A) + \frac{3}{2}Av$$

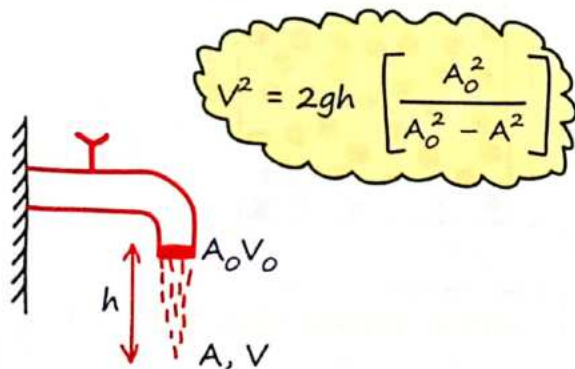
$$4 = 2.4 + 1.5v$$

$$4 - 2.4 = 1.5v$$

$$1.6 = 1.5v$$

$$v = \frac{1.6}{1.5} = 1.1 \text{ m/s}$$

Freely falling liquid:-

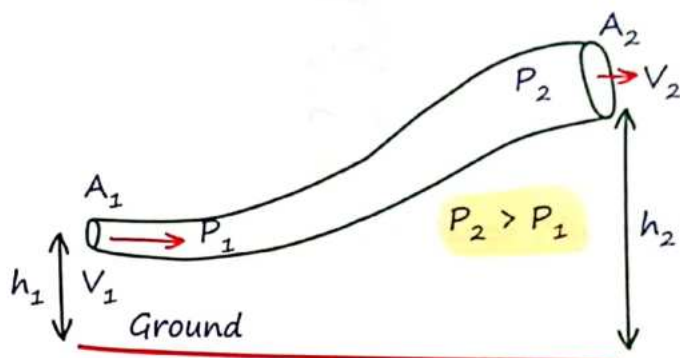


$$V^2 = 2gh \left[\frac{A_0^2}{A_0^2 - A^2} \right]$$

$$\text{Rate of Volume Flow} = AA_0 \sqrt{\frac{2gh}{A_0^2 - A^2}}$$

12 Bernoulli's equation

+ Bernoulli's Theorem states that for an ideal fluid (non-viscous, incompressible, and irrotational) flowing in a streamline manner, the total energy (pressure energy, kinetic energy, and potential energy) per unit volume of the fluid remains constant along a streamline.



$$P + \underbrace{pg h}_{\text{PE/V}} + \underbrace{\frac{1}{2} \rho V^2}_{\text{KE/V}} = \text{Const}^n$$

Divide by ρg .

$$\underbrace{\frac{P}{\rho g}}_{\text{Pressure head.}} + \underbrace{h}_{\text{Gravit}^n \text{ head.}} + \underbrace{\frac{V^2}{2g}}_{\text{Velocity head.}} = \text{Const}^n$$

Where:

- + P is the static pressure.
- + ρ is the density of the fluid.
- + v is the velocity of the fluid.
- + g is the acceleration due to gravity.
- + h is the height above a reference level.

Ramlal House

Wind is flowing outside then pressure inside P_1 and outside house P_2

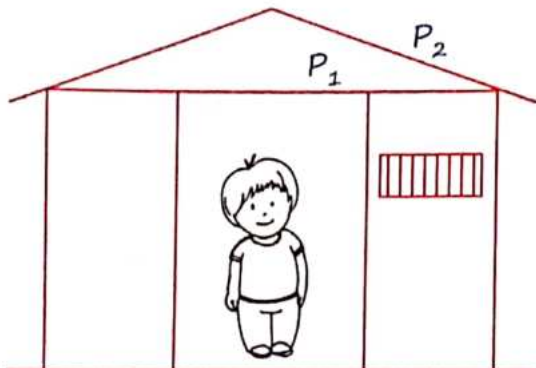
$$P_2 < P_1$$

Hence upthrust force will act on roof

$$F = (P_1 - P_2) \times \text{Area}$$

- * Another way to express this force using the wind speed (V) and the density of air (ρ) is:

$$F = \frac{1}{2} \rho V^2 A$$



Bernoulli's equation in horizontal flow:-

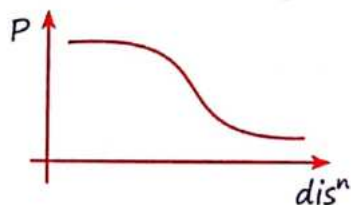
#



$$P + \frac{1}{2} \rho V^2 = \text{const}^n$$

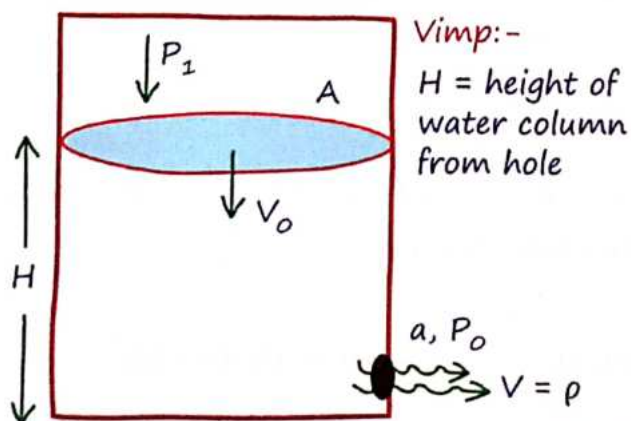
$$AV = \text{const}^n$$

Potential energy same on horizontal level



13 Velocity of Efflux

Closed container:



Vimp:-

H = height of water column from hole

$$V = \sqrt{\frac{2(P_1 - P_0)}{\rho} + 2gH}$$

$$P_1 = P_{\text{Top}}$$

$$\rho = \text{liquid}$$

$$P_0 = \text{atm P.}$$

$$AV_0 = av$$

Open Container (Torricelli चोचा)

- * Torricelli's Theorem states that the velocity of efflux (the speed at which a fluid exits through an orifice) is equivalent to the velocity that a freely falling object would acquire if it fell from the same height.

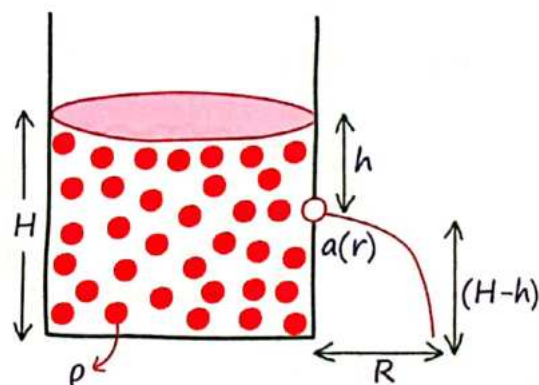
$$P_1 = P_0$$

$$V = \sqrt{2gh}$$

h:- height of liquid level from hole.

Velocity of efflux does not depend on density of liquid.

14 Range, Heights, Time of Flight of Efflux



$$\text{Rate of Volume Flow} = a \sqrt{2gh}$$

Time to reach base level:-

$$T_f = \sqrt{\frac{2(H-h)}{g}} \quad \left. \vphantom{T_f} \right\} \text{Take } H_{\text{vertical}}$$

Horizontal range:-

$$R = U_x T_f = \sqrt{2gh} \sqrt{\frac{2(H-h)}{g}}$$

$$R = \sqrt{4h(H-h)}$$

Time to Empty the Tank:

+ If the orifice is at the bottom of the tank, the time required to empty the tank is:

$$t = \frac{A}{a} \sqrt{\frac{2H}{g}}$$

Maximum horizontal range

+ The horizontal range of the liquid stream is maximum when the orifice is located at half the height of the liquid column i.e.,

$$h = \frac{H}{2} \quad \therefore R = \text{max?}$$

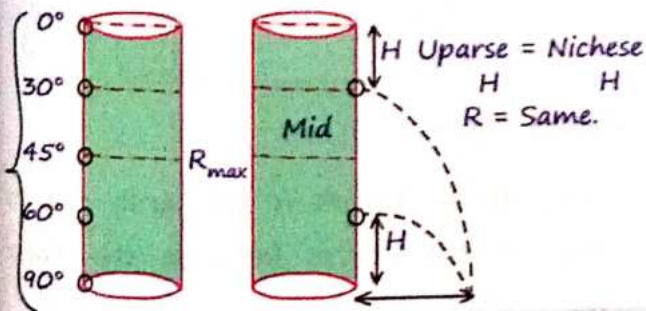
Therefore,

$$R_{\text{max}} = H$$

Just like Motion in 1-D

Angles and Range:-

+ At 45° , the range is maximized, illustrating the symmetry in projectile motion.



Force on Container due to efflux:-

$$F = \rho a v^2$$

$$* F = 2\rho a g H * \quad \therefore v = \sqrt{2gH}$$

ρ = density of liq.

H = Height of liquid column from Hole.

a = Area of orifice.

For massless container, " μ_{min} " to keep container at rest:-

a = hole

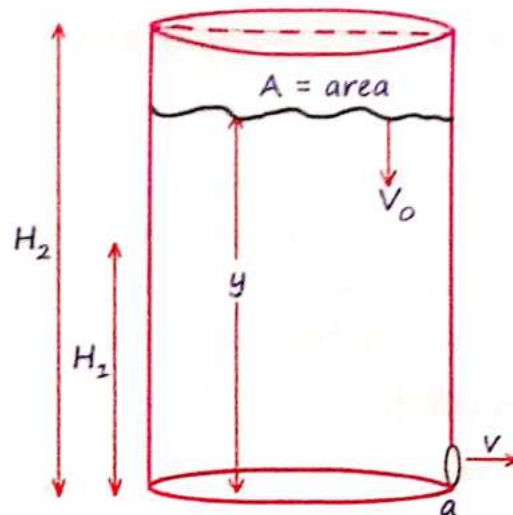
$\mu_{\text{mg}} = 2\rho a g H$ A = Container Area.

$$\mu_{\text{min}} = \frac{2a}{A}$$

Time Taken to Move Liquid from Height H_2 to H_1

$$V_0 = \frac{a}{A} \sqrt{2gy}$$

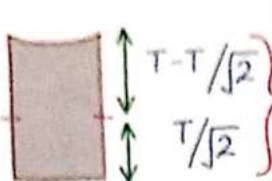
$$t = \frac{A}{a} \sqrt{\frac{2}{g}} (\sqrt{H_2} - \sqrt{H_1})$$



Time to Fall down:-

$$+ \text{ To bottom :- } t = \frac{A}{a} \sqrt{\frac{2H}{g}}$$

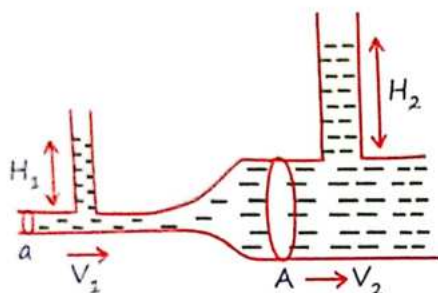
- Ratio from $h \rightarrow \frac{h}{2}$ & $\frac{h}{2} \rightarrow$ bottom

$$\frac{t_1}{t_2} = \frac{\sqrt{2}-1}{1}$$


1-D का उलट!

15 Venturimeter

- Measure rate of Volume Flow
- Based on Bernoulli's Principle.



$$V_2^2 = \frac{2g(H_2 - H_1) a^2 A^2}{A^2 - a^2}$$

* $aV_1 = AV_2 = V =$ Rate of Volume Flow:

$$P_1 + \frac{1}{2} \rho V_1^2 + 0 = P_2 + \frac{1}{2} \rho V_2^2 + 0$$

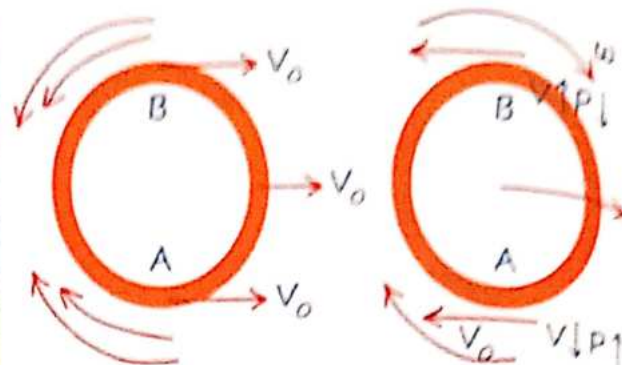
16 Dynamic Lift & Magnus Effect

Dynamic Lift:

- Dynamic lift is the force that acts on an object moving through a fluid (like air) due to differences in pressure created by the object's motion.

Magnus Effect:

- The Magnus Effect is the phenomenon where a spinning object moving through a fluid creates a force perpendicular to the direction of motion, due to pressure differences caused by the spin.



Football without Spinning

$$V_A = V_B = V_0 + V$$

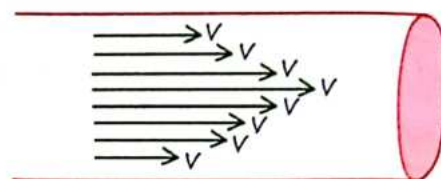
Football Spinning about vertical axis

$$V_A = V + V_0 - R\omega$$

$$V_B = V + V_0 + R\omega$$

17 Viscosity

- Viscosity is a measure of a fluid's resistance to flow. It describes how thick or sticky a fluid is.
- "1 Poissulle = 10 Poise."



- The diagram shows a fluid flowing through a pipe.
- The arrows represent the velocity (v) of the fluid particles at different layers within the pipe.
- Near the center of the pipe, the fluid moves faster, while near the walls of the pipe, the fluid moves slower due to friction between the fluid and the walls of the pipe.

Cohesive Forces:

- Viscosity in liquids is primarily due to the cohesive forces between the liquid molecules.

* In gases, viscosity is due to the diffusion of molecules.

Force between two liq:-

$$f = -\eta A \frac{\Delta V}{l} \quad \eta = \text{coeff. of viscosity.}$$

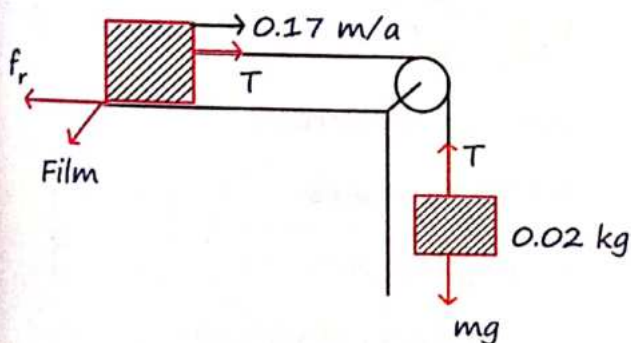
$$\eta = \frac{f}{A \left(\frac{\Delta V}{\Delta x} \right)} = \frac{\text{Shear Stress}}{\text{Velocity gradient}} = \frac{\text{Shear Stress}}{\text{Strain Rate}}$$

Q.4. A metal block of base area 0.2 m^2 is connected to a 0.02 kg mass via a string that passes over an ideal pulley as shown in figure. A liquid film of thickness 0.6 mm is placed between the block and the table. When released the block moves to the right with a constant speed of 0.17 m/s . The coefficient of viscosity of the liquid is: (2017-Gujarat)

- (a) $3.45 \times 10^3 \times \text{Pa-s}$
- (b) $3.45 \times 10^{-2} \times \text{Pa-s}$
- (c) $3.45 \times 10^{-3} \times \text{Pa-s}$
- (d) $3.45 \times 10^3 \times \text{Pa-s}$

Sol. (c) $F_r = T = mg$

$$\eta A \frac{\Delta v}{d} = mg$$



$$\eta = \frac{mgd}{A\Delta v} = \frac{2 \times 10 \times 6 \times 10^{-3}}{100 \times \frac{2}{10} \times 10 \times \frac{17}{100}} = \left(\frac{60}{17} \right) \times 10^{-3} \text{ Pa-s}$$

Difference Viscosity and Solid Friction:-

	Viscosity	(Solid Friction) NLM Friction
(a)	The viscous force is directly proportional to the surface area of contact of fluid layers.	(a) It is independent of the area of contact.
(b)	The viscous force is directly proportional to the relative velocity between the two layers in contact.	(b) It is independent of the relative velocity of the contact surface.
(c)	The viscous force does not depend on the normal reaction between the two layers of the fluid.	(c) It is directly proportional to the normal reaction between the surfaces in contact.
(d)	The viscous force on a solid moving in a fluid depends on the shape and size of the solid.	(d) It is independent of the shape and size of the bodies in contact.

18 Stoke's Law

Stoke's law describes the force of viscosity acting on a small spherical object moving through a fluid.

Only For Sphere.

Jab Force between Solid-liq:-

$$F = 6\pi\eta rV$$

Where:

η = coefficient of viscosity of the fluid.

r = radius of the spherical body.

V = terminal velocity of the body.

Terminal Velocity:-

The maximum constant velocity acquired by a body while falling through a viscous medium is called its terminal velocity. It is usually denoted by V_T

$$mg = F_B + f_r$$

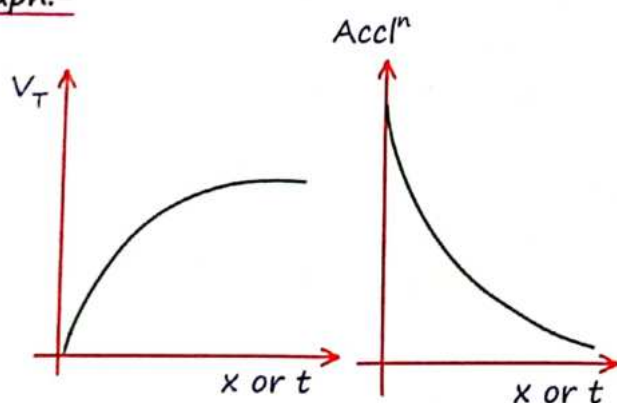
$$V_T = \frac{2}{9} \frac{r^2}{\eta} (\sigma - \rho)g.$$

Where,

r = radius of object, σ = density of object,

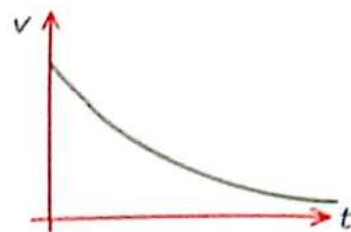
ρ = density of liquid.

Graph:-



* $V_T \propto r^2$:- The terminal velocity is directly proportional to the square of the radius of the body. That is why bigger rain drops fall with a larger velocity as compared to the smaller drops.

* Ball is thrown down with velocity greater than terminal velocity in viscous liq. then variation of " v " v/s " t "? (NEET-2022)



* Coalesce of Drops:-

* If V_T is the terminal velocity of the smaller drops and V'_T is the terminal velocity of the larger drop, then

$$V'_T = n^{2/3} V_T \quad R = n^{1/3} r$$

* Temperature dependence of η :-

For liquid

$$\eta_{liq} \propto \frac{1}{T}$$

For gas

$$\eta_{gas} \propto T$$

Poiseuille Equn:-

↳ Used to calculate the flow rate of Viscous liq.

↳ Bernoulli's theorem is not valid

Woh sirf ideal liquid keliye.

Fluid Current:-

$$Q = \frac{\Delta P}{\frac{8\eta l}{\pi r^4}} = \frac{\Delta P}{R_{\text{Fluid}}} \quad R_{\text{fluid of pipe}} = \frac{8\eta l}{\pi r^4}$$

ΔP = Press difference

r = radius of pipe.

l = length of pipe.

η = coefficient of viscosity

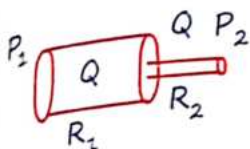
Series

$Q = \text{same}$
 $\Delta P = \text{Diff.}$

$$Q = \frac{\Delta P}{R_{eq}}$$

$$R_{eq} = R_1 + R_2$$

$$\Delta P = P_1 - P_2$$



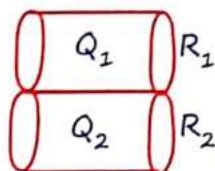
Parallel

$Q = \text{Diff.}$

$\Delta P = \text{same}$

$$Q = \frac{\Delta P}{R_{eq}}$$

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

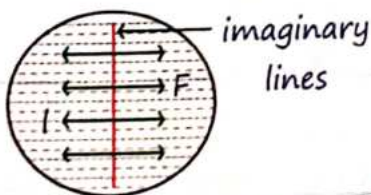


19 Surface Tension

Excess force required for lift

- Surface Tension is a property of liquid at rest by virtue of which a liquid surface gets contracted to a minimum area and behaves like a stretched membrane.
- Surface Tension of a liquid is measured by force per unit length on either side of any imaginary line drawn tangentially over the liquid surface, force being normal to the imaginary line
- It is a scalar quantity and depends only on nature of liquid but independent of area of surface.

$$S = \frac{F}{l}$$



Q.5. A liquid film is formed over a frame ABCD as shown in figure. Wire CD can slide without friction. The mass to be hung from CD to keep it in equilibrium is

(a) $\frac{TI}{g}$

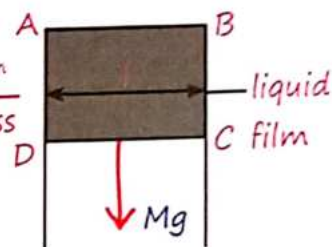
(b) $\frac{2TI}{g}$

(c) $\frac{g}{2TI}$

(d) $T \times 1$

Sol. $m = \frac{2sl}{g}$
 $mg = sl(2)$

$$\text{mass} = \left(\frac{\text{acc}^n}{\frac{F}{l} \times 1} \right) = \frac{\text{acc}^n}{\text{mass}}$$



Excess force required to lift: a

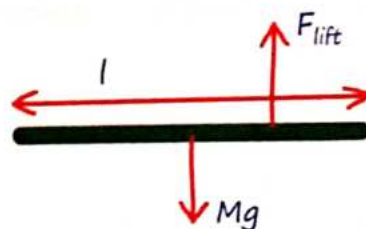
-Rod :- $S = \frac{F}{2l}$

-Disc :- $S = \frac{F}{2\pi r}$

-Ring :- $S = \frac{F}{2\pi r + 2\pi R}$

-Annular Disc :- $\uparrow$

Rod :-

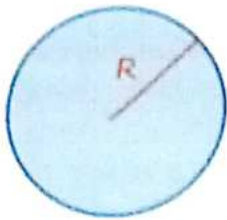


$$F_{\text{lift}} = F_{ST} + Mg$$

$$= 2Sl + Mg$$

20 Surface Energy

* $E = S.A.$



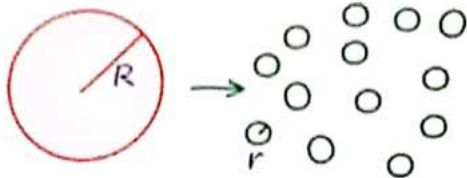
Sphere.

Surface area $= 4\pi R^2$

Volume $= \frac{4}{3} \pi R^3$

Circumference $= 2\pi R$

21 Splitting of Drops into Droplets

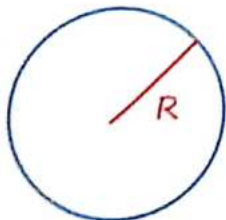


$R = n^{1/3} r$

$\Delta A = 4\pi [nr^2 - R^2]$; $\Delta A = 4\pi R^2 [n^{1/3} - 1]$

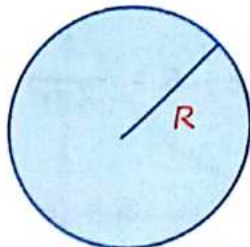
$\Delta A = 4\pi R^2 \left[\frac{R}{r} - 1 \right]$

Bubble



$A = 4\pi R^2 \times 2$

Drop



$A = 4\pi R^2$

Film: - Take Area double.

Energy Released when Droplets Combine to Form a Drop :-

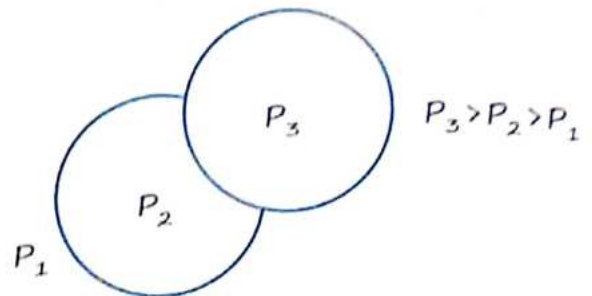
$\Delta E = E(n - n^{2/3}) = S4\pi r^2 (n - n^{2/3})$

$\frac{E_f}{E_i} = \frac{1}{n^{1/3}}$, % $E_{loss} = \left[\frac{1}{n^{1/3}} - 1 \right] \times 100$

$\Delta E = n4\pi r^2 - 4\pi R^2$

$\Delta E = 3VT \left[\frac{1}{r} - \frac{1}{R} \right]$ $V = \text{Vol of larger drop}$

Pressure is always high in concave side



Values of Excess Pressure :-

Drop
 $\Delta P = P_{in} - P_{out} = \frac{2S}{R}$

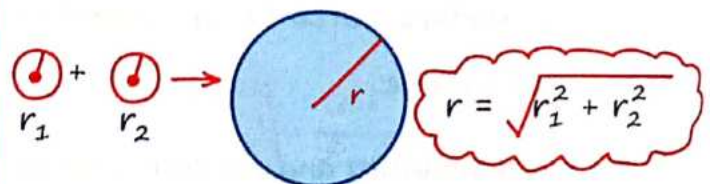
$P_{in} = P_{out} + \frac{2S}{R}$

Bubble
 $\Delta P = P_{in} - P_{out} = \frac{4S}{R}$

$P_{in} = P_{out} + \frac{4S}{R}$

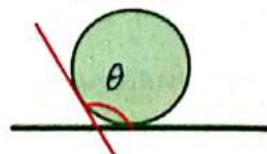
Radius of Coalesce :-

Two drops of radius r_1 & r_2 coalesce under isothermal condition.



Angle of Contact :-

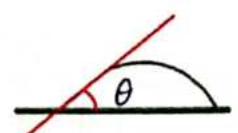
Obtuse



$\theta > 90^\circ$

Cohesion > Adhes.

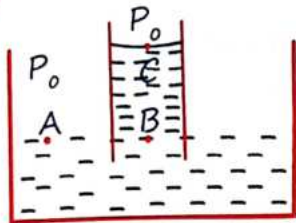
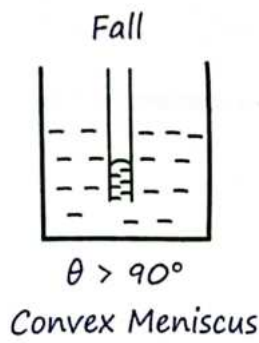
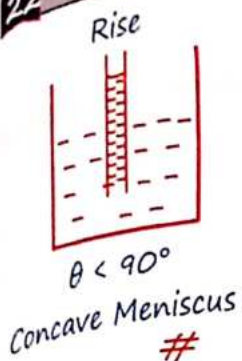
Acute



$\theta < 90^\circ$

Cohe. < Adhes.

22 Capillary Tube



$$P_A = P_B = P_0$$

Due to excess pressure

$$P_0 - \frac{2S}{R} = P_C \quad \dots(1)$$

Due to pressure variation in liquid

$$P_C + \rho gh = P_B = P_0 \quad \dots(2)$$

Putting value of P_C from (2)

$$P_0 - \frac{2S}{R} + \rho gh = P_0$$

$$\frac{2S}{R} = \rho gh \Rightarrow h = \frac{2S}{\rho g R}$$

Here, R is the radius of meniscus

If, r = radius of tube, then, $r = R \cos \theta$

$$\text{Therefore, } h = \frac{2S \cos \theta}{\rho g r}$$

+ Height of liq. rising in Capillary Tube

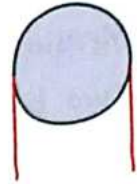
$$h = \frac{2S \cos \theta}{\rho g r} \rightarrow "g_{eff}"$$

$$M = \rho Ah = \frac{A 2S \cos \theta}{r g}$$

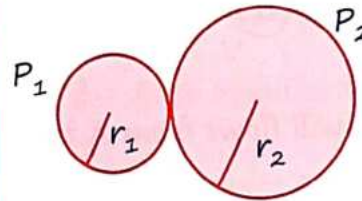
"Mass of liq. in tube"

+ Rise of liq. in tube of insufficient length :-

$$r \propto \frac{1}{h} \quad \text{don't overflow}$$



Radius of Interface



$$r = \frac{r_1 r_2}{r_1 - r_2}$$

+ Height/Depth of liquid $\propto 1/r$

+ Mass of liquid $\propto r$

+ Potential energy of liquid $\propto r^2$

+ If container with capillary in a freely falling lift, liquid rise upto complete length and does not overflow.

$$h \propto \cos \theta \propto \frac{1}{\theta}$$

+ For two different liquid if h , S and r same then find relation b/w density and contact angle.

$$h = \frac{2S \cos \theta}{\rho g r}$$

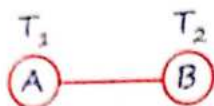
$$\cos \theta \propto \rho$$

$$\theta \propto \frac{1}{\rho}$$

MR* 6 धीमे कदमों पर यूँ मायूस ना हो,
पहाड़ कभी भाग कर नहीं चढ़े जाते!

1 Temperature

- Measure of hotness and coldness.
- Two body A at T_1 and B at T_2 put in contact.



If $T_1 > T_2$ then

S-1 Some temp^r will flows from A to

B

⇒ False

S-2 Some heat will flows from A to B

⇒ True

S-3 Heat will increase in B ⇒ False

S-4 Temp^r of A will decrease ⇒ True

- Heat can not be stored it can flows from Body A to B.

2 Measurement of temperature

$$\frac{\text{Temp}^r - \text{L.F.P}}{\text{U.F.P} - \text{L.F.P}} = \text{Constant}$$

$$\frac{^{\circ}\text{C} - 0}{100 - 0} = \frac{\text{F} - 32}{212 - 32} = \frac{\text{K} - 273}{373 - 273}$$

$$= \frac{\text{MR} - \text{L.F.P}}{\text{U.F.P} - \text{L.F.P}}$$

• Where

○ LFP = lower fixed point

○ UFP = upper fixed point

Note:-

For celcius scale

L.F.P = 0, U.F.P = 100

For Fahrenheit scale

L.F.P = 32, U.F.P = 212

For kelvin scale

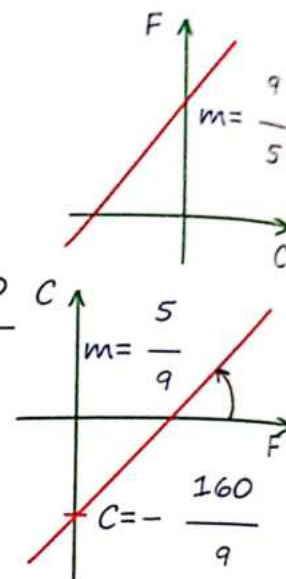
L.F.P = 273, U.F.P = 373

Relation in Fahrenheit & Celcius:-

$$F = \frac{9C}{5} + 32$$

$$C = \frac{5F}{9} - \frac{160}{9}$$

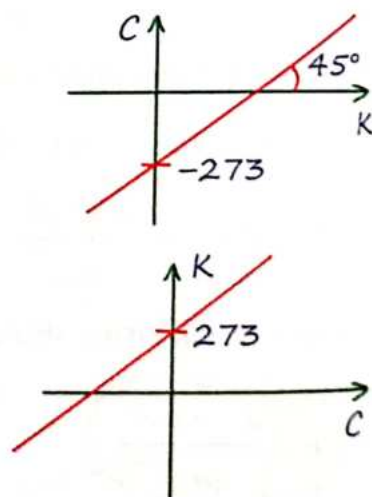
$$\frac{\Delta C}{5} = \frac{\Delta F}{9}$$



Relation between Kelvin & Celcius:-

$$C = K - 273$$

$$K = C + 273$$



• Relation between $^{\circ}\text{F}$ & Kelvin :-

$$F - 32 = \frac{9}{5} (K - 273)$$

Change in Temperature

$$\Delta C = \Delta K = \frac{5}{9} \Delta F$$

Q.1 The freezing point on MR° scale is 20° and boiling point 150°. A temperature of 60° thermometer will be read as.

Sol. Let x is reading at 60°C

$$\frac{x - 20}{150 - 20} = \frac{60 - 0}{100 - 0}$$

[°CX]

3 Construction of thermometer

Change in physical quantity per unit raise in temperature = constant

Resistance Thermometer:-

$$t = \frac{R_t - R_0}{R_{100} - R_0} \times 100$$

Pressure thermometer:-

$$t = \frac{P_t - P_0}{P_{100} - P_0} \times 100$$

Volume Thermometer:-

$$t = \frac{V_t - V_0}{V_{100} - V_0} \times 100$$

Length Thermometer:-

$$t = \frac{L_t - L_0}{L_{100} - L_0} \times 100$$

Q.2. Length of rod at 20°C is 10m and at 80°C is 40 m then find temperature when its length is 30 m.

Sol. Change in length per unit raise temperature = const

$$\frac{40 \text{ m} - 10 \text{ m}}{80^\circ\text{C} - 20^\circ\text{C}} = \frac{\ell - 10 \text{ m}}{t - 20}$$

$$\frac{1}{2} = \frac{30 - 10}{t - 20}$$

$$t = 60^\circ\text{C}$$

4 Thermal Expansion

Linear expansion:-

$$\Delta L = L_0 \alpha \Delta \theta ; L_2 = L_0 (1 + \alpha \Delta \theta)$$

Areal expansion:-

$$\Delta A = A_0 \beta \Delta \theta ; A_2 = A_0 (1 + \beta \Delta \theta)$$

Volume expansion:-

$$\Delta V = V_0 \gamma \Delta \theta ; V_2 = V_0 (1 + \gamma \Delta \theta)$$

Important relations:-

Bulk modulus and thermal coefficient of volume expansion.

$$\Delta P = \beta \frac{\Delta V}{V} = \beta \gamma \Delta \theta$$

β = Bulk Modulus of Elasticity

$$\frac{\alpha}{1} = \frac{\beta}{2} = \frac{\gamma}{3}$$

$$\alpha : \beta : \gamma = 1 : 2 : 3$$

+ For anisotropic crystal

$$\gamma = \alpha_x + \alpha_y + \alpha_z$$

For Isotropic $\alpha_x = \alpha_y = \alpha_z$

$$\gamma = 3\alpha$$

+ For Two Rod

$$\alpha_1 l_1$$

$$\alpha_2 l_2$$

* If difference in length of these two rod is independent of Temperature then

$$l_1 \alpha_1 = l_2 \alpha_2$$

Ram Lal Ne socha winter me gold buy karunga, summer me sold karunga tab length Increase ho Jayga ☺

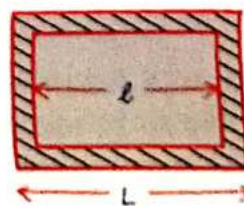
→ Ram Lal ko koi benefit nahi hoga

5 Cavity Problems

Photographic Enlargement.

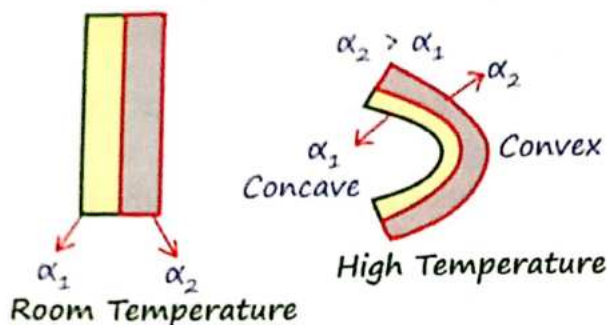
$$T \uparrow \rightarrow L \uparrow \quad l \uparrow$$

$$T \downarrow \rightarrow L \downarrow \quad l \downarrow$$



6 Bimetallic Strip

- A bimetallic strip is composed of two different metals with different coefficients of thermal expansion (α_1 and α_2).



Behavior at room temperature:

At room temperature, the strip remains straight because both metals expand or contract equally.

Behavior at High Temperature:

- When the temperature increases, the metal with the higher coefficient of thermal expansion (α_2) expands more than the metal with the lower coefficient (α_1).
- This differential expansion causes the bimetallic strip to bend.
- The side with the metal that has a higher expansion coefficient (α_2) becomes convex (curves outward).
- The side with the metal that has a lower expansion coefficient (α_1) becomes concave (curves inward).

7 Thermal Stress

$$\text{T.S.} = \frac{F}{A} = \alpha Y \Delta \theta$$

Y :- Young's Modulus

A rod of length ℓ and area A placed on smooth surface then due to increase in temperature thermal stress is zero.

Pendulum Clock:-

Note :-

Temperature $\uparrow$ = Clock Slow = Time loss
Temperature $\downarrow$ = Clock Fast = Time Gain

$$\text{Loss or gain in time, } \frac{\Delta T}{T} = \frac{1}{2} \frac{\Delta l}{l} = \frac{1}{2} \alpha \Delta \theta$$

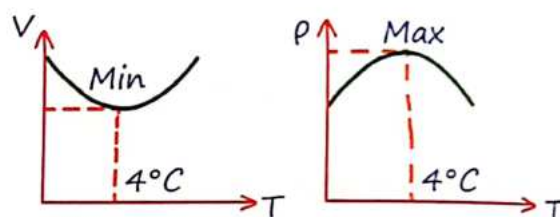
Variation of density with temperature

$$\rho_2 = \rho_0(1 - \gamma \Delta \theta)$$

Anomalous Behaviour of Water:-

Water shows unique behavior near 4°C , where its volume decreases to a minimum and density increases to a maximum.

- Volume vs. Temperature: Minimum volume at 4°C .
- Density vs. Temperature: Maximum density at 4°C .



8 Apparent Coefficient of Volume Expansion of Liquid

The apparent coefficient of volume expansion of a liquid is the observed expansion when the liquid is heated in a container.

Formula

$$\gamma_{\text{app}} = \gamma_{\text{real}} - \gamma_{\text{container}}$$

$$\gamma'_l = \gamma_{\text{liquid}} - \gamma_{\text{solid}}$$

- If $\gamma_l = \gamma_s$ then $\gamma'_l = 0$

Level Unchanged

- If $\gamma_l > \gamma_s$ then $\gamma'_l = +ve$

The liquid overflows because it expands more than the container.

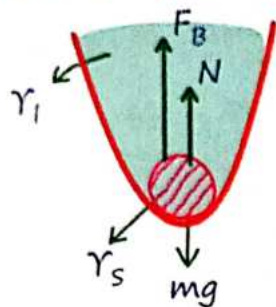
- If $\gamma_l < \gamma_s$ then $\gamma'_l = -ve$

The liquid level goes down because the container expands more than the liquid.

9 Effect of Expansion on Apparent Weight in Liquid

The apparent weight (N) of an object submerged in a liquid is the weight of the object minus the buoyant force exerted by the liquid.

$$N = mg - \rho_l V_s g$$



Scenarios Based on Coefficients of Expansion:

Case 1: $\gamma_l = \gamma_s$:

- If the coefficient of volume expansion of the liquid is equal to that of the solid, the apparent weight remains unchanged ($N = \text{same}$).

Case 2: $\gamma_l < \gamma_s$:

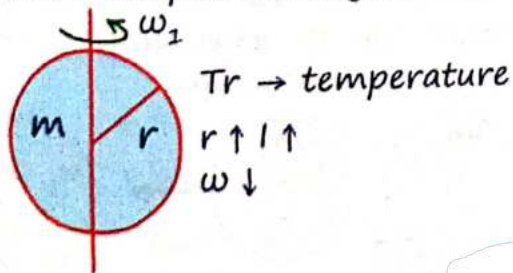
- If the coefficient of volume expansion of the liquid is less than that of the solid, the apparent weight decreases ($N = \text{dec.}$), and the submerged volume increases ($V_s \uparrow$).

Case 3: $\gamma_l > \gamma_s$:

- If the coefficient of volume expansion of the liquid is greater than that of the solid, the apparent weight increases ($N = \text{inc.}$), and the submerged volume decreases ($V_s \downarrow$).

Note:-

1) Effect of temperature on angular velocity



10 Meter Scale

- The true length of a rod can be determined using a meter scale by accounting for any thermal expansion.

Only one type will be asked :-

$$\text{True length of Rod} = \text{Reading taken} + \Delta L$$

$$\Delta L = L \alpha \Delta \theta$$

- where ΔL is the change in length due to thermal expansion.

11 Calorimetry

Calorimetry is the study of measuring the amount of heat exchanged

Specific Heat Capacity:-

Heat required to raise the temperature by unit degree $^{\circ}\text{C}$ of unit mass

$$Q = mS\Delta\theta$$

raise T^r of " m " Kg by $\Delta\theta$

Unit of $S = \text{J/kg kelvin}$

$$S = \frac{\phi}{m\Delta T}$$

If $\Delta T = 0$, $S = \infty$ (isothermal)

$$S = \frac{\phi}{m\Delta T}$$

If $\phi = 0$, then $S = 0$ (for adiabatic process)

Heat Capacity (C):-

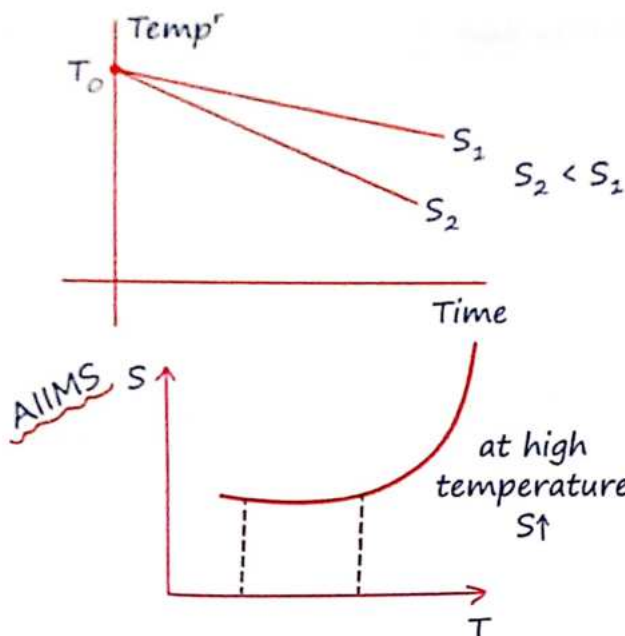
Heat required to raise temperature by unit of m mass.

$$C = mS \quad (\Delta\theta = 1^{\circ}\text{C})$$

raise T^r of object by 1°C

MR*

→ Joh Jaldi Garam hoga woh jaldi Thanda hoga. Uska specific heat capacity kam hai.



Molar heat Capacity [C]:-

$$Q = nC\Delta\theta$$

raise Temp^r of 1 mole Sub^s by 1°C

Note :-

$$S_w = 1 \text{ cal/gm}^\circ\text{C} = 4200 \text{ J/KgK}$$

$$S_{ice} = 0.5 \text{ cal/gm}^\circ\text{C} = 2100 \text{ J/KgK}$$

$$S_{steam} = 0.5 \text{ cal/gm}^\circ\text{C} = 2100 \text{ J/KgK}$$

$$1 \text{ cal} = 4.2 \text{ J}$$

Latent Heat:-

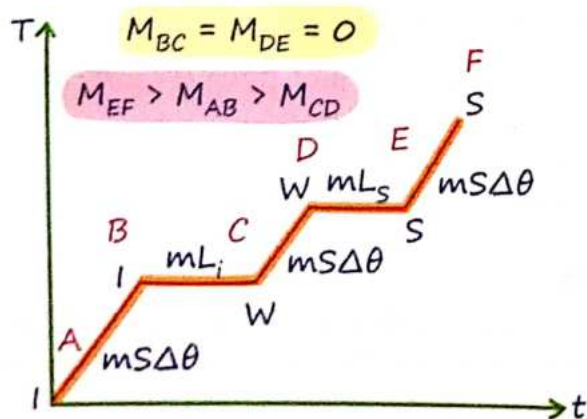
$$Q = mL$$

heat req. to change state.

$$L_{ice} = 80 \text{ cal/gm}$$

$$L_{steam} = 540 \text{ cal/gm}$$

12 Temp.-time graph



MR* feel

+ Water Can't exist below 0°C . Ice can't exist above 0°C . at 0°C both can exist. Mixture of (Ice + water) only Possible at 0°C . Zero se upar gya matlb sab pani ho gya, zero se Niche sab Ice.

+ Steam + water mixture only exist at 100°C .

+ Slope = $\frac{d\theta}{dT}$ Rate of heating

$$+ \frac{d\theta}{dt} = ms \frac{dT}{dt}, \quad \frac{d\theta}{dt} = \text{constant}$$

$$\frac{dT}{dt} \propto \frac{1}{S}$$

Q.3. 10 gm ice at 0°C mixed with 10 gm water at 40°C then mass of water in mixture?

$$\text{Sol. } Q_{ice} = mL = 800$$

$$Q_{water} = wt = 10 \times 40 = 400$$

only 5 gm ice will melt

$$m(\text{water}) = 15 \text{ gm.}$$

13 Principle of Calorimetry

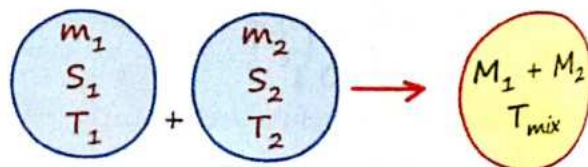
+ The principle of calorimetry states that in an isolated system, the heat lost by hot objects is equal to the heat gained by cold objects.

$$\text{Heat Loss} = \text{Heat Gain}$$

14 Mixture

Final Temperature of Mixture:-

+ When two liquids of different temperatures are mixed, the final temperature of the mixture can be determined using the principle of calorimetry.



Final Temperature of Mixture when two liquid of mass m_1 & m_2 of specific heat capacity S_1 and S_2 at temperature T_1 and T_2 mixed.

$$T_{\text{mix}} = \frac{m_1 S_1 T_1 + m_2 S_2 T_2}{m_1 S_1 + m_2 S_2}$$

Mixture of ice & water:-

MR*

- m gram Ice at 0°C mixed with W gram water at $T^\circ\text{C}$
- Bring both ice & water at common Temp^r and same phase.

Required :- $Q = mL$ (m gram Ice melt into m gram water at 0°C)

Supply :- $Q = WT = mS\Delta T$ (When W gram water cool down from T to 0°C)

1> $mL = WT$

$T_{\text{mix}} = 0^\circ\text{C}$ (amt)_{ice} = 0

(amt)_w = $W + m$

2> $mL > WT$

$T_{\text{mix}} = 0^\circ\text{C}$ (amt)_{ice} = $m - m'$

(amt)_w = $W + m'$

$$m' = \frac{WT}{L}$$

Amt. of ice converted to water.

3> $mL < WT$

$T = \frac{WT - mL}{m + W}$ *Paani hi Rahega.

Mixture of ice & steam:-

Required :- $Q = mL = 80m$

Supply :- $Q = WL + WS\Delta\theta = 640W$

1> $Q_{\text{Supply}} = Q_{\text{Req.}}$

$M_{\text{ice}} = 8W$

$$\frac{M_{\text{ice}}}{W_{\text{steam}}} = \frac{8}{1}$$

$T_{\text{mix}} = 0^\circ\text{C}.$

2> Ice $\xrightarrow{80m}$ water $\xrightarrow{100m}$ Water 100°C
 0°C 0°C

Steam $\xrightarrow{540W}$ Water 100°C
 100°C

$180m = 540W$

$$\frac{M_{\text{ice}}}{W} = \frac{3}{1}$$

MR*

	M_{ice}	:	W_{steam}
$T_{\text{mix}} \downarrow 100^\circ\text{C}$	1	:	1
	2	:	1
	3	:	1
$T_{\text{mix}} \downarrow 0^\circ \text{ to } 100^\circ\text{C}$	4	:	1
	5	:	1
	6	:	1
	7	:	1
s	8	:	1
	9	:	1

Q.4. 200 gm ice at -20°C is mixed with 500 gm water at 20°C , then find temperature of mixture and amount of water, ice in mixture

Sol. MR* Dono ko kisi ek phase me same temperature par le ke aao.

$Q_1 = ms\Delta T + mL = 2000 \text{ cal} + 16000 \text{ cal}$
 heat given to ice to melt $\Delta^\circ\text{C}$.

$Q_2 = ms\Delta t = 10000 \text{ cal}.$

Heat given by water when it fall from 20° to 0°C

$Q_1 > Q_2$ hence complete ice will not melt,
out of 10000 cal heat given by water
2000 cal used to increase temperature
and 8000 cal use to melt ice

$$m'L = 8000$$

$m' = 100$ ice will melt

$T_{\text{mix}} = 0^\circ$ because (ice + water) mixture

$$M_{\text{water}} = 600 \text{ gm}$$

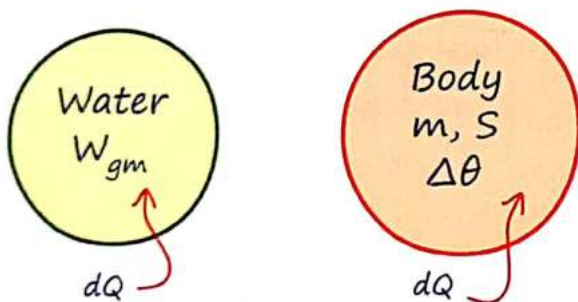
$$M_{\text{ice}} = 100 \text{ gm}$$

Q.5. 5 gm water at 30°C and 5 gm ice at -20°C mixed then temperature of mixture.

Sol. $T_{\text{mix}} = 0^\circ$ Requirement > supply

Water Equivalent:-

Woh liquid utnahi Heat lega ΔT Temp. Rise
Keliye jitna W_{gm} water le raha hh!
Toh Aapko uss Liquid Ko Na Assume Karke
Water Ko uss liquid Ki tarah Treat Karna hai!



$$\therefore mS\Delta\theta = W \times 1 \times \Delta\theta$$

$$W = mS$$

eg :- Water equivalent = 55g at 40°C
 $WT = 55 \times 40 = 2200 \text{ cal.}$

V. Imp Question :-

*A bullet of mass "m" moving with "u" hits an ice block of "M" gm kept on a frictionless floor & gets stuck in it. How much ice will melt if x% of the lost KE goes to ice? (initial temp of block & bullet = 0°C).

$$\left[\frac{1}{2} \frac{mMu^2}{M+m} \right] x\% = mL$$

+ Ice Formation :-



Formation of ice :-

0 to x : x to 2x : 2x to 3x

t : 3t : 5t

0 to x : 0 to 2x : 0 to 3x

t : 4t : 9t

$$t \propto x_2^2 - x_1^2$$

MR*

It's okay to feel up and down
Its normal and natural, don't
overthink, move forward and
work hard

1 Conduction

Heat flows from hot end to cold end, medium required but particles of medium simply oscillate but do not leave their position.

- + Slow process
- + Takes place in solid
- + Path may be zig-zag
- + Temperature of medium increases.

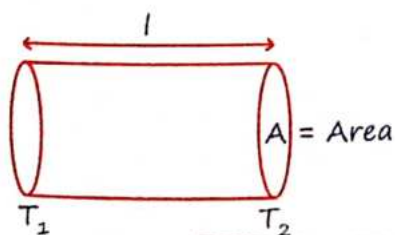
2 Convection

Medium required, each particle of medium absorbs heat and moves from hot end to cold end.

- + Slow process
- + Occurs in fluid not in solid
- + Temp^r of medium increases.

3 Radiation

- + Heat flows in the form of electromagnetic waves.
- + Medium is not required
- + Path straight line
- + No change in temp^r of medium.

4 Law of Thermal Conductivity

$$H = \frac{Q}{t} = \frac{KA\Delta T}{L}$$

K = Coefficient of Thermal Conductivity
(Material Property)

Q = Heat.

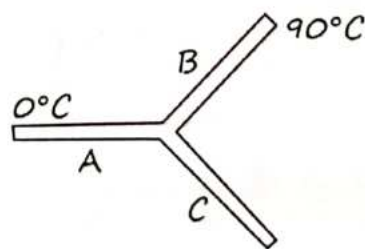
$$+ \frac{Q}{t} = KA \frac{dT}{dx}$$

Q.1. Three rods made of the same material and having the same cross section have been joined as shown in the figure. Each rod is of the same length. The left and right ends are kept at 0°C and 90° respectively.

The temperature of the junction of the three rods will be

- (a) 45°C (b) 60°C
(c) 30°C (d) 20°C

Sol.



$$T = \frac{T_1 + T_2 + T_3}{3}$$

$$T = \frac{0 + 90 + 90}{3} = \frac{180}{3} = 60$$

5 Heat Current

$$H = \frac{\Delta T}{R_T}, R_T = \frac{l}{KA}$$

R_T :- Thermal Resistance

6 Combination of Rod

Series Combination:-

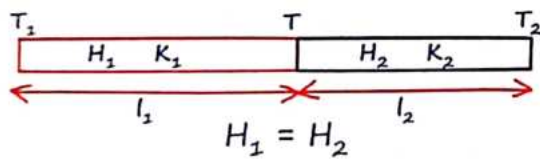
"R" add होगा

$$R_{eq} = R_1 + R_2 + R_3 \dots$$

$$\frac{l_{eq}}{K_{eq}} = \frac{l_1}{K_1} + \frac{l_2}{K_2} + \frac{l_3}{K_3}$$

Junction Temp:-

Rate of heat flow same in series combination.



$$\frac{K_1 (T_1 - T)}{l_1} = \frac{K_2 (T - T_2)}{l_2}$$

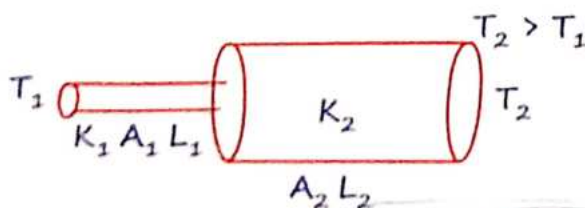
Parallel Combination:-

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

$$\frac{K_{eq} A_{eq}}{l} = \frac{K_1 A_1}{l} + \frac{K_2 A_2}{l} + \frac{K_3 A_3}{l}$$

7 Combination of Conductor

Series:-



$$R_{eq} = R_1 + R_2 = \frac{L_1}{K_1 A_1} + \frac{L_2}{K_2 A_2}$$

$$\frac{H}{dt} = \frac{\Delta T}{R_{eq}} = \frac{T_2 - T_1}{R_{eq}}$$

$$K_{eq} = \frac{L_1 + L_2}{\frac{L_1}{K_1} + \frac{L_2}{K_2}}$$

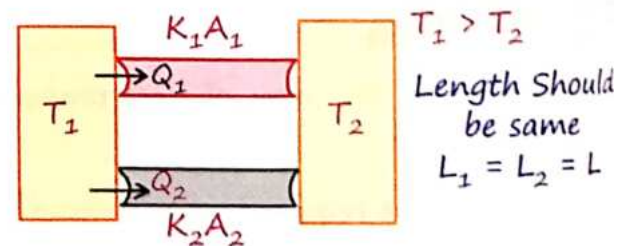
If $L_1 = L_2$

Then

$$K_{eq} = \frac{2K_1 K_2}{K_1 + K_2}$$

$$T_{mid} = \frac{\frac{K_1 A_1 T_1}{L_1} + \frac{K_2 A_2 T_2}{L_2}}{\frac{K_1 A_1}{L_1} + \frac{K_2 A_2}{L_2}}$$

Parallel:-



$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{K_{eq} [A_1 + A_2]}{L} = \frac{K_1 A_1}{L} + \frac{K_2 A_2}{L}$$

$$K_{eq} = \frac{K_1 A_1 + K_2 A_2}{A_1 + A_2}$$

if $A_1 = A_2 = A$

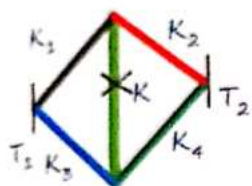
Then

$$K_{eq} = \frac{2K_1K_2}{K_1 + K_2}$$

$$H_{eq} = Q_1 + Q_2$$

$$= \frac{K_1(T_1 - T_2)A_1}{L} + \frac{K_2(T_1 - T_2)A_2}{L}$$

Wheat stone Bridge :-

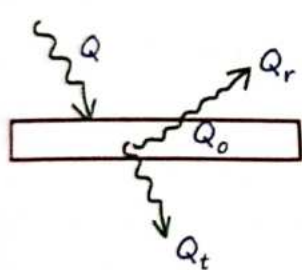


$$K_1 \cdot K_4 = K_3 \cdot K_2$$

$$\frac{K_1}{K_2} = \frac{K_3}{K_4}$$

8 Radiation

Black Body :-



$$Q = Q_o + Q_r + Q_t$$

$r + a + t = 1$

POWER

Absorptive $a = \frac{Q_a}{Q}$

Reflective $r = \frac{Q_r}{Q}$

Transmitive $t = \frac{Q_t}{Q}$

Emissive Power [Intensity] :-

$$E = \frac{Q}{At} = \frac{J}{m^2s} = \text{Watt/m}^2$$

Heat Transfer

Stefan's Law :-

$$E = \sigma T^4$$

$T = \text{Kelvin}$

$$\sigma = 5.67 \times 10^{-8} \frac{\text{Watt}}{m^2 K^4}$$

Emissivity :-

$$e = \frac{\text{Emissive Power of Normal body (E)}}{\text{Emissive Power of Black body (E)}}$$

For Black Body $(e) = 1$.

Emissive Power of Normal Body (E):-

$$E = eE = e\sigma T^4$$

$$P = \frac{Q}{t} = Ae\sigma T^4$$

Stefan-Boltzmann's Law:-

$$P_{loss} = P_{emit} - P_{absorb}$$

$$P_{loss} = \sigma eA [T^4 - T_o^4]$$

$$T_o = \text{Surr. Temp}^r, T = \text{body Temp}^r$$

Q.2. If the temperature of a body is increased by 50%, then the increase in the amount of radiation emitted by it will be

Sol. $\therefore$ Percentage increase in the amount of radiations emitted

$$\therefore \frac{E_1 - E_2}{E_1} \times 100 = \frac{(1.5T_1)^4 - T_1^4}{T_1^4} \times 100$$

$$\Rightarrow \frac{E_1 - E_2}{E_1} \times 100 = [(1.5)^4 - 1] \times 100$$

$$\frac{E_1 - E_2}{E_1} \times 100 = 400\%$$

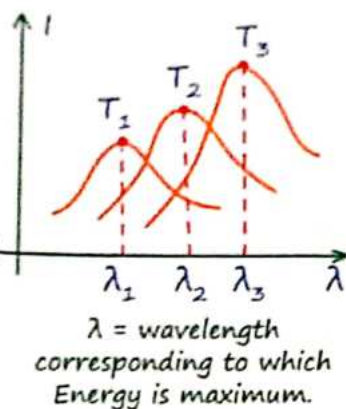
Wien's Law :-

$$\lambda_m T = b$$

$$\lambda_3 > \lambda_2 > \lambda_1$$

$$T_3 > T_2 > T_1$$

$$b = 2.9 \times 10^{-3} \text{ mK}$$



Newton's Law of Cooling :-

- Time 't' taken to fall temperature T_1 to T_2 where T_0 is the temperature of surrounding.

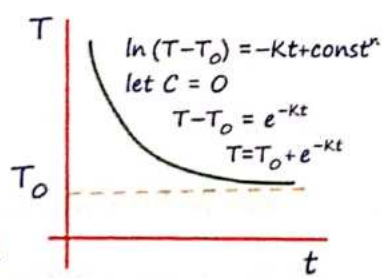
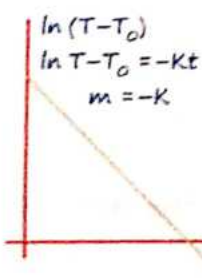
$$\frac{T_1 - T_2}{t} = K \left(\frac{T_1 + T_2}{2} - T_0 \right)$$

- Rate of cooling $\propto$ Temperature difference

$$90^\circ\text{C} \xrightarrow{t_1} 80^\circ\text{C} \xrightarrow{t_2} 70^\circ\text{C} \xrightarrow{t_3} 60^\circ\text{C}$$

$$\text{Time } t_1 < t_2 < t_3$$

(Time taken to fall temp^r for every 10°C)



Kirchoff's Law :-

A good absorber is a good emitter!

$$\Rightarrow \frac{E_\lambda}{a_\lambda} = E_\lambda \text{ (constant)}$$

Solar Constant:-

Total Thermal Energy falling per unit area per sec.

$$S = e\sigma T^4 \left[\frac{R^2}{r^2} \right]$$

e = emissivity of sun
 σ = Stephan's const
 T = Temp. of Sun.
 R = Radius of Sun.
 r = Distⁿ of sun & earth.

$$T_{\text{sun}} \propto \left[\frac{Sr^2}{e\sigma} \right]^{1/4}$$

The surface temp of sun found to be 5742K

Weisman-Fraz Law:-

Ratio of thermal conductivity and electrical conductivity at a temp^r is same for all body.

$$\frac{K}{\sigma T} = \text{Const}^n$$

$$K \propto \sigma$$

K = Thermal Conductivity

σ = Electrical Conductivity

MRⁿ

Jeet ke khatir junoon chahiye, ho ubal ayesa khoon chahiye, Aasman bhi ayega zameen par bas irado main jeet ki goonj chahiye. ♡

1 Thermodynamics

System:

Open → Mass exchange/Energy exchange

Closed → Mass not exchange/Energy exchange

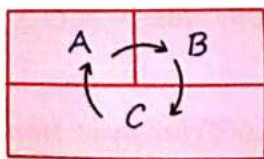
Isolated → Mass/energy not exchange

Thermal Equilibrium

If Heat transfer between 2 system is not in process i.e. their temperature is same then the 2-system are said to be in thermal equilibrium.

2 Zeroth Law of Thermodynamics

+ This law states that if two systems (A and B) are each in thermal equilibrium with a third system (C), then A and B are in thermal equilibrium with each other. This means that the temperature of A, B, and C are equal:



$T_A = T_B = T_C$ (They are in thermal contact)

3 Work done in thermodynamics

$$W = P \cdot \Delta V = \int P \cdot dV$$

Work done By gas

+ Vol. increase, expansion

+ $W = +ve$

On gas

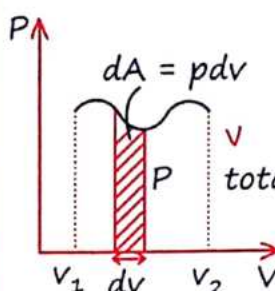
+ Vol decrease compression

+ $W = -ve$

4 P-V Graph → Indicator Diagram

+ This graph shows the work done in a cyclic process. The area enclosed by the loop in a P-V graph represents the work done.

Work = Area of Close loop in PV Graph



$$\text{total area} = \int_{V_1}^{V_2} P dv = \text{work done}$$

Work

Clock Wise

$W = +ve$

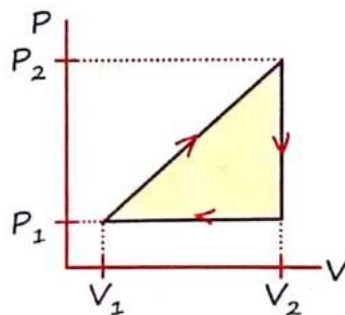
Anti clock Wise

$W = -ve$

+ Expansion → $\Delta V = +ve$ $W \uparrow$

+ Compression → $\Delta V = -ve$ $W \downarrow$

Q.1. In given fig. find workdone

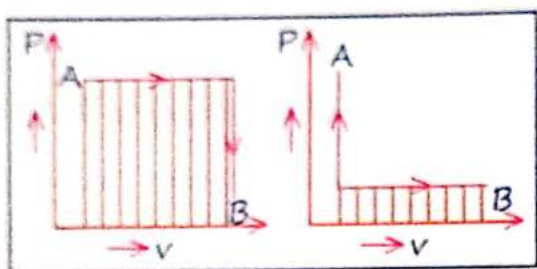


$$\text{Work} = + \frac{1}{2} (V_2 - V_1) (P_2 - P_1)$$

Q.2. In the following figure two indicator diagrams are shown. If the amounts of work done in them are W_1 and W_2 respectively, then

- (a) $W_1 > W_2$ (b) $W_1 < W_2$
 (c) $W_1 = W_2$ (d) $W_1 = W_2/4$

Sol. (a)



5 Internal Energy (U)

- Internal Energy (U) is the total energy contained within a system.
- It is path-independent, meaning it depends only on the state of the system, not how it got there.

$$U = KE + PE.$$

- It is temperature-dependent, meaning it changes with the temperature.

$$U \propto T$$

- Internal energy for one molecule

$$U = \frac{1}{2} K_B T = \frac{f}{2} K_B T \quad \left. \vphantom{U = \frac{1}{2} K_B T} \right\} \text{Due to 1 molecule}$$

- Internal energy for N molecule

$$U = \frac{f}{2} \frac{R}{N_A} T \times N = \frac{nfRT}{2} \quad \left. \vphantom{U = \frac{f}{2} \frac{R}{N_A} T \times N} \right\} \text{N molecule}$$

Where,

f = degree of freedom.

N = No. of molecules.

n = No. of moles

$$N_A = 6.023 \times 10^{23} \text{ mol}^{-1}$$

$$K_B = \frac{R}{N_A} = \text{Boltzmann Constant}$$

$$K_B = 1.38 \times 10^{-23} \text{ m}^2 \text{ kg s}^{-2} \text{ K}^{-1}$$

$$+ dU = nC_v dT$$

$$+ \text{Where, } n = \text{no. of mole, } \gamma = \frac{C_p}{C_v}$$

6 Degree of Freedom

- Degree of freedom is the number of independent ways in which molecules is free to move

1> **Monoatomic Gas** :- (Ne, He, Ar)

$$f = 3T + 0R$$

2> **Diatomic gas** :- (N_2 , O_2 , H_2)

$$f = 3T + 2R$$

3> **Triatomic Linear gas** :- (CO_2)

$$f = 3T + 2R$$

4> **Polyatomic gas/Triatomic Non-linear gas** :- (SO_2 , CH_4 , NH_3)

$$f = 3T + 3R$$

When Considering Vibrational Motion:-

- No effect for Monoatomic gas
- Diatomic = $f = 3T + 2R + 2V = 7$

Triatomic

$$\rightarrow \text{Linear } f = 3T + 2R + 2V = 7$$

$$\rightarrow \text{Non-Linear } f = 3T + 3R + 2V = 8$$

7 Heat Capacities

Specific heat Capacity (S) :-

$$S = \frac{dQ}{m dt}$$

Heat Capacity (C) :-

$$C = \frac{dQ}{dt}$$

Molar heat Capacity :-

$$C_m = \frac{dQ}{n dt}$$

Relation between Specific & Molar specific.

Heat capacity :-

$$* \boxed{C_m = M_{wt} S} \quad C_m = \frac{W}{n} S$$

* C_p & C_v denote the Specific Heat per unit mass of an ideal gas of Mwt "M" then :-

$$C_p - C_v = R \rightarrow \text{Molar Sp. heat}$$

$$S_p - S_v = \frac{R}{M} \quad (\text{Specific heat per unit mass})$$

Molar Heat Capacity:-

Constⁿ P

$$C_p = \frac{dQ}{dt n}$$

Constⁿ V

$$dQ = dU$$

$$C_v = \frac{fR}{2}$$

Note:-

$$\frac{C_p}{C_v} = \gamma$$

$$C_p - C_v = R$$

$$C_v = \frac{R}{\gamma - 1}$$

$$\boxed{\gamma = 1 + \frac{2}{f}}$$

MR* Table

Gas	DOF (f)	$C_v = \left(\frac{fR}{2}\right)$	$C_p = (C_v + R)$	$\gamma = \left(\frac{C_p}{C_v}\right)$
Monoatomic	3	$\frac{3R}{2}$	$\frac{5R}{2}$	$\frac{5}{3} = 1.66$
Diatomic	5	$\frac{5R}{2}$	$\frac{7R}{2}$	$\frac{7}{5} = 1.44$
Triatomic Linear	5	$\frac{5R}{2}$	$\frac{7R}{2}$	$\frac{7}{5} = 1.4$
Triatomic Non-linear	6	$3R$	$4R$	$\frac{4}{3} = 1.33$
Diatomic at high Temperature	7	$\frac{7R}{2}$	$\frac{9R}{2}$	$\frac{9}{7}$

8 Gas Mixture

$$(C_v)_{mix} = \frac{n_1 C_{v1} + n_2 C_{v2}}{n_1 + n_2}$$

$$f_{mix} = \frac{n_1 f_1 + n_2 f_2}{n_1 + n_2}$$

$$(C_p)_{mix} = \frac{n_1 C_{p1} + n_2 C_{p2}}{n_1 + n_2}$$

$$\gamma_{mix} = \frac{(C_p)_{mix}}{(C_v)_{mix}}$$

$$T_{mix} = \frac{n_1 f_1 T_1 + n_2 f_2 T_2}{n_1 f_1 + n_2 f_2}$$

$$\frac{n_1 + n_2}{\gamma_{mix} - 1} = \frac{n_1}{\gamma_1 - 1} + \frac{n_2}{\gamma_2 - 1}$$

9 1st Law of Thermodynamics

$$dQ = dU + dW$$

Based on Energy Conservation

dQ	dU	dW
Given = + To gas	$T \uparrow = +$ of gas	by gas = +
Taken = - from gas	$T \downarrow = -$ of gas	on gas = -

10 Gas laws and thermodynamic process

Gay-Lussac Law:-

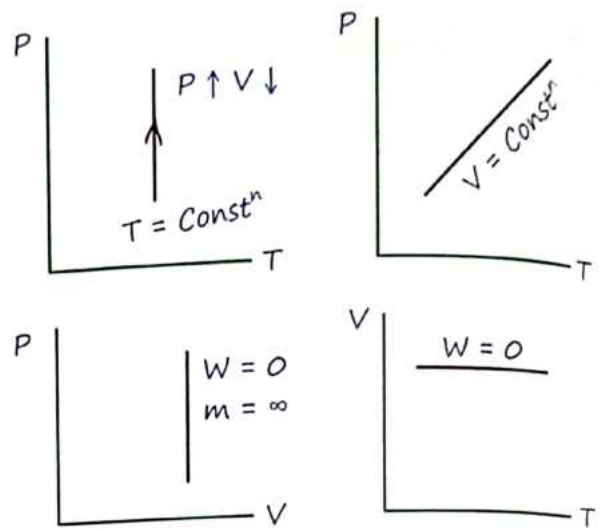
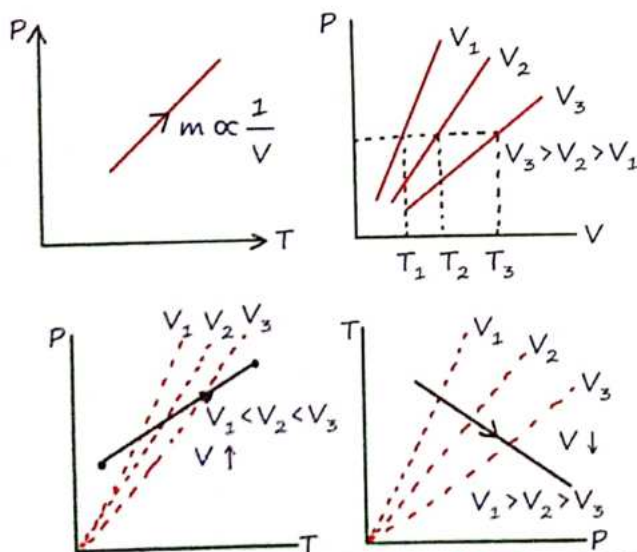
At constant volume, the pressure on a given substance is directly proportional to absolute temperature, i.e.,

$$V = \text{Const}^n \quad (P \propto T)$$

Isochoric Process !

- Work done = 0
- Heat added, $Q = \Delta U = nC_V \Delta T$
- For a monatomic ideal gas, the heat added can also be expressed as $Q = \frac{nR\Delta T}{2}$

Graph:-



Charles Law:-

Charles' Law states that the volume of a given mass of gas is directly proportional to its absolute temperature, provided the pressure remains constant.

$$\text{i.e., } P = \text{Const}^n \quad (V \propto T)$$

Isobaric Process !

When the pressure is constant, the thermodynamic process is isobaric and the work done by the gas in this process is given by the

$$W = P\Delta V = P(V_2 - V_1)$$

Fraction of Heat goes in Work :-

In an isobaric process, not all the heat added to the system goes into doing work. Some of it goes into increasing the internal energy of the gas.

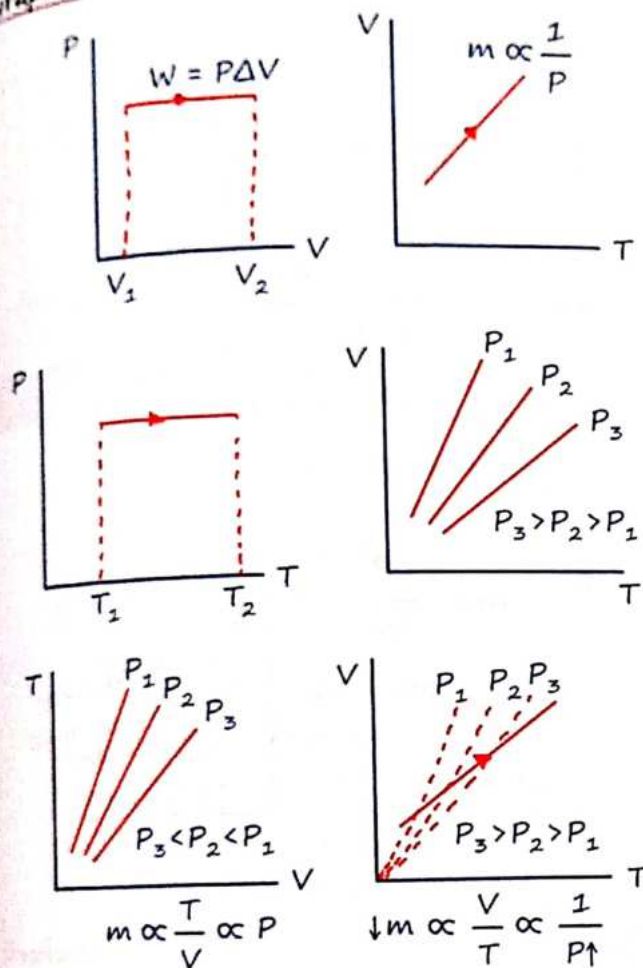
- Fraction of Heat that goes into Work done

$$\frac{W}{Q} = 1 - \frac{1}{\gamma}$$

Fraction of Heat that goes into Internal Energy:

$$\frac{dU}{dQ} = \frac{1}{\gamma}$$

Graph:-



Boyle's Law:-

- Boyle's Law states that the pressure of a given mass of gas is inversely proportional to its volume, provided the temperature remains constant.

$$T = \text{Const}^n \quad PV = \text{Const}^n$$

Isothermal Process !

- An isothermal process is a thermodynamic process in which the temperature of the system remains constant.

Characteristics:-

- This process is reversible, meaning it happens very slowly to ensure equilibrium at all stages.
- The heat added to the system is entirely used for doing work.
- $\Delta U = 0 \rightarrow$ Hum jitna Kam Kareng sab heat mein jayega!
- Molar heat capacity for an isothermal process is theoretically infinite because any amount of heat can be added without changing the temperature. $C_m = \frac{dQ}{n\Delta T} = \infty$

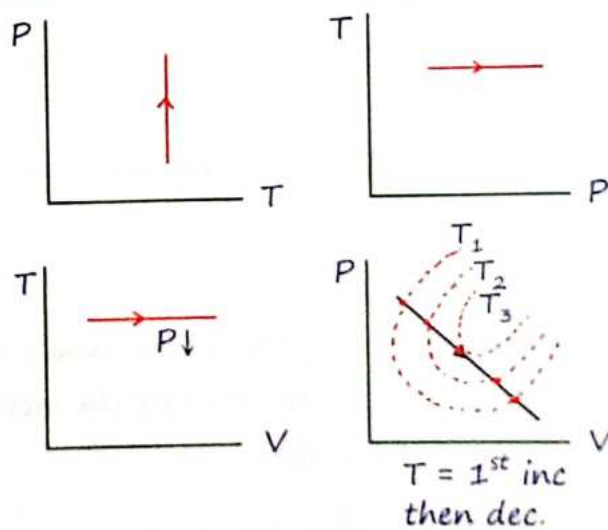
Work done in Isothermal process

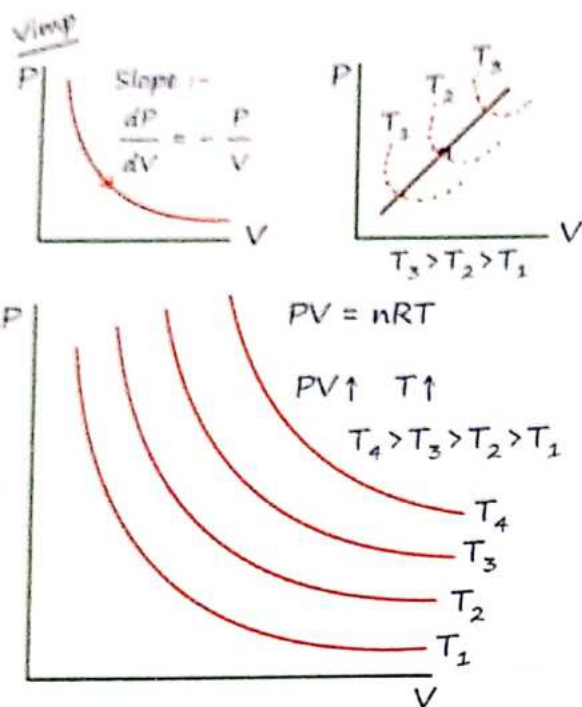
$$\text{Work} = nRT \left[\ln \left(\frac{V_2}{V_1} \right) \right]$$

$$W = 2.303 nRT \left[\log_{10} \left(\frac{V_2}{V_1} \right) \right] = dQ$$

$$W = 2.303 nRT \left[\log_{10} \left(\frac{P_1}{P_2} \right) \right] = dQ$$

Graph:-





11 Adiabatic Process

- In an adiabatic process, the system is perfectly insulated, so no heat enters or leaves the system. i.e.,

$$\Delta Q = 0 \quad PV^\gamma = \text{Constn}$$

$$\gamma = \text{Adiabatic Coefficient} = \frac{C_p}{C_v}$$

Process characteristics:-

Sudden Process:

- Adiabatic processes are typically very rapid, meaning the system does not have time to exchange heat with its surroundings.

Work Done (dW):

- In an adiabatic process, the work done by or on the system changes the internal energy of the system.

$$dW = -dU$$

Example: Tyre burst

Expansion :- $V \uparrow$ $W = +ve$

$$U = -ve \quad T \downarrow$$

Compression :- $V \downarrow$ $W = -ve$

$$U = +ve \quad T \uparrow$$

$$C_m = \frac{dQ}{n\Delta T} = 0$$

Entropy, $S = 0$

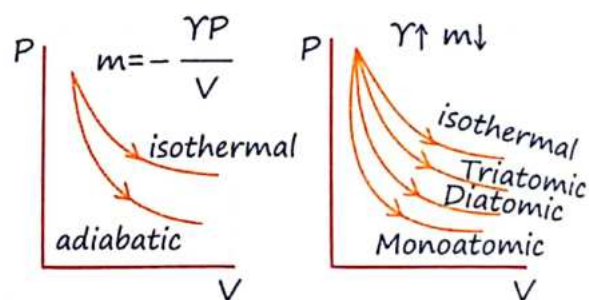
Note :-

$$dW = -nC_v \Delta T = \frac{nR(T_2 - T_1)}{1 - \gamma}$$

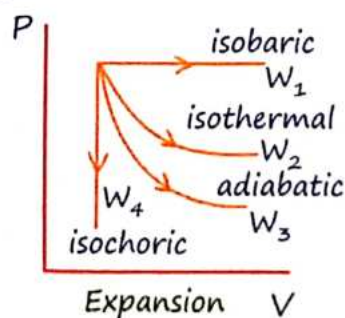
$$dW = \frac{P_2 V_2 - P_1 V_1}{1 - \gamma}$$

$$PV^\gamma = TV^{\gamma-1} = P^{1-\gamma} T^\gamma = \text{Constant}$$

Graph :-



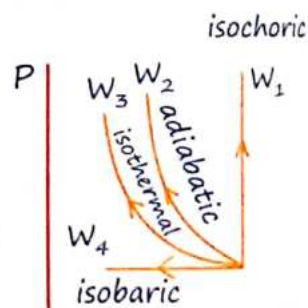
Wimp



Expansion V

$$W_1 > W_2 > W_3$$

$$W_4 = 0$$



Compression V

$$W_2 > W_3 > W_4$$

$$W_1 = 0$$

Note :-

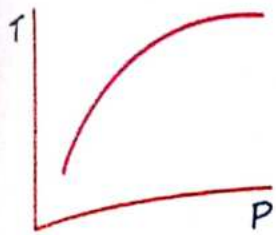
$$\text{Adiabatic Elasticity} = \text{Bulk Modulus} = \gamma P$$

$$P^{1-\gamma} T^{\gamma} = \text{Const}^n$$

$$T = P^{\frac{\gamma-1}{\gamma}}$$

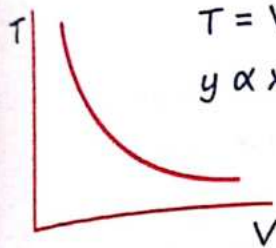
$$\gamma > 1$$

Power of $P < 1$



$$T = V^{1-\gamma}$$

$y \propto x^{-1}$ Rectangular hyperbola



12 Polytropic Process

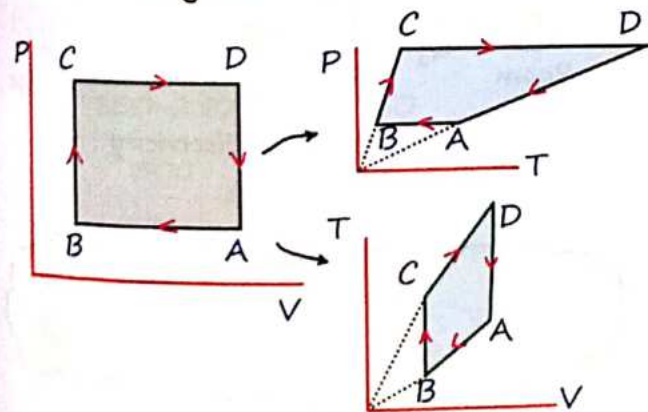
$$PV^x = TV^{x-1} = P^{1-x} T^x = \text{const}^n$$

$$W = \frac{nR(T_2 - T_1)}{1-x} \quad C_m = C_v + \frac{R}{1-x}$$

$$C_m = \frac{R}{\gamma-1} + \frac{R}{1-x}$$

Graph Conversions :-

With the help of P-V diagram draw P-T and T-V diagram



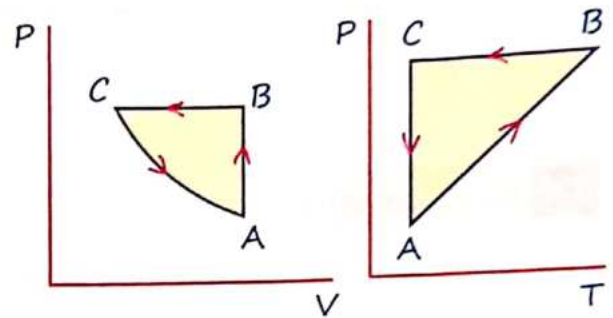
$$A \rightarrow B = P = \text{Const}^n \quad V \downarrow \quad T \downarrow$$

$$B \rightarrow C = V = \text{Const}^n \quad P \uparrow \quad T \uparrow$$

$$C \rightarrow D = P = \text{Const}^n \quad V \uparrow \quad T \uparrow$$

$$D \rightarrow A = V = \text{Const}^n \quad P \downarrow \quad T \downarrow$$

With the help of P-V diagram draw P-T

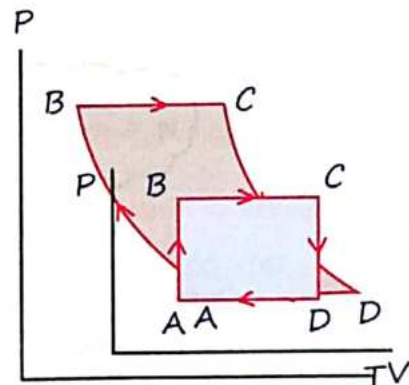


$$A \rightarrow B \quad P \uparrow \quad T \uparrow \quad V = \text{Const}^n$$

$$B \rightarrow C \quad V \downarrow \quad T \downarrow \quad P = \text{Const}^n$$

$$C \rightarrow A \quad P \downarrow \quad V \uparrow \quad T = \text{Const}^n$$

With the help of P-V diagram draw P-T



$$A \rightarrow B \quad T = \text{Const}^n \quad P \uparrow \quad V \downarrow$$

$$B \rightarrow C \quad P = \text{Const}^n \quad T \uparrow \quad V \uparrow$$

$$C \rightarrow D \quad T = \text{Const}^n \quad P \downarrow \quad V \uparrow$$

$$D \rightarrow A \quad P = \text{Const}^n \quad T \downarrow \quad V \downarrow$$

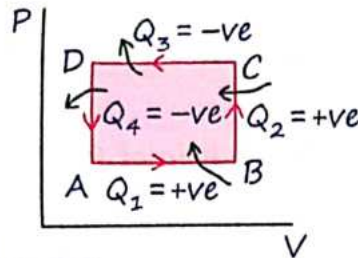
13 Efficiency (η)

$$\eta = \frac{\text{Out put}}{\text{Input}}$$

$$\eta = \frac{WD}{Q_{\text{given}}}$$

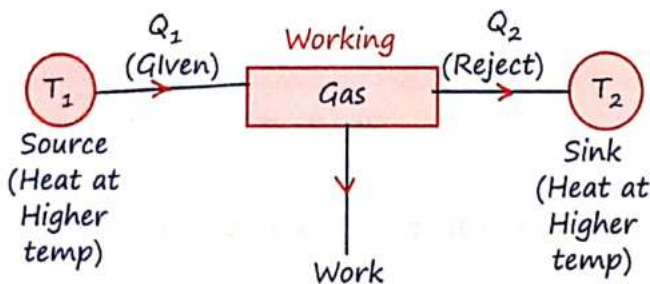
$$\eta = \frac{A_{\text{req}}}{Q_1 + Q_2}$$

$$\eta = \frac{A_{\text{req}}}{\underbrace{nC_p \Delta T}_{AB} + \underbrace{nC_v \Delta T}_{BC}}$$



14 Heat Engine

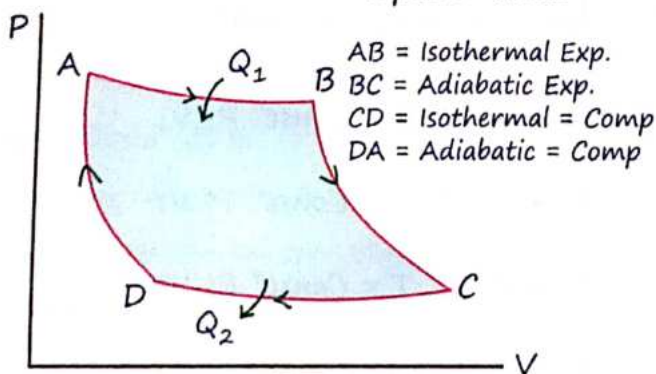
It is a device which converts heat engine into mechanical energy.



$$Q_1 = W + Q_2 \quad \eta = \frac{Q_1 - Q_2}{Q_1}$$

Carnot Engine:-

Output :- Work
Input :- Heat



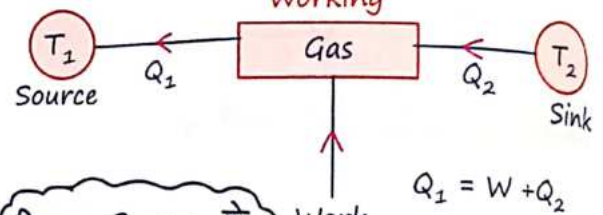
$$\text{Carnot Theorem :- } \frac{Q_1}{Q_2} = \frac{T_1}{T_2}$$

$$\eta = \frac{\text{Work}}{Q_1} \quad \eta = 1 - \frac{Q_2}{Q_1}$$

$$\therefore \eta = 1 - \frac{T_{\text{Kam}}}{T_{\text{Jyada}}}$$

Heat Pump :-

Output : Heat
Input : Work



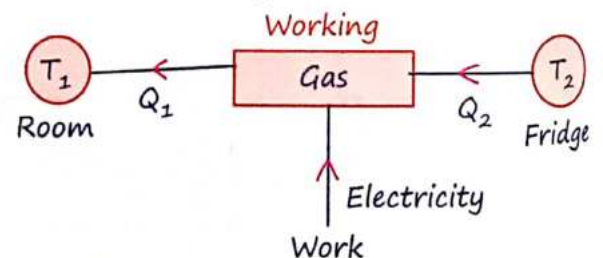
Pump Engine से उल्टा Kaam Krta hai !

$$\eta = \frac{Q_1}{W} = \frac{1}{\eta_{\text{engine}}}$$

Refrigerator :-

Output :- Heat
Input :- Work

• Same as Pump.



$$\beta = \frac{Q_2}{W} = \frac{Q_2}{Q_1 - Q_2} = \frac{T_2}{T_1 - T_2} = \frac{\text{Kam}}{\text{diff.}}$$

$$\beta = \frac{1-\eta}{\eta} = \frac{1}{\eta} - 1 \quad \underline{\underline{Vimp}}$$

Note:-

Heat engines and refrigerator are deleted from the syllabus

Q.3. The efficiency of a Carnot heat engine is $\frac{1}{3}$. Find the coefficient of performance of Carnot refrigerator when both heat engine and refrigerator are working between similar source and sink.

Sol. Efficiency of heat engine is, $\eta = \frac{1}{3}$

The relation between β & η when same Carnot engine is used,

$$\Rightarrow \beta = \frac{1-\eta}{\eta}$$

$$\therefore \beta = \frac{1 - \frac{1}{3}}{\frac{1}{3}} = 2$$

Q.4. A Carnot engine works as a refrigerator in between 250 K and 300 K. If it acquires 750 calories from heat source at low temperature, then what is the heat generated at higher temperature. (in calories)?

$$\begin{aligned} \text{Sol. } \eta &= \frac{T_1 - T_2}{T} = \frac{Q_1 - Q_2}{Q_1} \\ &= \frac{300 - 250}{300} = \frac{Q - 750}{Q} \\ \Rightarrow \frac{50}{300} &= 1 - \frac{750}{Q} \\ \Rightarrow \frac{750}{Q} &= 1 - \frac{1}{6} = \frac{5}{6} \end{aligned}$$

$$\therefore Q = \frac{750 \times 6}{5}$$

$$= 900$$

Q.5. A system is taken from state A to state B along two different paths 1 and 2. If the heat absorbed and work done by the system along these two paths are Q_1, Q_2 and W_1, W_2 respectively, then

(a) $Q_1 = Q_2$

(b) $W_1 = W_2$

(c) $Q_1 - W_1 = Q_2 - W_2$

(d) $Q_1 + W_1 = Q_2 + W_2$

Sol. Internal energy does not depend on path. Heat & work depends (c) $Q_1 - W_1 = Q_2 - W_2$ is correct.

Q.6. In a given process, $dW = 0, dQ < 0$, then for the gas:

(a) Temperature increases

(b) Volume decreases

(c) Pressure decreases

(d) Pressure increases

Sol. $dW = 0 \quad dQ < 0$

$$dU + dW = dQ$$

$$dU = dQ$$

$$dU < 0$$

Temperature decrease

Volume constant

$$P \propto T$$

Pressure decreases

Q.7. If 32 gm of O_2 at $27^\circ C$ is mixed with 64 gm of O_2 at $327^\circ C$ in an adiabatic vessel, then the final temperature of the mixture will be:

Sol. For adiabatic process, gain of heat = loss of heat =

$$\text{or, } m_1 S(T - T_1) = m_2 S(T_2 - T)$$

$$\text{or, } m_1(T - T_1) = m_2(T_2 - T)$$

$$\text{or, } 32 \times (T - 27) = 64 \times (327 - T)$$

$$\text{or, } T - 27 = 2(327 - T)$$

$$\text{or, } T - 27 = 654 - 2T$$

$$\text{or, } 3T = 681$$

$$\text{or, } T = 227^\circ C$$

Q.8. If W_1 is the work done in compressing an ideal gas from a given initial state through a certain volume isothermally and W_2 is the work done in compressing the same gas from the same initial state through the same volume adiabatically, then:

$$(a) W_1 = W_2 \quad (b) W_1 < W_2$$

$$(c) W_1 > W_2 \quad (d) W_1 = 2W_2$$

Sol. (b)

Q.9. During an experiment an ideal gas is found to obey an additional law $VP^2 = \text{constant}$. The gas is initially at a temperature T and volume V . When it expands to a volume $2V$, the temperature becomes

Sol. Ideal gas: $VP^2 = \text{constant}$

Again $PV = nRT$ from equation of state.
Hence, $VP \times P = \text{constant}$ i.e. $nRT \times P = \text{constant}$

$$\text{Again, } P = \frac{nRT}{V}$$

$$\therefore \frac{(nRT)^2}{V} = \text{constant}$$

$$\frac{T^2}{V} = \text{constant}$$

Thus volume V when expanded to $2V$, temperature T_2

$$T_2 = \sqrt{\frac{2V}{V}} T_1 = 2\sqrt{T_1}$$

Q.10. A Carnot engine having an efficiency of $\frac{1}{10}$ as a heat engine, is used as a refrigerator. If the work done on the system is 10 J, the amount of energy absorbed from the reservoir at lower temperature is:

Sol. Coefficient of performance of a refrigerator

$$\beta = \frac{1 - \eta}{\eta}$$

$$\beta = \frac{1 - \frac{1}{10}}{\frac{1}{10}} = 9$$

Also $\beta = \frac{Q_L}{W}$ (where W is the work done)

or

$$Q_L = \beta \times W = 9 \times 10 = 90 \text{ J}$$

Q.11. During an adiabatic process, the pressure of a gas is found to be proportional to the cube of its temperature. The ratio of C_p/C_v for the gas is equal to:

Sol. $P \propto T^3$,

$$PV = nRT$$

$$P \propto T^3$$

$$P \propto (PV)^3$$

$$P^2 V^3 = \text{constant}$$

$$PV^{\frac{3}{2}} = \text{constant}$$

$$\gamma = \frac{3}{2}$$

Q.12. A monoatomic gas at pressure p_1 and

V_1 is compressed adiabatically to $\frac{1}{8}$ th its original volume. What is the final pressure of the gas?

Sol. Correct option is (a)

$$\text{It is given that } \frac{C_p}{C_v} = \gamma = \frac{5}{3}$$

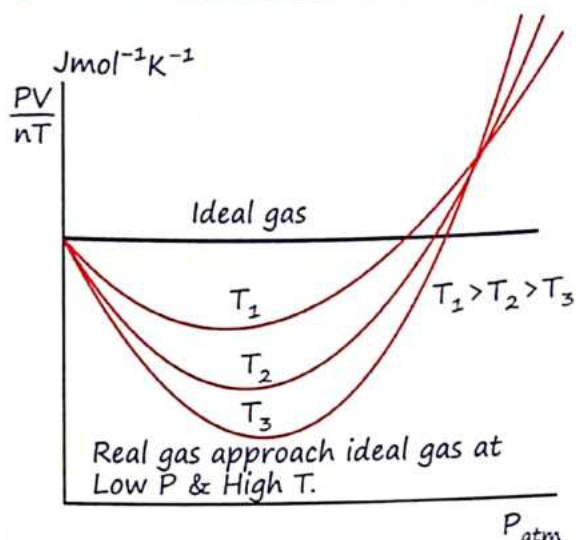
For an adiabatic process,

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

$$\Rightarrow \frac{P_2}{P_1} = \left(\frac{V_1}{V_2} \right)^{5/3}$$

$$= \left(\frac{8}{1} \right)^{5/3} = 32$$

15 Kinetic Theory of Ideal Gas



Ideal Gas Behavior:

- For an ideal gas, the relationship between pressure (P), volume (V), and temperature (T) is given by the ideal gas law: $PV = nRT$
- In the graph, this relationship is represented by a horizontal line where $\frac{PV}{nT}$ equals the ideal gas constant.

Real Gas Behavior:

- Real gases deviate from ideal gas behavior, especially at high pressures and low temperatures. This deviation is depicted by the red curves in the graph.
- The red curves show how real gases approach the behavior of an ideal gas under different conditions.
- Kinetic theory of gases is based on the molecules picture of matter. According to which :-

- (1) A given amount of gas is a mixture of very large number of identical molecules of the order of Avogadro's number.

- (2) The molecules are moving randomly in all directions.
- (3) At ordinary temperature and pressure, the size of the molecules is very small as compared to the distances between them. Thus, the interaction between them is negligible. Hence, they move according to Newton's laws of motion.
- (4) The molecules do not exert any force of attraction or repulsion on each other, except during collisions.
- (5) The collisions of molecules against each other or with the walls of the container are perfectly elastic. Such that the momentum and the kinetic energy of the molecules are conserved during collisions, though their velocities change.

Q.13. Select the appropriate property of an ideal gas

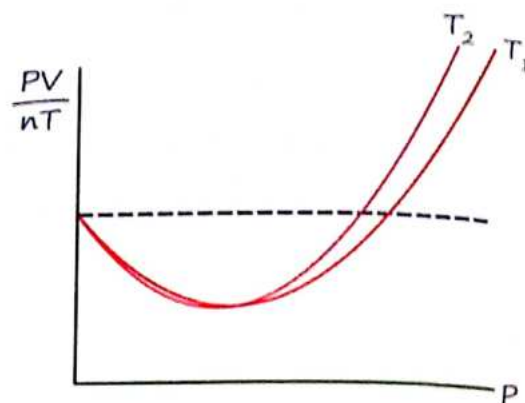
- (a) Its molecules are infinitesimally small/
- (b) There are no forces of interaction between molecules/
- (c) It strictly obeys the ideal gas laws/
- (d) All of these

Sol. (d)

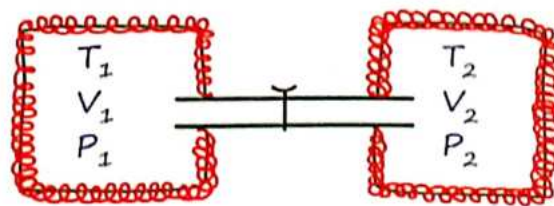
Q.14. The figure shows the plot of $\frac{PV}{nT}$ vs P , for oxygen gas at two different nT temperatures then

- (a) $T_1 > T_2$
- (b) $T_2 > T_1$
- (c) $T_1 = T_2$
- (d) May be (a) and (b)

Sol. (a)



Mixing of gases



The valve joining the two vessels is opened :-

$$T_{\text{mix}} = ?$$

$$T_{\text{mix}} = \frac{n_1 T_1 + n_2 T_2}{n_1 + n_2} = \frac{\frac{P_1 V_1}{R} + \frac{P_2 V_2}{R}}{\frac{P_1 V_1}{RT_1} + \frac{P_2 V_2}{RT_2}}$$

$$T_{\text{mix}} = \frac{(P_1 V_1 + P_2 V_2) T_1 T_2}{P_1 V_1 T_2 + P_2 V_2 T_1}$$

Pressure of ideal gas :-

$$P = \frac{1}{3} n m V_{\text{rms}}^2 \quad n = \text{no. density} = \frac{N}{V}$$

$$V_{\text{rms}} = \sqrt{\frac{3P}{nm}} \quad m = \text{mass of each molecule}$$

Kinetic interpretation of temperature :-

$$KE = \frac{3RT}{2}$$

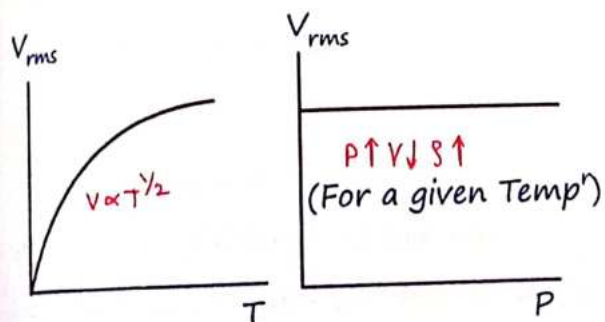
Energy density

$$\frac{KE}{V} = \frac{3P}{2}$$

Rotational KE :-

$$KE = \frac{fRT}{2N_A} \quad KE = \frac{fKT}{2}$$

$$V_{rms} = \sqrt{\frac{3KT}{m}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3P}{\rho}} \quad V_{rms}$$



Average speed of the molecule of gas:-

$$V_{avg} = \sqrt{\frac{8RT}{\pi M}} = \sqrt{\frac{8}{3\pi}} V_{rms} \quad V_{avg}$$

$$V_{avg} = 0.92 V_{rms}$$

Most probable speed of the molecule of gas:-

$$V_{mp} = \sqrt{\frac{2RT}{M}} \quad V_{MP}$$

Q.15. Four molecules of a gas have speeds 1, 2, 3 and 4 km/s. The value of the r.m.s. speed of the gas molecules is

(a) $\frac{1}{2}\sqrt{15}$ km/s (b) $\frac{1}{2}\sqrt{10}$ km/s

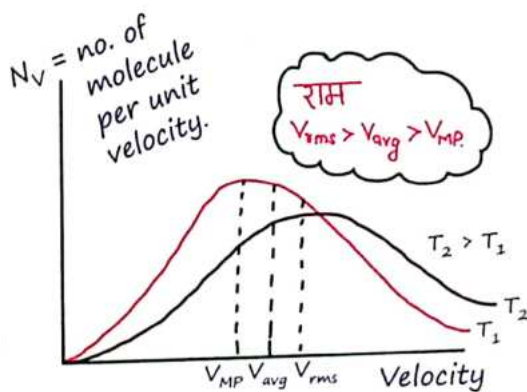
(c) 2.5 km/s (d) $\sqrt{\frac{15}{2}}$ km/s

Sol.

$$V_{r.m.s} = \sqrt{\frac{1^2 + 2^2 + 3^2 + 4^2}{4}} = \sqrt{\frac{1 + 4 + 9 + 16}{4}}$$

$$= \sqrt{\frac{30}{4}} = \sqrt{\frac{15}{2}}$$

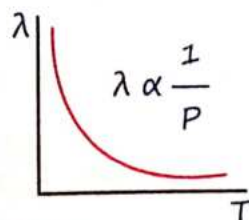
Maxwell Distribution Curve :-



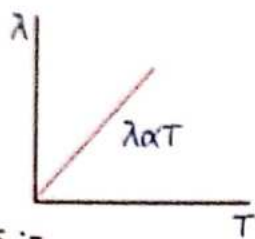
Area = $\int N_v dv$ = no. of molecules.
 Temp^r independent.

Mean free path :-

$$\lambda = \frac{1}{\sqrt{2} \pi n d^2} \quad n = \text{number density} = \frac{N}{V}$$



$$\lambda = \frac{RT}{\sqrt{2\pi N_A P}} \quad d = \text{diameter} = 2r$$



Density of gas :-

$$d = \frac{PM}{RT} = \frac{Pm}{K_B T}$$

Assertion and reason based questions:-

A → A liquid is filled in container which is moving with high speed does not have higher temperature.

R → Temp^r of liquid related to internal energy, not to K.E. of liquid.

Sol.:- Both are true and correct explanation.

A → A mixture of petrol and air when ignited is not in equilibrium state.

R → Its temperature and pressure not uniform.

Sol.:- Both are correct and correct explanation.

A → ΔQ is extensive.

R → It is propotional to total mass of system.

Sol.:- Both are true and correct explanation.

Simple Harmonic Motion

Periodic Motion

(A motion which Repeat itself after a regular interval of time)

(repeat about mean)

(repeat not about mean)

Oscillation

Not Oscillation

large Amplitude

- Small amplitude
- Energy conserved
- $F \propto -x^n$

Not S.H.M

Simple harmonic Motion

- + Vibration $\rightarrow$ Oscillation with high frequency.
- + Periodic but not oscillation
 - $\rightarrow$ all uniform circular motion
 - $\rightarrow$ earth around sun
- + All SHM is oscillatory and periodic
- + All oscillatory is periodic but need not S.H.M
- + All periodic need not to be oscillatory & S.H.M.

2 S.H.M

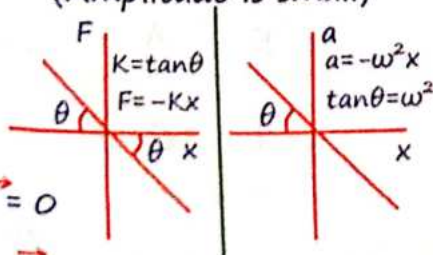
(Amplitude is small)

$$F \propto -x$$

$$a = -\omega^2 x$$

$$\frac{d^2 x}{dt^2} + \omega^2 x = 0$$

$$a \propto -x$$



Equation of SHM:-

$$x = A \sin(\omega t + \phi)$$

$$v = \frac{dx}{dt} = A\omega \cos(\omega t + \phi)$$

$$a = \frac{dv}{dt} = -A\omega^2 \sin(\omega t + \phi)$$

Where,

A = Amplitude

ω = Angular Frequency

$$= \frac{2\pi}{T} = 2\pi f$$

ϕ = initial phase, x = displacement from mean position.

Note:-

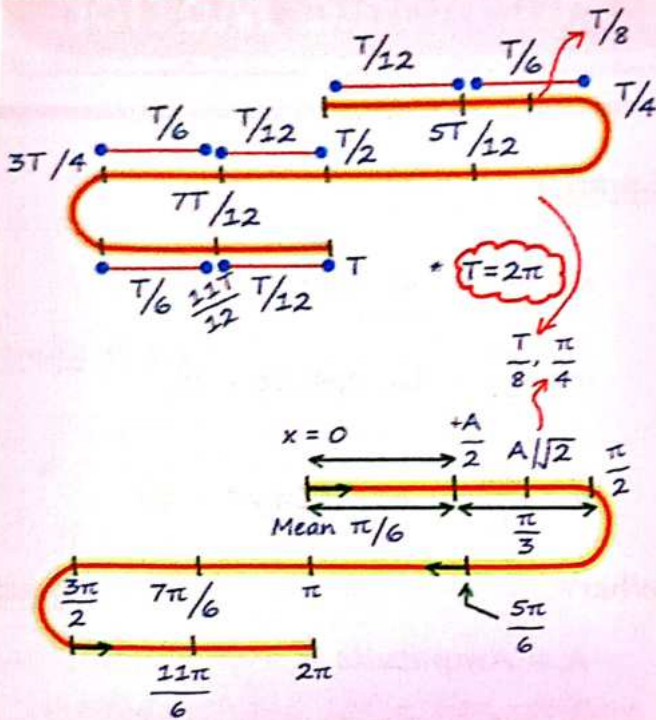
- + $\phi = 0$ for mean se start
- + $\phi = \pi/2$ for extreme se start
- + $\phi = \pi/6$ for half of extreme se start
- + $\phi = \pi/4$ for $x = \frac{A}{\sqrt{2}}$ se start
- + $a \propto -x^n$ $n = \text{even} = \text{Translational}$

$n = \text{odd but not } 1.$

$\rightarrow$ "Oscillation but not SHM"

$$a \propto -x^1 \Rightarrow \text{SHM!}$$

Fire Concept MR*



3 Motion From Mean position

Position,

$$+ x = A \sin \omega t$$

velocity,

$$+ v = A\omega \cos(\omega t)$$

$$+ v = A\omega \sqrt{1 - \sin^2 \omega t}$$

$$+ v = \omega \sqrt{A^2 - x^2} \quad \omega = \sqrt{\frac{k}{m}}$$

Acceleration:-

$$a = -A\omega^2 \sin \omega t, \quad a = -\omega^2 x$$

Note:-

Mean

$$x = 0$$

$$v_{\max} = A\omega$$

$$a = 0$$

Extreme

$$x = A$$

$$v = 0$$

$$a_{\max} = -\omega^2 A$$

MR*

Mai tere piche, wo mere piche hai re kism
mai tum se na mil pau, wo mere se na m
pay

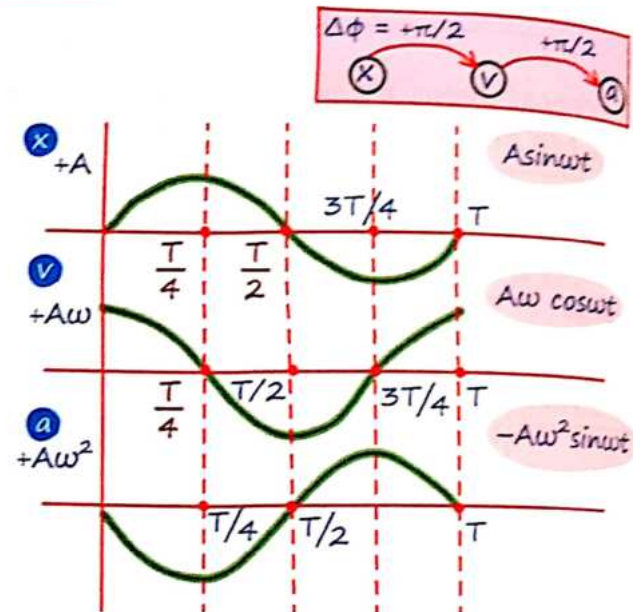
Mai → Velocity

Wo → Position

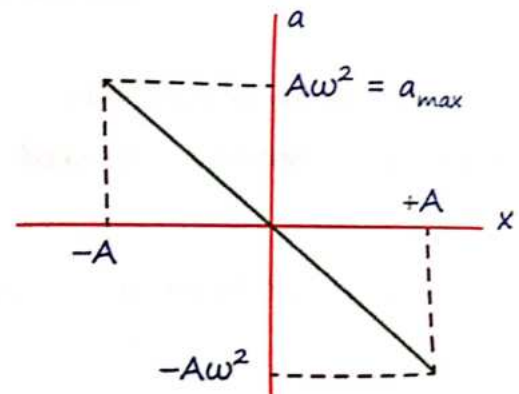
turn → accⁿ
😊

4 Graphs

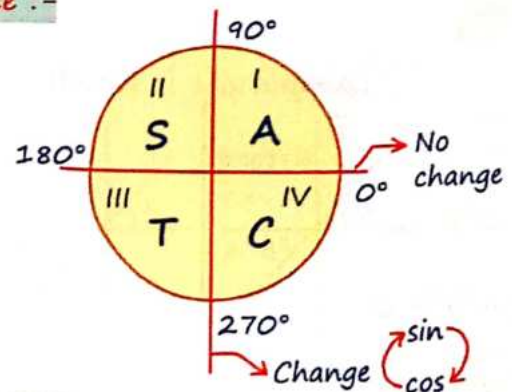
x - t:-

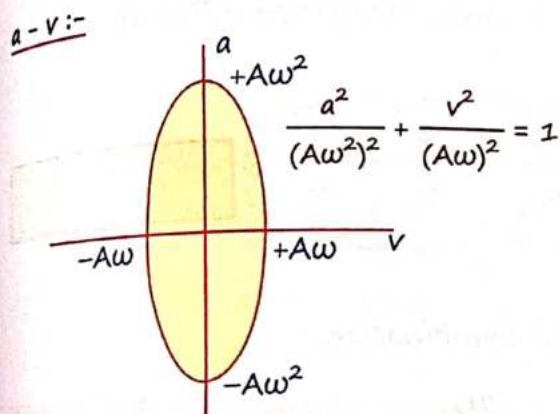
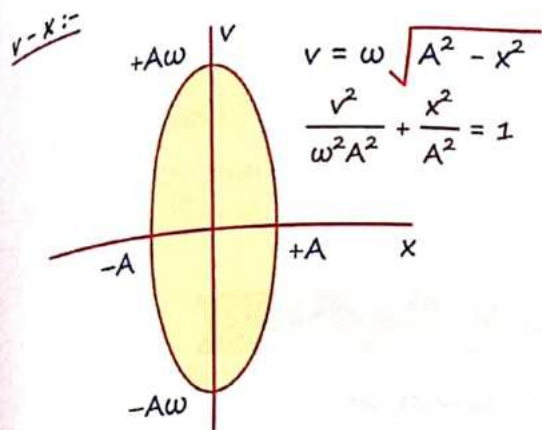


a = -ω²x:-



Note :-





Important

Time period (TP) of a particle executing SHM along straight line with its velocity at position x_1 and x_2 from mean are V_1 and V_2 & Then TP:-

$$T = 2\pi \sqrt{\frac{x_2^2 - x_1^2}{v_1^2 - v_2^2}}$$

5 Energies

Kinetic energy, $= \frac{1}{2} mv^2$

we know $m = \frac{K}{\omega^2}$

Kinetic energy:-

$$KE = \frac{1}{2} K(A^2 - x^2) = \frac{1}{2} KA^2 \cos^2(\omega t + \phi)$$

Potential energy:-

$$PE = \frac{Kx^2}{2} + U_0 = \frac{1}{2} KA^2 \sin^2 \omega t + U_0$$

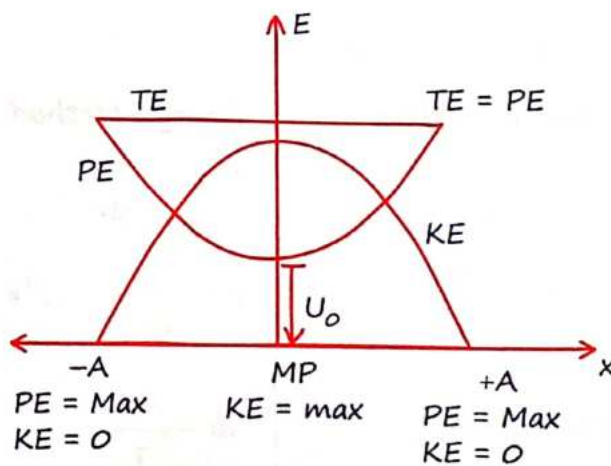
Total energy:-

$$TE = \frac{1}{2} KA^2 + U_0 = \text{constant through out the motion}$$

+ $K = \text{Force Const}^n = m\omega^2$

Energy graphs:-

The graph visually represents how kinetic and potential energies vary with displacement in simple harmonic motion. The total energy of the system remains constant, and energy oscillates between kinetic and potential forms as the particle moves between the mean and extreme positions.



+ At the mean position (MP):

- KE is at its maximum.
- PE is zero.
- Total Energy (TE) = Maximum Kinetic Energy.

* At the extreme positions ($\pm A$):

- PE is at its maximum.
- KE is zero.
- Total Energy (TE) = Maximum Potential Energy.

MR*

Physical Quantity	Time Period	Frequency
Position	T	f
Velocity	T	f
Speed.	T/2	2f
Acceleration	T	f
KE	T/2	2f
PE	T/2	2f
KE-PE	T/2	2f
KE-PE	T/4	4f
ME = KE + PE	∞	0

Kitne time lagega Repeat Krne mein

Kitne bar given time में अपने Pattern को Repeat करेगा.

Time period (T.P) of SHM:-

Force Method

$$F = -Kx$$

$$a = -\frac{Kx}{m} = -\omega^2 x$$

$$\omega = \frac{2\pi}{T}$$

Energy Method

$$F = -\frac{dU}{dr}$$

$$a = \frac{F}{m} = -\omega^2 x$$

$$\omega = \frac{2\pi}{T}$$

TP of Spring Mass System:-

$$T = 2\pi \sqrt{\frac{M}{K}}$$

($T_{\text{Horizontal}} = T_{\text{vertical}}$)

$F = \text{const}^n$ की औकात नहीं है की SHM करवाए! independent of "g" & shape of object

MR*

- * Equilibrium से जब तुम्हने दिया Mass को तो वो जो F extra आया है वो Oscillate करवाएगा
- * Eqⁿ likhkar Equilibrium का $Kx_0 = mg$ Put किया तो कटा था कटा है और करेगा!

6 Combination of Spring

Series Combination:-

F = same, elongation different

$$\frac{1}{K_{eq}} = \frac{1}{K_1} + \frac{1}{K_2} \quad T^2 = T_1^2 + T_2^2$$

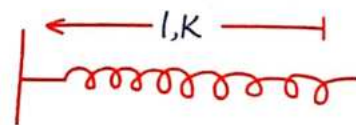
Parallel Combination:-

F = different, elongation = same

$$K_{eq} = K_1 + K_2 \quad \frac{1}{T^2} = \frac{1}{T_1^2} + \frac{1}{T_2^2}$$

Cutting of Spring:-

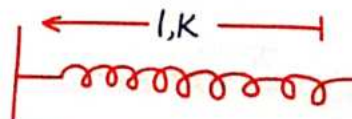
Spring constant $K \propto \frac{1}{\text{Length of spring}}$



cut in two equal part then



Q.1. Cut into three part of ratio 1:2:3 the ratio of spring constant.



$$l_1 : l_2 : l_3 = 1 : 2 : 3 = x : 2x : 3x$$

$$K_1 : K_2 : K_3 = \frac{1}{x} : \frac{1}{2x} : \frac{1}{3x}$$

$$K_1 : K_2 : K_3 = 6 : 3 : 2$$

$$K_1 = 6K \quad K_2 = 3K \quad K_3 = 2K$$

eg:-
cut

$$K \propto \frac{1}{l}$$

$$l_1 : l_2 : l_3 = 1 : 3 : 5$$

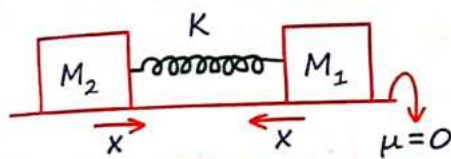
$$\text{join } l_{\text{eq}} = x : 3x : 5x \quad l = 9x$$

$$x = \frac{l}{9}$$

$$\text{Parallel :- } K_{\text{eq}} = \frac{9K}{1} + \frac{9K}{3} + \frac{9K}{5}$$

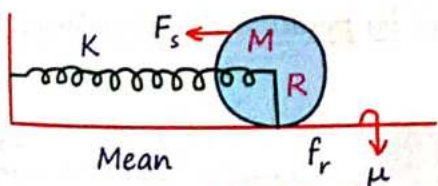
$$= 9K + 3K + \frac{9K}{5} = \frac{69K}{5}$$

Reduced mass Concept :-



$$T = 2\pi \sqrt{\frac{M_1 M_2}{(M_1 + M_2)K}}$$

Rotation + Translation wale Que. :-

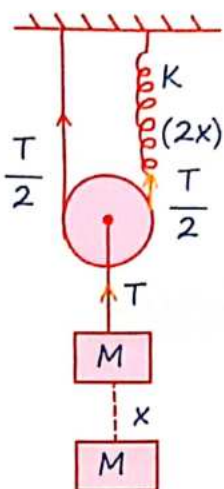


$$T = 2\pi \sqrt{\frac{M + I/R^2}{K}} \quad I = \text{Moment of inertia}$$

$$T = 2\pi \sqrt{\frac{M}{K}} \text{ if } \mu = 0$$

7 Constrain Motion

1>



Since, forces are balancing, therefore

$$K(2x) = \frac{T}{2}$$

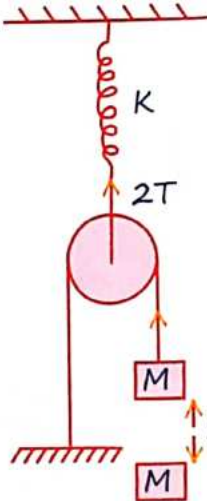
$$\therefore T = 4Kx$$

$$a = \frac{4K}{M}x (\because T = Ma)$$

Comparing with $a = \omega^2 x$

$$\omega = \sqrt{\frac{4K}{M}} \Rightarrow T = 2\pi \sqrt{\frac{M}{4K}}$$

2>

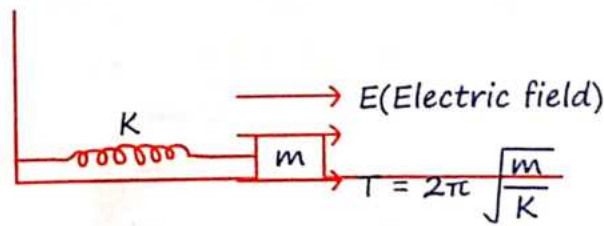


$$2T = \frac{Kx}{2}$$

$$T = \frac{Kx}{4}$$

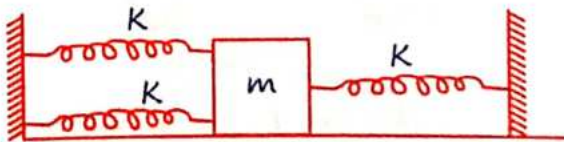
$$T = 2\pi \sqrt{\frac{4m}{k}}$$

3>

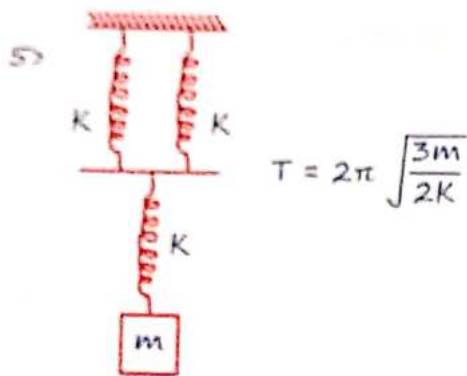


$$T = 2\pi \sqrt{\frac{m}{K}}$$

4>



$$T = 2\pi \sqrt{\frac{m}{3K}}$$



8 Time Period of Simple Pendulum

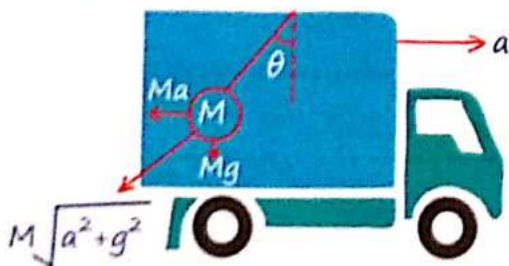
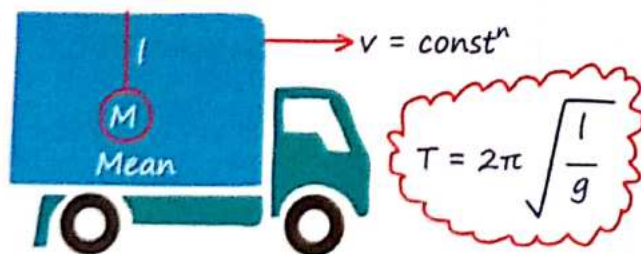
$$T = 2\pi \sqrt{\frac{l}{g_{\text{eff}}}}$$

• Special Cases :

Lift :-

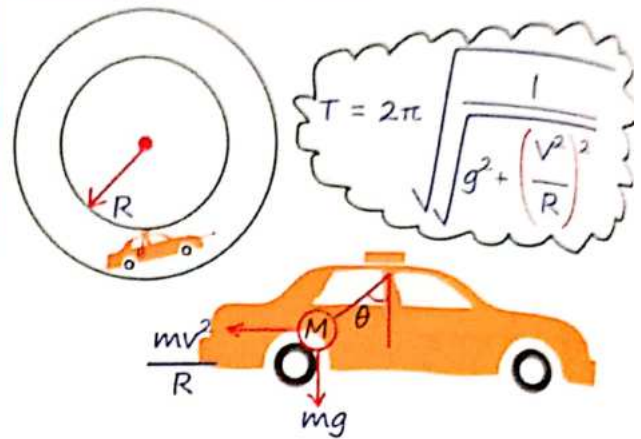
- Up :- $T = 2\pi \sqrt{\frac{l}{g+a}}$
- Down :- $T = 2\pi \sqrt{\frac{l}{g-a}}$
- Free Fall :- $T = \infty$

Car :-

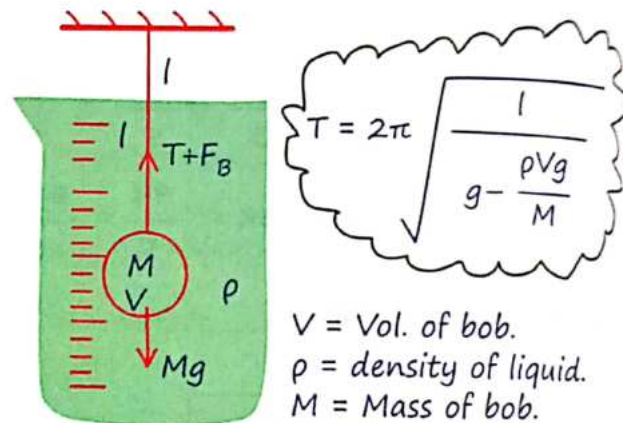


$$T = 2\pi \sqrt{\frac{l}{a^2 + g^2}}$$

Car at Circular Track :-



Pendulum in liquid :-

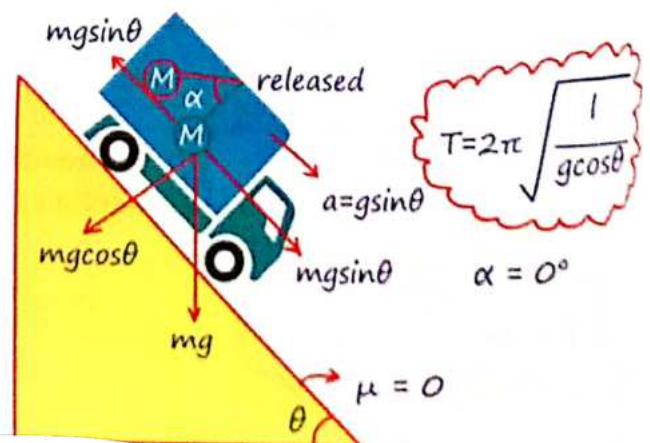


$$T = 2\pi \sqrt{\frac{l}{g \left(1 - \frac{\rho}{\sigma}\right)}}$$

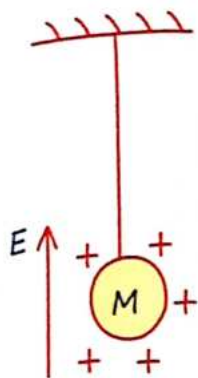
$\sigma = \text{density of bob.}$

Pendulum जो किसी Fluid में डाला तो उसका T.P. बढ़ेगा!

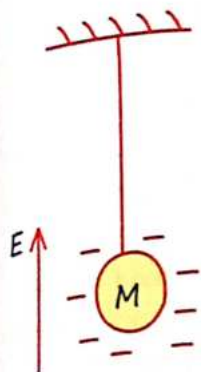
Angle made by pendulum with ceiling is 90° .



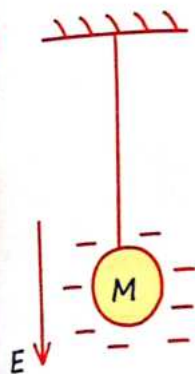
T.P. in Electric field:-



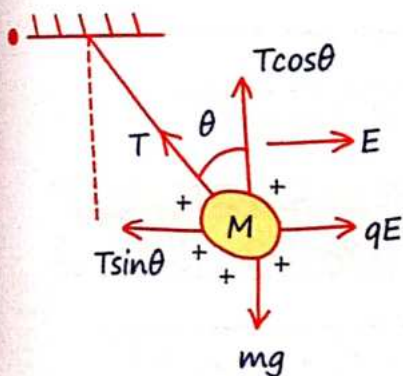
$$T = 2\pi \sqrt{\frac{l}{g - \frac{qE}{M}}}$$



$$T = 2\pi \sqrt{\frac{l}{g + \frac{qE}{M}}}$$



$$T = 2\pi \sqrt{\frac{l}{g - \frac{qE}{M}}}$$



$$T = 2\pi \sqrt{\frac{l}{g^2 + \left(\frac{qE}{M}\right)^2}}$$

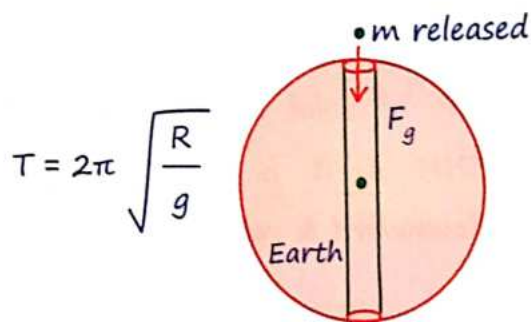
T.P. when length of Simple Pendulum is very

large :-

$*T = 2\pi \sqrt{\frac{1}{g\left(\frac{1}{l} + \frac{1}{R}\right)}}$	+ $l \gg R$ $T = 2\pi \sqrt{\frac{R}{g}}$ $T = 84.6 \text{ min.}$
--	---

$*l = R$ $T = 2\pi \sqrt{\frac{R}{2g}}$	+ $l \ll R$ $T = 2\pi \sqrt{\frac{l}{g}}$
--	--

A mass released inside a tunnel through the Earth would oscillate back and forth through simple harmonic motion with the time period.



9> Second Pendulum :

length = 1m Time period = 2 sec

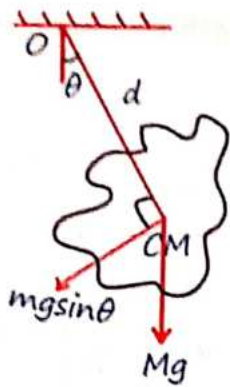
$f = 0.5 \text{ Hz}$

TP of Physical Pendulum :-

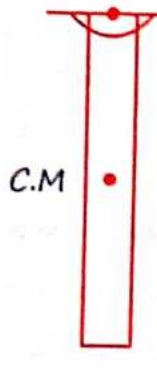
$$T = 2\pi \sqrt{\frac{I}{mgd}}$$

I = M.O.I of object wrt point of suspension.

d = distⁿ of O & CM !



Vertical Rod hinged about one end



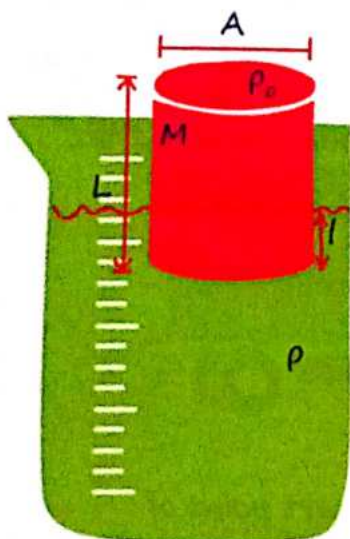
$$I = \frac{ML^2}{3}$$

$$T = 2\pi \sqrt{\frac{ML^2}{3mg \frac{L}{2}}}$$

$$T = 2\pi \sqrt{\frac{2L}{3g}}$$

* Special Case :-

1> T.P. of Solid Cylinder performing SHM if it is slightly displaced downward & released :-



l = length of cylinder inside liquid.

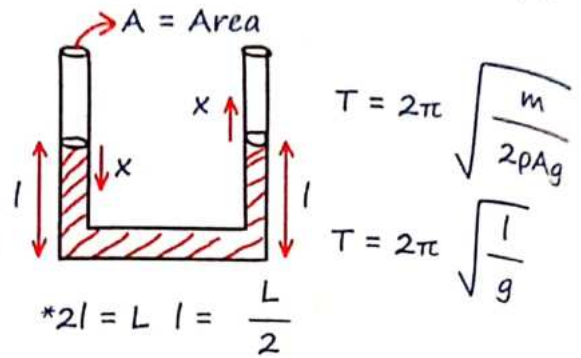
$$T = 2\pi \sqrt{\frac{m}{\rho A g}}$$

$$T = 2\pi \sqrt{\frac{\rho_0 L}{\rho g}}$$

$$\rho l = \rho_0 L$$

$$T = 2\pi \sqrt{\frac{l}{g}}$$

2> Oscillation of liquid column (I) :-



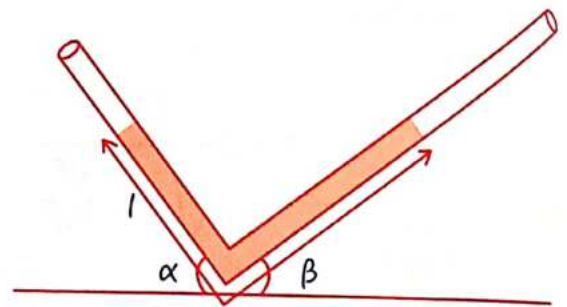
m = mass of liquid

ρ = density of liquid

A = Area of liquid column.

L = Total length of liquid column.

Oscillation of liquid column (II) :-



L = total length of water column

$$T = 2\pi \sqrt{\frac{2L}{g(\sin\alpha + \sin\beta)}}$$

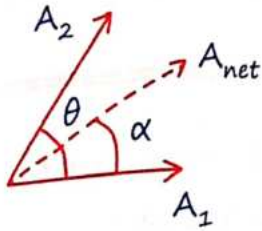
9 Superposition of SHM of two object Oscillating in Same Direction

$$A_{\text{net}} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos\theta}$$

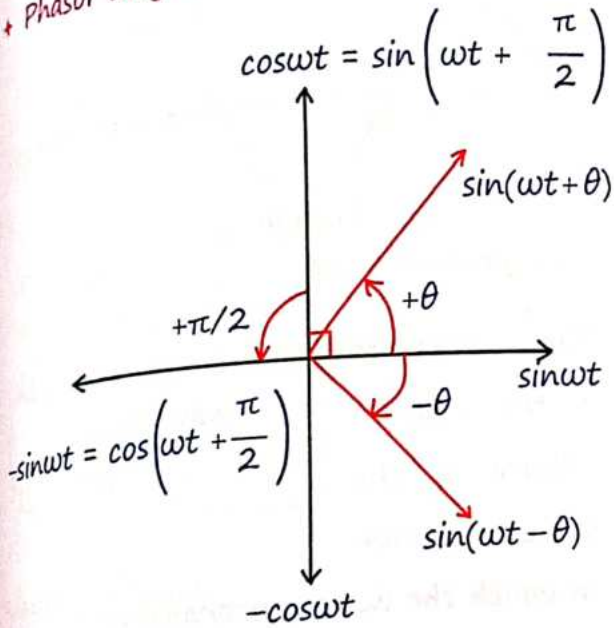
θ = initial phase difference.

$$\tan\alpha = \frac{A_2 \sin\theta}{A_1 + A_2 \cos\theta}$$

$\alpha = \text{Angle b/n } A_{\text{net}} \text{ \& } 1^{\text{st}} \text{ SHM}$



* Phasor diagram :-



Equation of S.H.M of 1 st	Equation of S.H.M of 2 nd	Equation of Superimposed S.H.M
$x_1 = A \sin(\omega t)$	$x_2 = A \cos(\omega t)$	$x = \sqrt{2} A \sin(\omega t + \pi/4)$

$x_1 = A \sin(\omega t)$	$x_2 = A \cos(\omega t + \pi/6)$	$x = A \sin(\omega t + \pi/3)$
$x_1 = A \sin(\omega t - \pi/6)$	$x_2 = A \cos(\omega t - \pi/3)$	$x = \sqrt{3} A \sin(\omega t)$
$x_1 = A \sin(\omega t)$	$y_1 = A \cos(\omega t)$	<u>Circle</u> $x_1^2 + y_1^2 = A^2$
$x_1 = A \sin(\omega t)$	$y_2 = A \sin(\omega t)$	<u>Straight line</u> $x_1 = y_2$
$x_1 = A \sin(\omega t + \pi/3)$	$x_2 = A \cos(\omega t + \pi/6)$	$x = \sqrt{3} A \cos(\omega t)$

Q.2. Amplitude becomes half in 4-oscillation then find amplitude after 16-oscillation.

Sol. $A = A_0 e^{-n}$ $A = A_0 e^{-16}$

$$\frac{A_0}{2} = A_0 e^{-4} \quad = A_0 e^{-4 \times 4}$$

$$e^{-4} = \frac{1}{2} \quad = A_0 \left(\frac{1}{2}\right)^4$$

$$e^{-4} = \frac{1}{2} \quad A = \frac{A_0}{16}$$

MR*

‘ Failure is the first step of success. ’

Tum wave si ho main tumhara medium,
Tum aati ho meri zindagi mein Mera use krti
ho, main apne mean position ke about aage
piche oscillate krta rehjata hu tumhare liye,
aur tum Mera use krke aage nikal jaati ho...

WAVE

Mechanical Wave

Medium ✓

≠ Sound, Wave
on string, wave
on water.

Electromagnetic Waves:

Travel without needing
any material medium
to move through. They
can move through
a vacuum (empty
space). Examples of
electromagnetic waves
include light waves and
radio waves.

1 Mechanical Waves

1> Transverse Wave:-

- The particles of the medium vibrate at right angles to the direction of propagation of wave.

Particle upar
niche wave aage.

Ex. :- String Wave

Non-Mech. Wave

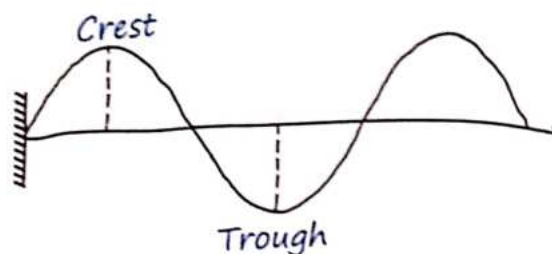
Medium ✗

≠ EM Wave
i.e., Light
Wave.

Matter Waves:

Linked with
tiny particles
like electrons
and protons.
These waves are
important in
understanding
the behavior of
these fundamental
particles.

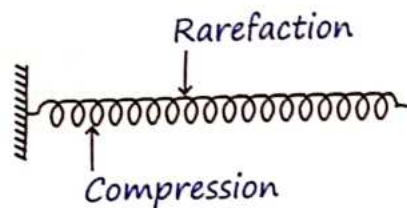
Transverse waves travel in the form of
crests and troughs).



2> Longitudinal Wave :-

- The particles of the medium vibrate in the same direction in which the wave is propagating.

Particle age-piche aur
wave aage.
Ex. :- Sound.



- Longitudinal waves travel in the form of compressions and rarefactions.

2 Equation of Propagating Harmonic Wave

SHM ek particle Ka oscillation hai wave
hazaro lakho particle Ka oscillation hai!

$$Y = A \sin (\omega t \pm Kx \pm \phi)$$

Where, A = Amplitude of particle.

ω = Angular frequency

ϕ = initial phase [t = 0 x = 0]

Angular wave no., $K = \frac{2\pi}{\lambda}$
 wave no., $\bar{v} = \frac{1}{\lambda}$

Longitudinal Wave :- $\updownarrow \updownarrow$

Wave in "y" $y = A \sin [Ky + \omega t + \phi]$ Particle in "y"
 Dono **Same**

Transverse Wave :- $\leftrightarrow \leftrightarrow$

PARTICLE is moving in "x" $x = A \sin (Ky + \omega t)$ WAVE oscillates in "y"
 Dono **different**

$$y = A \sin 2\pi \left[\frac{x}{\lambda} + \frac{t}{T} \right]$$

Velocity of Wave :-

$$V_{\text{wave}} = \frac{\omega}{K} = \frac{\lambda}{T} = \lambda f$$

Note:-

$y = A \sin (Kx - \omega t)$ } ek $\oplus$ dusra $\ominus$!
 Wave is moving in $+x$ -axis.

$y = A \sin (Kx + \omega t)$ } dono $\oplus$
 Wave is moving in $-x$ -axis.

$V_p = A\omega \cos (Kx + \omega t)$

$$(V_p)_{\text{max}} = A\omega$$

Relation Between wave Velocity & maximum Particle Velocity:-

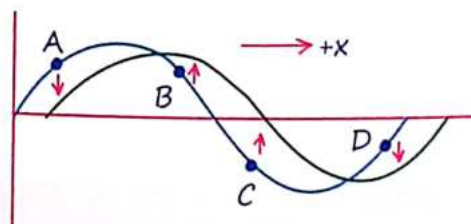
$$*(V_p)_{\text{max}} = AKV_{\text{wave}}$$

Estimation of Particle Whether IT'LL Go UP/DOWN:-

$$V_{\text{particle}} = -V_{\text{wave}} \times \text{Slope of wave}$$

MR*

Na Tumhe Formula likhna na slope dhekhna direct answer bs wave jis direction mein travel Kr rahi hai uss direction mein **thoda shift Krdo**!



Condition of Wave equation:-

$$\frac{d^2y}{dt^2} = \frac{\omega^2}{K^2} \cdot \frac{d^2y}{dx^2} \quad \left| \quad \frac{d^2y}{dx^2} = \frac{d^2y}{dt^2} \cdot \frac{1}{V^2} \right.$$

+ "y" $\rightarrow$ Finite hona chahiye at all position of "x":

$$y = f(ax + bt) = \frac{f}{(ax + bt)}$$

Important Relation :-

$$\frac{2\pi}{\phi} = \frac{\lambda}{\Delta x} = \frac{T}{\Delta t}$$

Q.1. The maximum particle velocity is 3 times the wave velocity of a progressive wave. If A is the amplitude of oscillating particle, find phase difference between two particles of separation x.

Sol. The maximum particle velocity = 3 time wave velocity

$$A\omega = 3v$$

$$\therefore \omega = \frac{3v}{A}$$

As we know

$$v = \lambda f$$

$$\therefore \lambda = \frac{v}{f}$$

$$\text{or } \lambda = \frac{2\pi v}{\frac{3v}{A}}$$

$$\therefore \lambda = \frac{2\pi A}{3}$$

$$\phi = \text{phase difference} = \frac{2\pi}{\lambda} \times x$$

$$= \left(\frac{2\pi}{\frac{2\pi A}{3}} \right) x$$

$$= \frac{3x}{A}$$

Q.2. The velocity of waves in a string fixed at both ends is 2 m/s. The string forms standing waves with nodes 5.0 cm apart. The frequency of vibration of the string in Hz is

Sol. Here the distance between the two nodes is half of the wavelength

$$\frac{\lambda}{2} = 5.0 \text{ cm} \Rightarrow \lambda = 10 \text{ cm}$$

$$\text{Hence } n = \frac{v}{\lambda} = \frac{200}{10} = 20 \text{ Hz}$$

3 Velocity of transverse Wave in A String

$$\text{Strain} = \frac{\Delta l}{l} = \alpha \Delta T$$

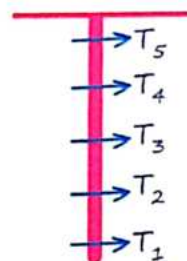
α = coefficient of linear expansion

$$v = \sqrt{\frac{Y\alpha\Delta T}{\rho}}$$

$\mu = M/L$ Y = Young Modulus

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{T}{\rho A}} = \sqrt{\frac{\text{Stress}}{\rho}} = \sqrt{\frac{Y \cdot \text{Strain}}{\rho}}$$

$v \propto \sqrt{T}$ $T_5 > T_4 > T_3 > T_2 > T_1$
 Jaise-Jaise Upar jaoge
 $T \uparrow \therefore v_{\text{wave}} \uparrow$

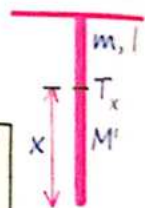


MR*

$$m' = \mu x \quad T_x = m'g = \mu gx$$

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{gx}$$

General Point

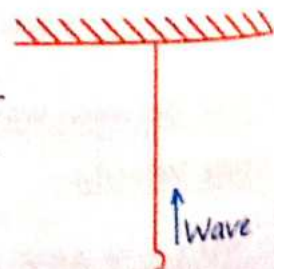


$\therefore v$ & a of wave at $L/2$:-

$$v = \sqrt{\frac{gL}{2}} \quad a = \frac{g}{2} = \text{Const}^n$$

* Time taken to reach top point

$$T = 2 \sqrt{\frac{L}{g}} = \sqrt{\frac{4L}{g}}$$

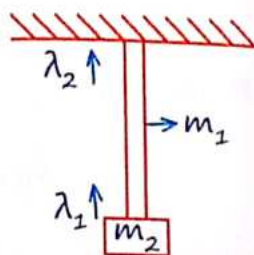


MR+ Ratio of wave length of transvers wave at bottom to top

$V = \lambda f \rightarrow$ frequency same

$\lambda \propto V \propto \sqrt{\text{tension}}$

$$\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{m_2}{m_1 + m_2}}$$



4 Sound Wave

Sound is a type of energy that makes it possible for us to hear things. It travels in waves, and these waves move in a pattern where particles of the medium (like air) vibrate back and forth in the same direction as the wave. This type of wave is called a longitudinal wave.

Speed of :-

1> Sound :- $V_s > V_{liq} > V_{gas}$

2> Light :- $V_{vacuum} > V_{gas} > V_{liq} > V_s$

Sound waves can Reflect, Refract, Interfere and Diffract but cannot be polarized. Polarization is a property that only transverse waves (like light waves) can have.

Range :-

○ Infrasonic Sound :- $f < 20 \text{ Hz}$, $\rightarrow$ cannot hear

○ Audible Sound :- $20 \text{ Hz} \leq f \leq 20 \text{ KHz}$, can hear

○ Ultrasonic Sound :- $f > 20 \text{ KHz}$, can't hear

5 Speed of Sound

$$V_s = \sqrt{\frac{\gamma}{\rho}}$$

$$V_G \text{ or } V_L = \sqrt{\frac{\beta}{\rho}}$$

ρ = Density of gas.

γ = Young modules β = Bulk modules

V_G or V_L = speed of sound in gas or liquid.

Newton's Formula :- (isothermal Process)

$$V_{G \& L} = \sqrt{\frac{P}{\rho}}, \beta = P$$

$$V = 280 \text{ m/s}$$

Typical speed of sound

Laplace's Correction :- (Adiabatic Process)

$$V_{G \& L} = \sqrt{\frac{\gamma P}{\rho}}, \gamma = 1 + \frac{2}{f}, \beta = \gamma P$$

$$V = \sqrt{\frac{\gamma R T}{M}}$$

$$V \propto \sqrt{T} \propto \frac{1}{\sqrt{M}}$$

Where,

γ : Adiabatic index.

R : Universal gas constant.

T : Absolute temperature.

M : Molar mass of the gas.

Note :-

+ $V_{\text{moist air}} > V_{\text{dry air}}$

+ $\rho_{\text{moist air}} < \rho_{\text{dry air}}$

+ $V_{\text{rms}} > V_{\text{sound in gas}}$

+ Speed of sound in gas varies with temperature.

Let speed of sound is V_0 at 0°C . Speed of sound is V_t at $T^\circ\text{C}$. Then find the relation between them

$$V_t = V_0 \left[1 + \frac{t}{546} \right] \left\{ \begin{array}{l} \text{Sab } ^\circ\text{C mein} \\ \text{Chalega} \end{array} \right.$$

$$\Delta V = \frac{V_0 t}{546} = 0.61t$$

○ Change in Velocity of Sound is **0.61 m/s**

+ Per unit raise in temperature i.e. **0.18%**

Sound Wave Travel Due to Pressure & Density

Variation:-

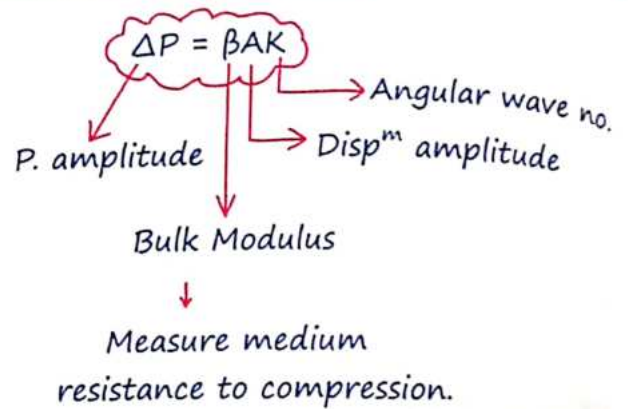
- + In compressions, the pressure and density of the air molecules are at their highest. In rarefactions, the pressure and density are at their lowest. This alternating pattern of compressions and rarefactions enables sound waves to propagate through a medium.

	Compression	Rarefaction
Pressure (P)	Maximum (high pressure)	Minimum (low pressure)
Density (ρ)	Maximum (high density)	Minimum (low density)
Displacement (S)	Minimum	Maximum

Note:-

ϕ between disp^m (s) & Pressure wave in sound wave is $\pi/2$.

Relation Between Pressure Amplitude & Displacement Amplitude:-



Intensity of Wave :-

- + Intensity (I) is the amount of energy (E) passing through a unit area (A) per unit time (t).

$$I = \frac{E}{At} = \frac{1}{2} \rho V A^2 \omega^2$$

$$+ \text{ Unit} = \frac{\text{J}}{\text{m}^2 \text{sec}} = \frac{\text{watt}}{\text{m}^2}$$

Where,

I: Intensity of the medium
 wave
 V: Velocity of the
 wave
 E: Energy
 A: Area
 A: Amplitude of the
 wave
 t: Time
 w: Angular frequency
 ρ : Density of the

Energy Density of Wave:-

- + Energy density (u_e) is the amount of energy stored in a unit volume of the medium through which the wave is propagating.

$$u_e = \frac{\text{Energy}}{\text{Volume}} = \frac{I(\text{Intensity})}{V(\text{Velocity})} = \frac{1}{2} \rho A^2 \omega^2$$

$$\text{Where, } I = \frac{1}{2} \rho V A^2 \omega^2$$

MR* $\rightarrow$ direction se feel karo

Intensity Variation Due to Diff. Sources:-

Point source	Linear Source	Planar Source
$I \propto \frac{A^2}{r^2}$	$I \propto \frac{A^2}{r}$	$I \propto r^0$ $A \propto r^0$

6 Loudness of Sound Wave

- The reference intensity level, I_0 , is the quietest sound that the average human ear can hear, typically:

$$I_0 = 10^{-12} \frac{\text{Watt}}{\text{m}^2}$$

- The sound level or loudness (L) in decibels (dB) is calculated using the following formula:

$$L = 10 \log_{10} \left(\frac{I}{I_0} \right) \text{ dB}$$

$$+ 1 \text{ Bel} = 10 \text{ dB}$$

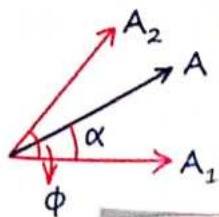
MR*

Change in loudness:

$$\Delta L = 10 \log_{10} \left(\frac{I_f}{I_i} \right) = 10 \log_{10} \left(\frac{I_{\max}}{I_{\min}} \right)$$

7 Principle of Superposition of Waves

- Angle (α) Between Resultant and One of the Components: $\tan \alpha = \frac{A_2 \sin \phi}{A_1 + A_2 \cos \phi}$



- Resultant amplitude

$$A^2 = A_1^2 + A_2^2 + 2A_1A_2 \cos \phi$$

- Intensity of resultant wave.

$$I = I_1 + I_2 + 2\sqrt{I_1}\sqrt{I_2} \cos \phi \quad \therefore I \propto A^2$$

$$+ \frac{I_{\max}}{I_{\min}} = \frac{(\sqrt{I_1} + \sqrt{I_2})^2}{(\sqrt{I_1} - \sqrt{I_2})^2} = \frac{(A_1 + A_2)^2}{(A_1 - A_2)^2}$$

$$+ \frac{I_{\max}}{I_{\min}} = \frac{\left(\sqrt{\frac{I_1}{I_2}} + 1 \right)^2}{\left(\sqrt{\frac{I_1}{I_2}} - 1 \right)^2} = \frac{\left(\frac{A_1}{A_2} + 1 \right)^2}{\left(\frac{A_1}{A_2} - 1 \right)^2}$$

8 Interference

Aaise doh wave Ka superposition jiska λ & f same hai !

Constructive	Destructive
$A_{\max} = A_1 + A_2$	$A_{\min} = A_1 - A_2$
$I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2$ $\cos \phi = 1$ $\phi = 0, 2\pi, 4\pi, 6\pi,$ $\phi = 2n\pi$	$I_{\min} = (\sqrt{I_1} - \sqrt{I_2})^2$ $\cos \phi = -1$ $\phi = \pi, 3\pi, 5\pi, 7\pi,$ $\phi = (2n+1)\pi$
$\Delta x = 0, \lambda, 2\lambda, 3\lambda,$ $\Delta x = n\lambda$	$x = \frac{\lambda}{2}, \frac{3\lambda}{2}, \frac{5\lambda}{2}, \dots$ $\Delta x = (2n+1)\frac{\lambda}{2}$

Two wave same amplitude A_0 :-

Constructive

$$A_{\max} = 2A_0$$

$$I_{\max} = (2\sqrt{I})^2 = 4I$$

Destructive

$$A_{\min} = 0$$

$$I_{\min} = 0$$

Q.3. The intensity ratio of the two interfering beams of light is β . What is the value of

$$[(I_{\max} - I_{\min}) / (I_{\max} + I_{\min})]^2?$$

(a) $\sqrt{\beta}$ (b) $2\sqrt{\beta} / (1 + \beta)$

(c) $2 / (1 + \beta)$ (d) $(1 + \beta) / 2\sqrt{\beta}$

Sol. $\Rightarrow \frac{2\sqrt{\beta}}{1 + \beta}$

Apply MR*

If $\beta = 1$, then $\frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = 1$

9 Reflection of Transverse Waves in String

+ Transverse waves can reflect (or refract) when they encounter a boundary or a change in medium. This reflection can occur when a wave travels from a rarer (less dense) medium to a denser medium or vice versa.

(String और Sound का Same hai)

(a) Rarer $\rightarrow$ Denser :-

	Reflected	Transmitted
Frequency, f	f	f
Speed (v)	Same	Decrease $U \downarrow$
Wavelength (λ)	Same	Decrease $\lambda \downarrow$
Phase diff. (ϕ)	π	0 (No phase change)

(b) Denser $\rightarrow$ Rarer :-

	Reflected	Transmitted
Frequency, f	f	f
Speed (v)	Same	Increase $U \uparrow$
Wavelength (λ)	Same	Increase $\lambda \uparrow$
Phase diff. (ϕ)	0	0 (No phase change)

10 Stationary Wave

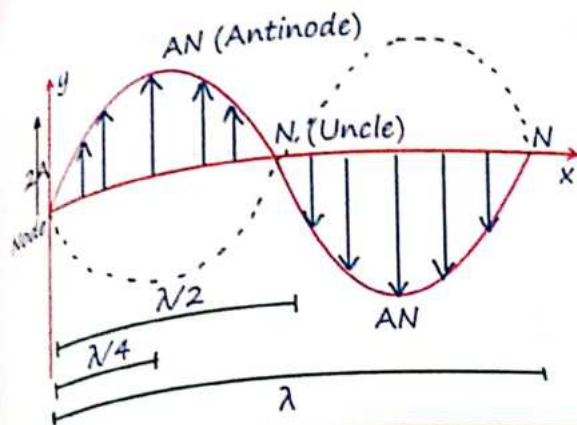
When two waves with similar properties (same frequency and amplitude) travel in opposite directions within a confined medium, their superposition creates a new type of wave. This resulting wave appears to stand still, rather than traveling through the medium. This phenomenon is known as a stationary wave or standing wave.

Aaise doh wave ka superposition jiska sab kuch same hoga bs direction opposite

General equation:-

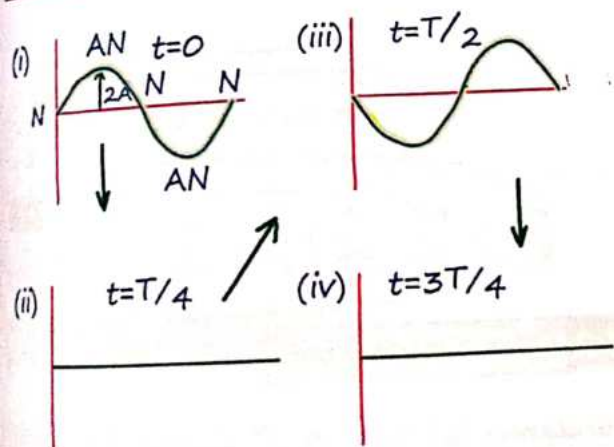
$$y = \underbrace{2A \sin(Kx)}_{\text{Amplitude}} \underbrace{\cos(\omega t)}_{\text{SHM.}}$$

Where, $K = \frac{2\pi}{\lambda}$, $\omega = \frac{2\pi}{T}$



	Node	Antinode
Condition:	$\phi = Kx = n\pi$	$\phi = Kx = (2n+1)\frac{\pi}{2}$
Position:	$x = \frac{n\lambda}{2}$	$x = (2n+1)\frac{\lambda}{4}$
Occur at:	integral multiples of half wavelength	odd multiples of quarter wavelength

Time evolution of stationary wave:



* A flat line, indicating that the wave's displacement is zero throughout the medium

Time	Description
$t=0$	The wave has its initial form with nodes and antinodes positioned as per the general equation.

$t = \frac{T}{4}$ and $t = \frac{3T}{4}$	The wave passes through its equilibrium position, with maximum speed of particles at these points.
$t = \frac{T}{2}$	The wave returns to a form similar to its initial form but inverted, with nodes and antinodes remaining fixed in position.

+ Different equation of Stationary Wave :-

$$Y = 2A \sin(Kx) \cdot \cos(\omega t)$$

$$Y = 2A \cos(Kx) \cdot \sin(\omega t)$$

$$Y = 2A \sin(Kx) \cdot \sin(\omega t)$$

$$Y = 2A \cos(Kx) \cdot \cos(\omega t)$$

Q.4. If node is formed at origin then amplitude which is at between node and antinode.

Sol. Midway between node and Antinode

$$x = \frac{\lambda}{8}, \quad x = 2A \sin\left(\frac{2\pi}{\lambda} \times \frac{\lambda}{8}\right)$$

$$= 2A \sin \frac{\pi}{4} = \sqrt{2} A$$

Difference Between Stationary & Progressive

Wave:-

Stationary wave	Progressive wave
1> Particle at Node $\rightarrow$ rest	1> No Particle at rest.
2> No Transfer of Energy and momentum.	2> Transfer of E & P occurs.
3> Sabka "A" diff.	3> "A" same.
4> All in same phase between N.	4> All are in diff. phase

Stationary wave

5> All particles cross mean position with diff. speed at same time.

Progressive wave

5> All particles cross MP with same speed at diff. time.

11 Formation of Stationary Wave in String Sonometre Wire

Frequency:-

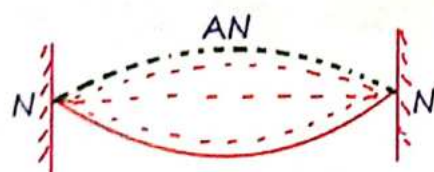
$$f = \frac{V}{2L} = \frac{1}{2L} \sqrt{\frac{T}{\mu}} = \frac{1}{2L} \sqrt{\frac{T}{\rho A}}$$

$$f = \frac{1}{2L} \sqrt{\frac{\text{Stress}}{\rho}}$$

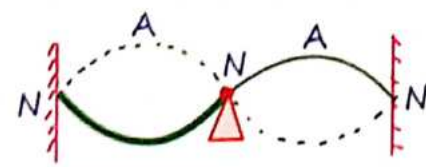
Where,

- + V: Wave speed
- + L: Length of the string
- + T: Tension in the string
- + μ : Linear mass density
- + A is the cross-sectional area

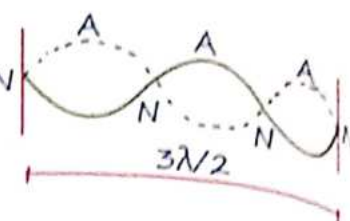
(a) Fundamental or 1st harmonic :-

$$f' = \frac{V}{\lambda} = \frac{V}{2L}$$


(b) 2nd harmonic or 1st overtone :-

$$f = \frac{V}{\lambda} = \frac{V}{L} = 2f'$$


(c) 3rd harmonic or 2st overtone:-

$$f = \frac{V}{\lambda} = \frac{3V}{2L}$$
$$f = 3f'$$


Vimp

n - harmonics or (n-1) overtone :-

$$f = n \left(\frac{V}{2L} \right)$$

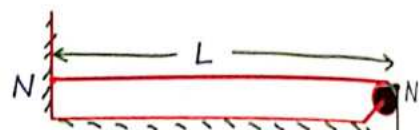
If, No. of antinode = n, then
No. of Node = n + 1

Difference between any two consecutive harmonics :-

$$\Delta f = \frac{V}{2L} = f$$
 NEET

+ Sonometer Wire :-

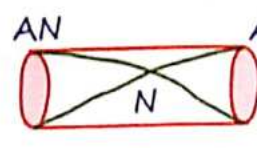
μ = linear mass density of wire.



$$f' = \frac{nV}{2L} = \frac{n}{2L} \sqrt{\frac{T}{\mu}} = \frac{n}{2L} \sqrt{\frac{mg}{\mu}}$$

12 Open Organ Pipe

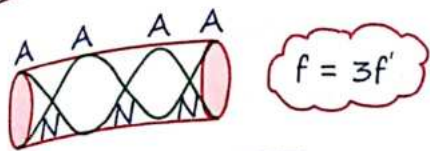
Fundamental Freq. or 1st harmonic :-


$$f' = \frac{V}{2L}$$

2nd harmonic or 1st overtone :-


$$f = 2f'$$

3rd harmonic or 2nd overtone :-

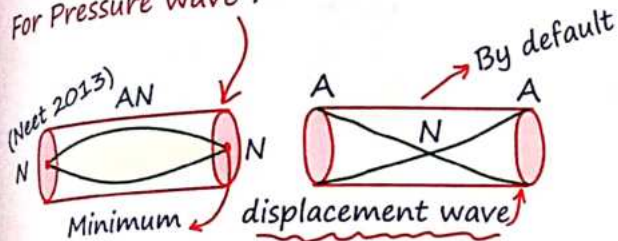


$$* f = \frac{nV}{2L} = \frac{n}{2L} \sqrt{\frac{\gamma P}{\rho}} = \frac{n}{2L} \sqrt{\frac{\gamma RT}{M}}$$

+ n = harmonic $(n-1)$ = overtone
No. of Node = n No. of Antinode = $n+1$

Wimp $f_1 : f_2 : f_3 = 1 : 2 : 3 : \dots$ integral

For Pressure wave :-



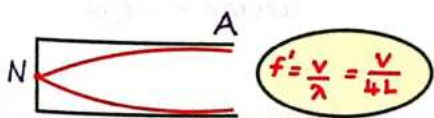
$\therefore$ Disp^m wave & Pressure wave have $\phi = \pi/2$.

Note:-

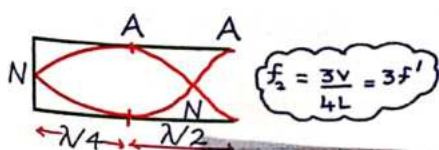
- + Antinode = Maximum pressure variation.
- + Nodes = Minimum pressure variation.

13 Closed Organ Pipe

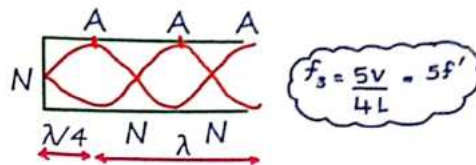
Fundamental or 1st harmonic :-



3rd harmonic or 1st overtone :-



5th harmonic or 2nd overtone :-



$(2n+1)$ = Harmonic n = Overtone

$$* f = (2n+1) \frac{V}{4L} \quad f_1 : f_2 : f_3 = 1 : 3 : 5$$

Difference between any two consecutive harmonics :-

$$\Delta f = 2f' = \frac{2V}{4L} \quad \underline{\underline{NEET}}$$

The MR*

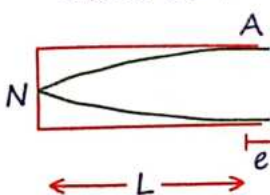
Frequency Ka ratio likhdo aur uss ratio mein dhekho Konse wave form Ki baat chalrahi jiss wave form Ki baat chal rahi hogi wahi tumhara N & AN hoga.

$$f_1 : f_3 : f_5 : f_7 = 1 : 3 : 5 : 7$$

2 nd overtone	3 rd overtone
3 wave form	4 wave form
3N & 3AN	4N & 4AN

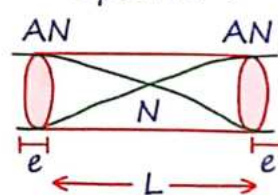
End Correction:-

Closed OP :-



$$f = \frac{(2n+1)V}{4(L+e)} \quad e = 0.6r$$

Open OP :-

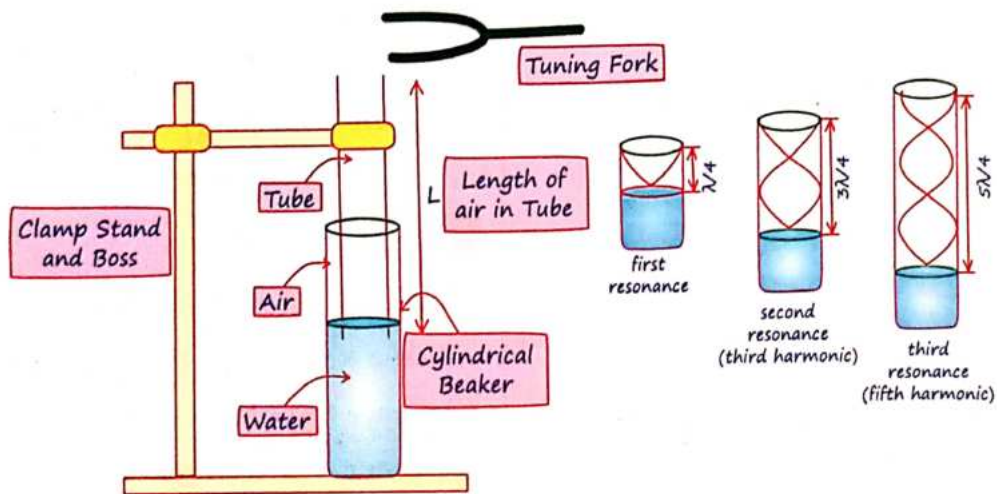


$$f = \frac{nV}{2(L+2e)} \quad e = 0.6r$$

Q.5. The length of an open organ pipe is twice the length of another closed organ pipe. The fundamental frequency of the open pipe is 100 Hz. The frequency of the third harmonic of the closed pipe is.

Sol. The fundamental frequency of open pipe
 $= f_o = \frac{v}{2l} = 100 \text{ Hz}$
 Hence the third harmonic of closed organ pipe is $\frac{3v}{4(l/2)}$
 $= 300 \text{ Hz}$

14 Resonance Tube



$$f_1 = \frac{v}{4L_1} = f_T \quad f_2 = \frac{3v}{4L_2} = f_T$$

"Resonance"

NEET

$$\frac{v}{4L_1} = \frac{3v}{4L_2}$$

$$L_2 = 3L_1^*$$

+ End Correction using resonance Tube :-

$$e = \frac{l_2 - 3l_1}{2}$$

Beats!

Aaisi doh wave jinka "A" same hai lekin "f" slightly different. $\Delta f < 10$.

$$Y = \underbrace{2A}_{\text{Amplitude}} \left[\underbrace{\sin\left(\frac{\omega_1 + \omega_2}{2}t\right)}_{\text{oscill}^n} \cdot \underbrace{\cos\left(\frac{\omega_1 - \omega_2}{2}t\right)}_{\text{oscill}^n} \right]$$

Ye Amplitude Ko decide Karega !

$$\text{Angular freq of } A = \frac{\omega_1 - \omega_2}{2}$$

$$\begin{aligned} \text{Angular freq of } I &= \omega_1 - \omega_2 \\ \text{Angular beat freq} &= \omega_1 - \omega_2 \end{aligned}$$

Vimp

$$f_B = f_1 - f_2$$

* Tuning Fork :-

Sharp :- $f \uparrow$

Waxing :- $f \downarrow$

$$\text{Waxing :- } \frac{I_{\max}}{I_{\min}} = \frac{(A_1 + A_2)^2}{(A_1 - A_2)^2}$$

$$\text{Waning :- } \frac{I_{\max}}{I_{\min}} = \frac{(A_1 + A_2)^2}{(A_1 - A_2)^2}$$

Q.6. Two tuning forks when sounded together produced 4 beats/sec. The frequency of one fork is 256. The number of beats heard increases when the fork of frequency 256 is loaded with wax. The frequency of the other fork is

Sol. $f_A - f_B = 4 \text{ Hz}$

$$f_A = 256 \text{ Hz}$$

$$f_B = ??$$

$$\rightarrow 260 \text{ Hz}$$

$$\rightarrow 252 \text{ Hz}$$

(Possible)

In waxing of A f_A will decrease but frequency difference is increasing hence 260 Hz will be answer.

Q.7. A tuning fork A produces 2 beats per second when sounded with a tuning fork of frequency 200 Hz. When A is loaded with wax the beats stop. What is the frequency of fork A?

Sol. There are 2 beats per second

$$\therefore f_1 - f_2 = 2$$

$$\Rightarrow 200 - f_2 = \pm 2$$

$$f_2 = 200 \pm 2$$

$$f_2 = 202 \text{ or } f_2 = 198$$

When loaded with wax the beats stop since the frequency decreases on loading.

$$\therefore f_2 = 202 \text{ Hz}$$

1 Charge

Charge is a fundamental property of matter that causes it to experience a force when placed in an electric or magnetic field. It is the property by which objects exert an electrostatic force of repulsion or attraction on each other.

+ There are two Types of Charges:

Positive Charge (+): due to the deficiency of electrons

Negative Charge (-): due to the excess of electrons

Behaviour of charge under different condition:-

Charge	Electric field	Magnetic field	EM Wave
Rest	✓	✗	✗
$V = \text{Const}^n$	✓	✓	✗
Accelerated	✓	✓	✓

Properties of Charge:-

- + Scalar, Conserved, Quantized
 $Q = ne$, $n = \text{integer}$
- + Invariant: does not depend upon speed:-
- + Charge can't exist without mass but mass can exist without charge.
- + Same charge $\rightarrow$ repel (may attract)
- + Opposite charge $\rightarrow$ must attract
- + SI unit $1\text{ C} = 3 \times 10^9\text{ esu}$. $1\text{C} = \frac{1}{10}\text{ emu}$.
- + Smallest unit frankline = 1 esu
- + Largest unit faraday, 1 faraday = 96500 C
- + One charge may attract other Neutral

Surety check of a charge body is repulsion:-

Attraction can occur between charged and neutral objects due to polarization effects, but repulsion only occurs between like charges.

Quarks :- Does not exist in free state

up	$+\frac{2e}{3}$
down	$-\frac{1}{3}e$

Q.1 Which of the following charge is possible:

- (a) $\frac{1\text{C}}{100}$ (b) $\frac{1e}{50}$
(c) $4.8 \times 10^{-21}\text{C}$ (d) $1.56 \times 10^{-13}\text{C}$

Sol. Use $Q = ne$ (a) is possible

Note: Charge को Sharp Point पसंद है:-

2 Method of Charging a body

Charging of Body by conduction:-

- + Conduction involves transferring charge through direct contact between materials.
- + For conductor only

$$Q \propto R. [R \text{ Radius}]$$

Ex When two conductors with charges Q_1 & Q_2 and Radius R_1 & R_2 are connected: If Q_1' is the new charge on first conductor then

$$\text{New Charge} = \frac{\text{Uska } R}{\text{Total } R} [Q_{\text{Total}}]$$

$$Q_1' = \frac{R_1}{R_1 + R_2} [Q_1 + Q_2]$$

Charging of body by friction :-

- Friction involves rubbing two different materials together, causing a transfer of electrons.
- For insulator only equal and opposite charge on two rubbing object

Charging by induction :-

- Induction involves bringing a charged object close to another object without direct contact, causing a redistribution of charges in the second object.
- For conductor and dielectric $\rightarrow$ equal or lesser charge of positive nature induced.

Charge density :-

Charge on Linear object	Charge on Areal object	Charge on Volumetric object
$dq = \lambda dl$	$dq = \sigma dA$	$dq = \rho dV$
$\lambda = \frac{dq}{dl} \text{ C/m}$	$\sigma = \frac{dq}{dA} \text{ C/m}^2$	$\rho = \frac{dq}{dV} \text{ C/m}^3$
$\uparrow$ Linear charge density	$\uparrow$ Areal charge density	$\uparrow$ Charge density

3 Gold leaf experiment

- Device used to detect charge, not use to measure charge.
- Method involves conduction or induction.
- Diverge angle $\propto$ charge of leaf.

4 Coulomb's Law

- Valid for point or spherical charge symmetry
- The force of interaction between two point charges is directly proportional to the product of the charges and Inversely proportional to the square of the distance between them.

$$F = \frac{kq_1q_2}{r^2} \quad \epsilon_0 = 8.85 \times 10^{-12} \frac{C^2}{Nm^2}$$

$$K = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \frac{Nm^2}{C^2} \quad \text{Mag: - Formula } \ddagger \quad \text{Dir: - Buddhi } \ddagger$$

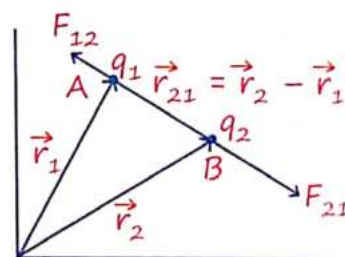
* Another Medium between Charges :-

$$K = \epsilon_r = \frac{\epsilon_m}{\epsilon_0} \quad \left(F' = \frac{F_0}{K} \right) \text{ dec. to } K \text{ times its initial value.}$$

- Conservative, long range, follow inverse square law $\left(F \propto \frac{1}{r^2} \right)$, central, mediated by photon.
- Electrostatic force on q_2 due to q_1 does not depends upon medium or presence of other charge $\rightarrow$ True.
- Net electrostatics force on q_2 depends upon medium or presence of other charge $\rightarrow$ True.

Coulomb's Law in Vector Form :-

$$\vec{F}_{21} = \frac{Kq_1q_2}{|\vec{r}_{21}|^2} \hat{r}_{21} = \frac{Kq_1q_2}{|\vec{r}_{21}|^3} \vec{r}_{21}$$



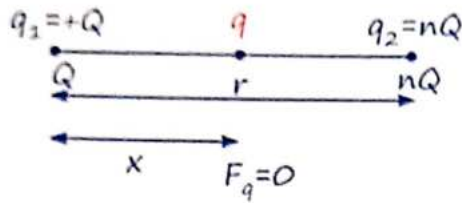
Q.2 Two identical charge 'q' repel each other with a force of 100 N, one of the charge is increased by 10% and decreased by 10% then new force of repulsion at the same distance?

$$\text{Ans. } q'_1 = 110\%q, q'_2 = 90\%q$$

$$F = \frac{Kq_1q_2}{r^2}, F' = \frac{Kq'_1q'_2}{r^2} \Rightarrow F' = 99N$$

Neutral Point :-

- Find position of 3rd charge where force on that charge will be zero



Like Charge

$$x = \frac{d}{\sqrt{n} + 1}$$

$$n = \frac{\text{Bada charge}}{\text{Chota charge}}$$

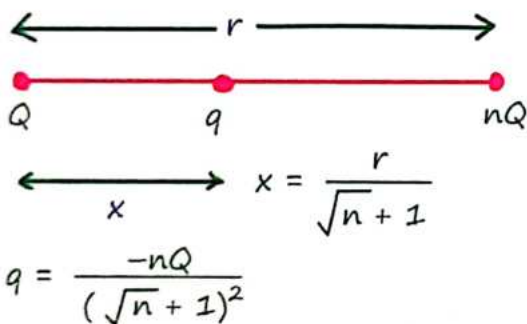
Unlike Charge

$$x = \frac{d}{\sqrt{n} - 1}$$

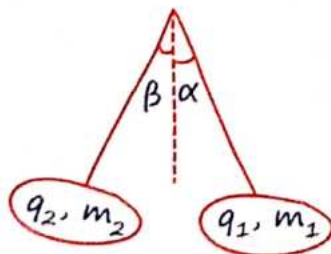
Note:-

If we Divide charge Equally, they repel each other with $F_{\max}$

Q.3 Find Position & Value of "q" so that System will be in Equilibrium:-



Pendulum Problem :-



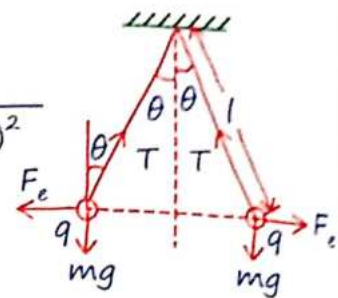
- Value of α and β depends on charges and masses but ratio of α and β does not depends on charges and depends only on masses

$$\frac{\tan \alpha}{\tan \beta} = \frac{m_2}{m_1}$$

- If $m_1 = m_2$ then $\alpha = \beta$
- If $m_1 > m_2$ then $\alpha < \beta$

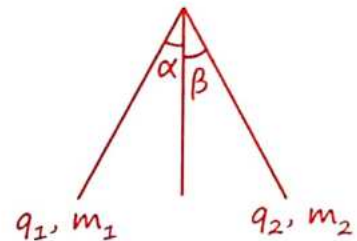
$$\tan \theta = \frac{F_e}{mg}$$

$$T = \sqrt{F_e^2 + (mg)^2}$$



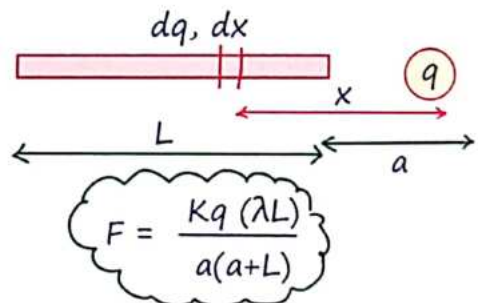
Q.4 If $q_1 > q_2$ but $m_1 = m_2$ then which of the following is correct?

- (a) $\alpha = \beta$ (b) $\alpha > \beta$ (c) $\alpha < \beta$



Ans. (a)

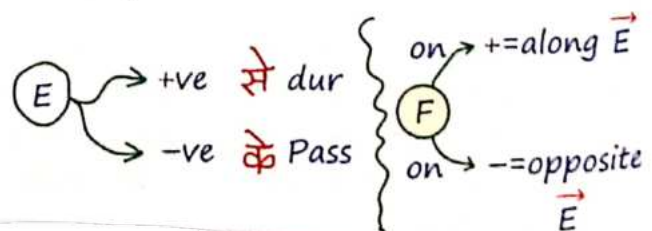
Force on Rod Due to Point Charge :-



5 Electric Field Intensity

- Electrostatic force per unit positive charge.

$$\vec{E} = \frac{\vec{F}}{q} = \text{N/C. Vector!}$$

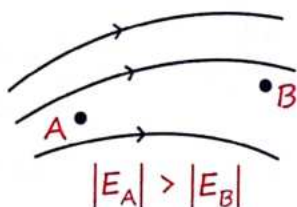


6 Electric field lines

- Electric field lines are imaginary lines or curves that represent the direction and strength of the electric field in a region of space. At any point on the line, the tangent to the line gives the direction of the electric field vector at that point.



- The density of the lines at any point indicates the magnitude of the electric field: closer lines mean a stronger field.



Properties of electric field lines:-

- Electric field lines originate from positive charges and terminate at negative charges.
- Electric field lines never cross each other. If they did, it would imply two different directions of the electric field at the same point, which is impossible.
- Electric field lines do not form closed loops. They begin at a positive charge and end at a negative charge, or extend to infinity in the case of isolated charges.
- In regions where there are no charges, electric field lines are smooth, continuous curves without an

Superposition Principle:

- When dealing with multiple charges, each pair of charges interacts independently.
- The total force on any charge is found by adding up all the individual forces exerted on it by every other charge.

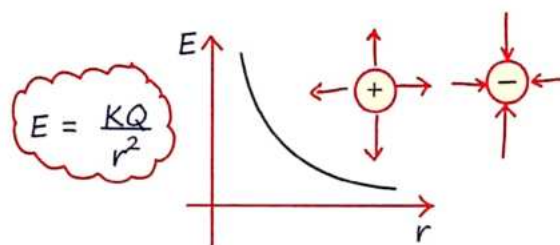
Mathematically

Consider a system of n point charges;

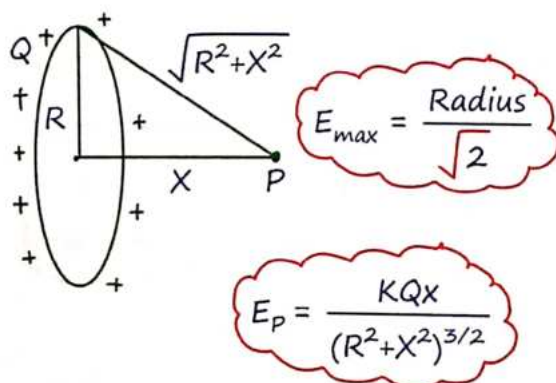
$q_1, q_2, q_3, \dots, q_n$ distributed in space. Let the charges $q_2, q_3, \dots, q_n$ exert forces $\vec{F}_{12}, \vec{F}_{13}, \dots, \vec{F}_{1n}$ on charge q_1 . The total force on charge q_1 ($\vec{F}_1$) is given by:

$$\vec{F}_1 = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 q_2}{r_{12}^2} \hat{r}_{12} + \frac{q_1 q_3}{r_{13}^2} \hat{r}_{13} + \dots + \frac{q_1 q_n}{r_{1n}^2} \hat{r}_{1n} \right)$$

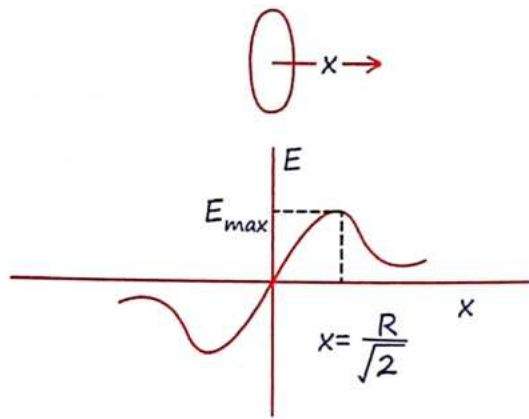
Electric field due to Point charge :-



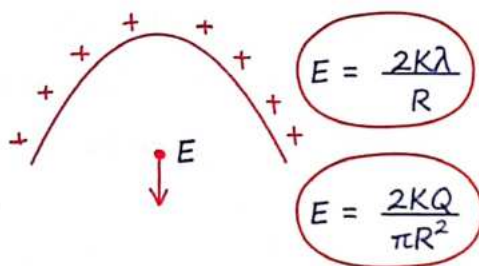
Electric field due to Ring :-



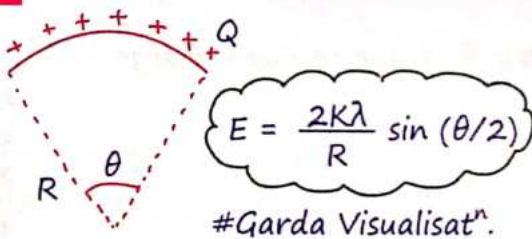
+ Graph B/w electric field and distance



Electric field due to A Half ring :-



MR*



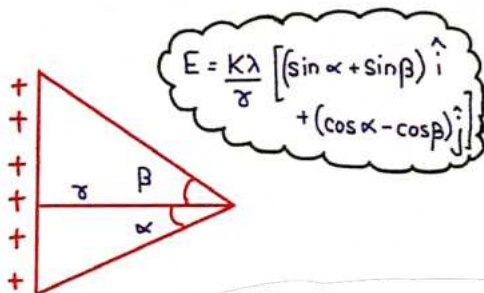
if $\theta = \pi$ (Half Ring)

then, $E = \frac{2K\lambda}{r}$

if $\theta = 2\pi$ (Ring)

then $E = 0$

Electric Field Due to Line Charge :-



+ Infinite - line: $-E = \frac{2K\lambda}{r}$

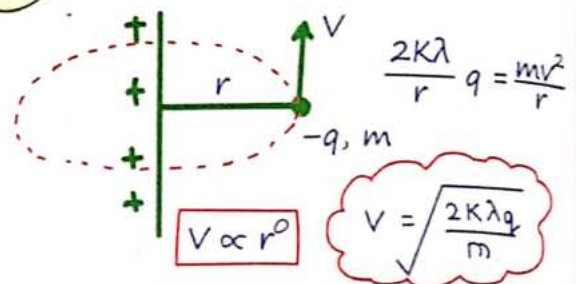
+ Semi- Infinite line: $-E = \frac{K\lambda}{r} (\hat{i} - \hat{j})$

+ Finite line: $-E = \frac{K\lambda}{r} [(\sin \alpha + \sin \beta) \hat{i} + (\cos \alpha - \cos \beta) \hat{j}]$

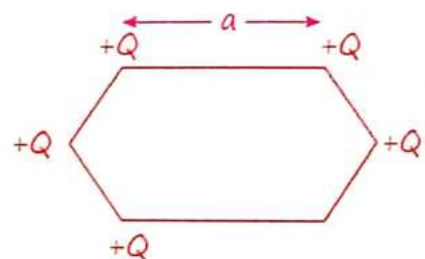
+ Finite line at 'r' when $(\alpha = \beta)$: $-E = \frac{2K\lambda}{r} \sin \alpha \hat{i}$

IMP Case

*Charge $(-q, m)$ moving around infinite line charge with speed v .



Q.5 Find electric field at centre ?



Sol. $E = \frac{KQ}{a^2}$

MR*

Put $+Q$ and $-Q$ charge on the corner, where no charge is present.

Motion of Charged Particle in Uniform E.F. :-

Charge is drop/released :-

Acceleration, $a = \frac{qE}{m}$, Velocity, $v = \frac{qEt}{m}$

Momentum, $p = qEt$

Distance covered, $s = \frac{1}{2} \frac{qEt^2}{m}$

Kinetic energy, $K.E = \frac{1}{2} \frac{q^2 E^2 t^2}{m}$

- Positron, $P^+ = e, m$. Deuteron = $e, 2m$.
- α -particle = $2e, 4m$.

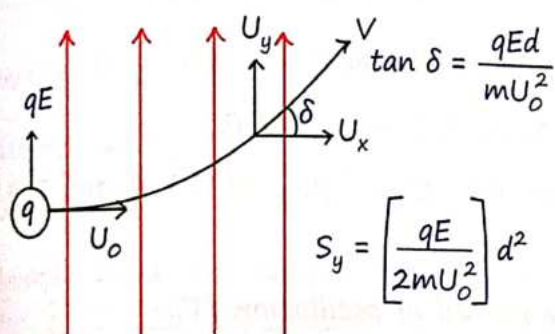
Charge is projected with " U_0 " in dirⁿ of E.F. :-

$a = \frac{qE}{m} = \text{Const}^n$ $\star V = U_0 + \frac{qEt}{m}$

$\star \Delta P = qEt$

$\star S = U_0 t + \frac{1}{2} \frac{qEt^2}{m}$ $\star P_t = m \left[U_0 + \frac{qEt}{m} \right]$

3> Charge is projected $\perp$ er to E.F. :-



Ex Proton, Deuteron and α -particle projected in electric field perpendicular to it, then find ratio of deviation if:

(i) They projected with same speed

$\delta_P : \delta_D : \delta_\alpha = 2 : 1 : 1$ $\boxed{\delta \propto q/m}$

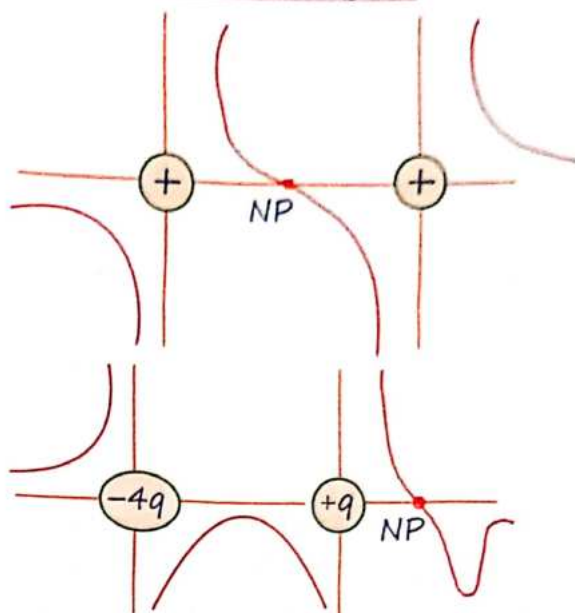
(ii) They are projected with same K.E.

$\delta_P : \delta_D : \delta_\alpha = 1 : 1 : 2$ $\boxed{\delta \propto q}$

(iii) Projected with same momentum

$\delta_P : \delta_D : \delta_\alpha = 1 : 2 : 8$ $\boxed{\delta \propto qm}$

Graph of Electric field with R due to Combination of Point Charge :-



7 Electric Dipole

$\star$ Two equal and opposite charge placed at small distance.

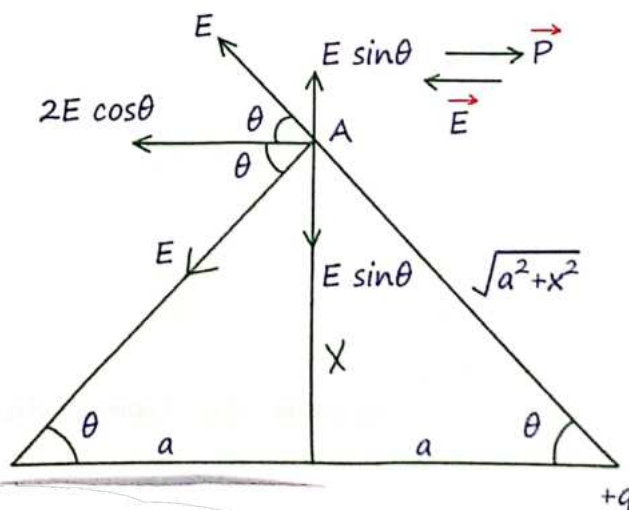
$\star$ Unit = C.m, Net charge = 0, Ideal dipole = small dipole

$\vec{P} = \left[\text{Mag. of } Q. \right] \times \left[\text{Dist}^n \text{ between two charge} \right]$

$\star$ Direction of dipole moment -ve to +ve charge.

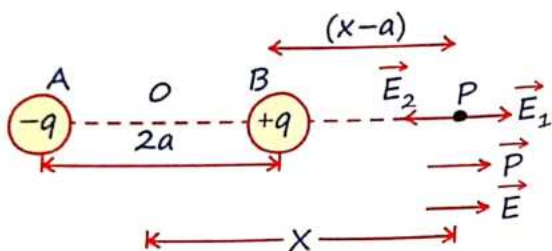
Equatorial line :-

$\vec{E} = \frac{-K\vec{P}}{(a^2 + x^2)^{3/2}}$ $\vec{E} = \frac{-K\vec{P}}{x^3}$



Axial line :-

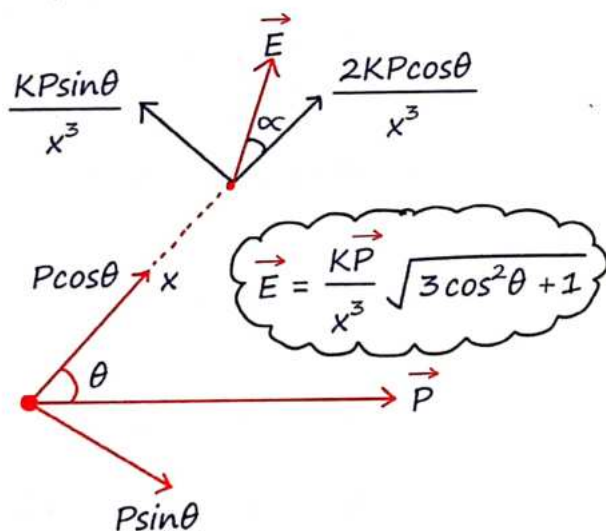
$$\vec{E} = \frac{2KPx}{(a^2 - x^2)^2} \quad \vec{E} = + \frac{2KP}{x^3}$$



$$\therefore \frac{E_{\text{equi}}}{E_{\text{axial}}} = \frac{1}{2}$$

Axial पर E.F. Same line mein hogi aur E.F. equatorial line pr $\perp^{\text{er}}$ hogi aur dipole Ke opposite hogi.

At an angle θ from ideal dipole:-

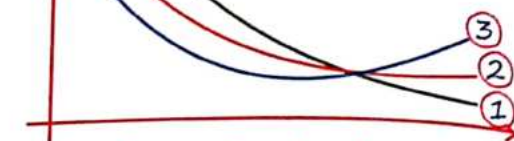


$\therefore$ Angle between $\vec{E}$ & $\vec{P} = \theta + \alpha$

$$\tan \alpha = \frac{KP \sin \theta / x^3}{2KP \cos \theta / x^3} = \frac{\tan \theta}{2}$$

$\theta = \tan^{-1}(\sqrt{2})$ The Angle from dipole at which $\vec{E} \perp \vec{P}$

- 1> Force b/w two dipole $\propto 1/r^4$
- 2> Force b/w point charge & dipole $\propto 1/r^3$
- 3> Force b/w two Charge $\propto 1/r^2$



*Jiska Power $\uparrow$ woh sabse niche? x

8 Electric oscillation in electric field

When an electric dipole is placed in a uniform electric field, it experiences a torque that tends to align the dipole with the field direction. This setup can lead to oscillatory motion if the dipole is displaced from its equilibrium position.

Torque on an Electric Dipole:-

$$\vec{\tau} = \vec{P} \times \vec{E} = PE \sin \theta \quad \text{Vector!}$$

θ = Angle between E & P .

Torque hamesha dipole को E.F. की dirⁿ में Align करना चाएगा!

Time period of oscillation (T):-

$$T = 2\pi \sqrt{\frac{I}{PE}}$$

I = Moment of Inertia

T = Time period of oscillation

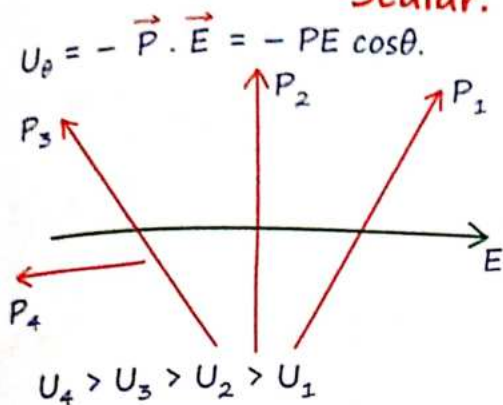
P = Electric dipole moment

E = Electric field.

- + Electric field on the axis of electric dipole is always parallel to electric dipole moment $\rightarrow$ False
- + Electric field on the equatorial line of dipole is anti-parallel to dipole $\rightarrow$ True

Electrostatic P.E. Stored in Dipole in Uniform Electric field :-

Scalar.



$\theta = 0^\circ$	$\theta = 90^\circ$	$\theta = 180^\circ$
$\vec{P} \parallel \vec{E}$	$\vec{P} \perp \vec{E}$	$\vec{P}_{anti} \parallel \vec{E}$
$F = 0$	$F = 0$	$F = 0$
$\tau = 0$	$\tau = PE$	$\tau = 0$
$U_{min} = -PE$	$U = 0$	$U_{max} = PE$

+ Work Done to Rotate dipole $W = \Delta U = U_f - U_i$

+ Work done by E.F. to rotate dipole $w = -\Delta U$

*Special Case :-

+ Work Done to Rotate from Stable to Unstable :- $U = 2PE$.

अगर जहासे Rotate किया वही पर वापस लाया तो

W.D. = 0!

9 Electric Flux

$$\phi = \text{flux} = \vec{E} \cdot \vec{A} = EA \cos \theta$$

θ = Angle between $\vec{E}$ & $\vec{A}$

→ Gives the idea of electrostatic energy passing through given area.

→ Counting of field lines passing through given cross-section area

→ Scalar

→ Unit (Volt - meter) & $(Nm^2 C^{-1})$

$$\rightarrow d\phi = \oint \vec{E} \cdot d\vec{A} \text{ variable electric field.}$$

Flux :- Aaisi line joh area ko aarpar ched banakr jarahi hai !

$$\begin{aligned} \text{3-D body :- } \phi_{in} &= -ve \phi_{out} = +ve \phi_{Total} \\ &= \phi_{in} + \phi_{out} \end{aligned}$$

$$\text{Uniform E.F. :- } (\phi_{Total})_{close \text{ surface}} = 0.$$

Q.6 A charge Q placed on the corner of square plate of side 'L' then find flux through that square plate → zero.

10 Gauss Law

$$\phi = \oint \vec{E} \cdot d\vec{A} = \frac{q_{enclosed}}{\epsilon_0}$$

ϕ = सिर्फ inside charge से आरगा

E = inside और outside दोनों से !

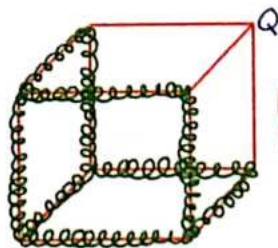
→ Flux from close surface does not depends shape, size of surface and location of charge inside surface and charge outside the surface.

→ Always valid

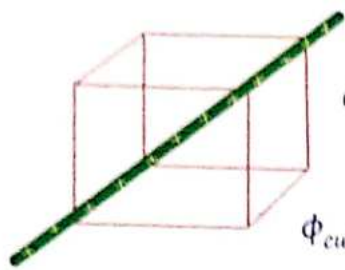
→ Only applicable to calculate electric field for symmetric charge distribution

Special Case :-

+ Flux through any one of the faces not touching the corner charges :-



$$\phi = \frac{1}{3} \left[\frac{Q}{8\epsilon_0} \right]$$



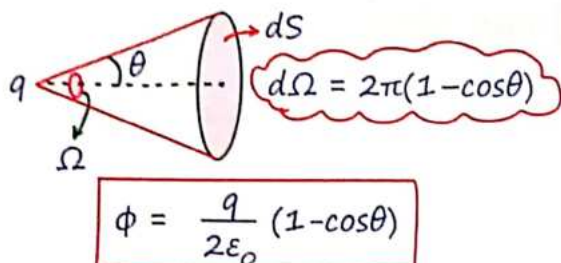
$$\phi_{\max} = \frac{\sqrt{3} \lambda}{\epsilon_0}$$

$$\phi_{\text{cube}} = \frac{\lambda l}{\epsilon_0}$$

Aaise Question mein pahile length nikalo, charge distribution dekho

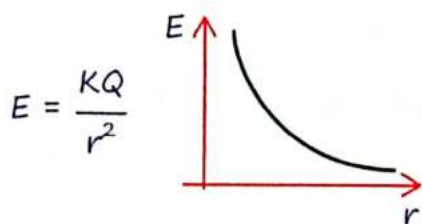
$$\phi = \frac{\lambda l}{\epsilon_0} \text{ lagado !}$$

+ Relation between plane angle & solid angle :-

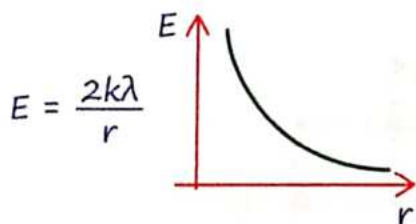


11 Application of gauss law

E.F. Due to point charge :-



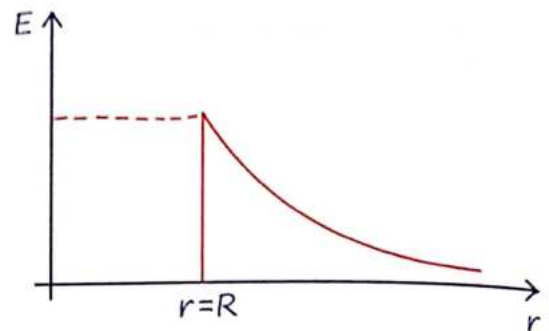
Infinite line charge :-



Infinite hollow/solid conducting or hollow non-conducting cylinder:-

$$E = \frac{2K\lambda}{x} = \frac{\sigma R}{\epsilon_0 r}$$

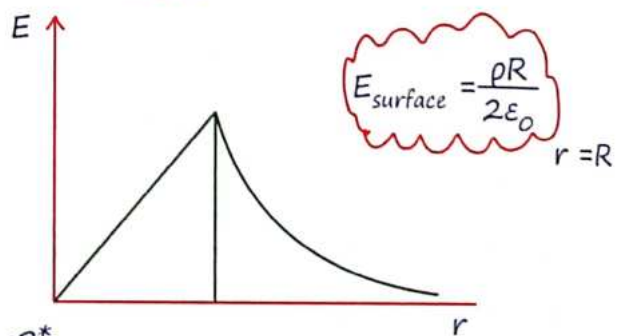
radius
outside point



Solid non-conducting cylinder :-

$$E_{\text{out}} = \frac{\rho R^2}{2\epsilon_0 r}$$

$$E_{\text{in}} = \frac{\rho r}{2\epsilon_0}$$



MR*

$$q_{\text{in}} = \lambda l = \sigma [2\pi R l] = \rho [\pi R^2 l]$$

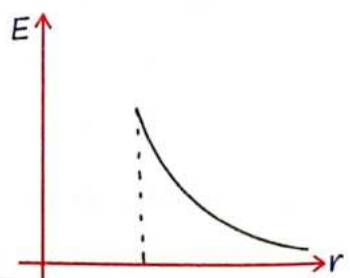
$$E = \frac{2K\lambda}{r}$$

E.F. Due to hollow conducting/N.C. Solid conducting sphere :-

$$E_{\text{in}} = 0$$

$$E_{\text{out}} = \frac{KQ}{r^2}$$

$$E_{\text{sur}} = \frac{KQ}{R^2}$$

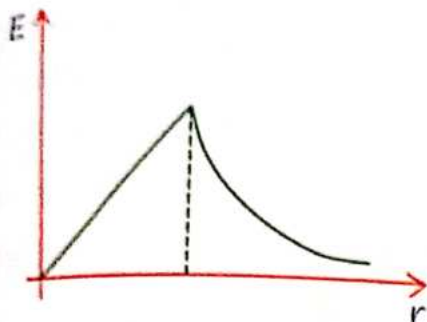


E.F. due to solid non-conducting sphere :-

$$E_{in} = \frac{KQr}{R^3} \quad E_{out} = \frac{KQ}{r^2}$$

$$\rho = \frac{Q}{\frac{4}{3}\pi R^3}$$

$$E_{out} = \frac{KQ}{R^2} \quad E_{in} = \frac{\rho r}{3\epsilon_0}$$



Spherical Shell:-

(like Hollow Sphere).

$$E_{out} = \frac{KQ}{r^2} \quad E_{sur} = \frac{KQ}{R^2} \quad E_{in} = 0$$

Conducting Shell :-

E.F. inside isolated Conductor is zero.

E.F. Inside cavity of non-conducting solid sphere :-

distn b/w centre of sphere to centre of cavity

$$\vec{E}_p = \frac{\rho \vec{r}}{3\epsilon_0} \text{ (uniform) } \vec{r} = \vec{r}_0 - \vec{r}_1$$

from Sphere from Cavity

E.F. Due to infinite large non conducting plate/sheet :-

$$E = \frac{\sigma}{2\epsilon_0} = \frac{Q}{2A\epsilon_0}$$

Kuch na bola jaye toh by default Non-conducting consider Karo!

E.F. Due to conducting plate :-

$$E = \frac{\sigma}{\epsilon_0} = \frac{Q}{2A\epsilon_0}$$

Charge Ke Terms में दोनो sheet Ka Formula same है!

Electric field due to charge disc:-

$$E = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{x}{\sqrt{R^2 + x^2}} \right)$$

Charge Distribution :-

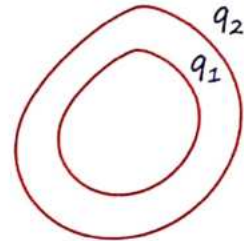
$$\text{Charges on outer surface of plate} = \frac{\text{Total Charge}}{2}$$

$$\begin{array}{c}
 \begin{array}{|c|} \hline 2Q \\ \hline \end{array} \quad \begin{array}{|c|} \hline 3Q \\ \hline \end{array} \Rightarrow \begin{array}{|c|c|} \hline \frac{5Q}{2} & \frac{-Q}{2} \\ \hline \end{array} \quad \begin{array}{|c|c|} \hline \frac{+Q}{2} & \frac{5Q}{2} \\ \hline \end{array} \\
 \\
 \begin{array}{c} \frac{5Q}{2} - \frac{Q}{2} = \underline{\underline{2Q}} \\ \frac{5Q}{2} + \frac{Q}{2} = \underline{\underline{3Q}} \end{array}
 \end{array}$$

○ Charge distribution on concentric sphere :-

$$q \left(\text{final charge on outer surface of outer sphere} \right) = q_1 + q_2$$

$$\text{○ Charge on inner surface of outer sphere} = q_2 - q$$



MR*

‘मुश्किल नहीं है कुछ दुनिया में,
तू जरा हिम्मत तो कर।
ख्वाब बदलेंगे हकीकत में,
तू जरा कोशिश तो कर॥’

1 Electric Potential

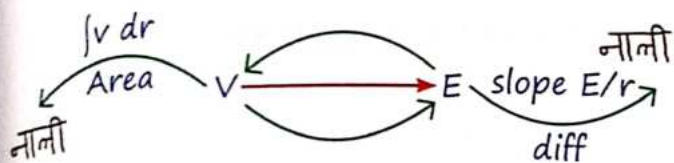
Work done in bringing unit positive charge from infinity to the point without change in K.E. is called electric potential at that point. (or) Negative work done by electric force in bringing unit positive charge from infinity to that point.

+ depends upon reference frame

+ unit: volt, J/C, weber/sec, N-m/C

Potential :-

$$\Delta V = - \int \vec{E} \cdot d\vec{r} = \text{Area of } -E/r \text{ graph}$$



$$E = - \frac{dV}{dr} \text{ (diff)}^n$$

$$E = -(\text{slope of } V/r \text{ graph})$$

$$\Delta V = - \int_{r_A}^{r_B} \vec{E} \cdot d\vec{r} = - \int_{r_A}^{r_B} \frac{F_{CF}}{q} \cdot dr = - \frac{W_{CF}}{q}$$

+ Scalar

+ Ref. at ∞ , $V = 0$.

+ Electric potential energy is defined only in a conservative field.

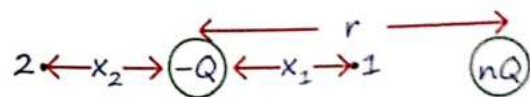
$$\# E = - \frac{dV}{dr} = - [\text{Slope of } V-r \text{ graph}]$$

$$E = - \left[\frac{\partial V}{\partial x} \hat{i} + \frac{\partial V}{\partial y} \hat{j} + \frac{\partial V}{\partial z} \hat{k} \right]$$

+ Potential decrease in the direction of Electric Field.

+ Potential due to positive charge at some distance must be positive $\rightarrow$ False

+ Due to positive charge potential may be +ve, -ve, zero depends on reference $\rightarrow$ True



At two point potential will be zero one, b/w the charges and 2nd left of smaller charge ($V = 0$ karne ke liye dono charge opposite nature ke hona Chahiye)

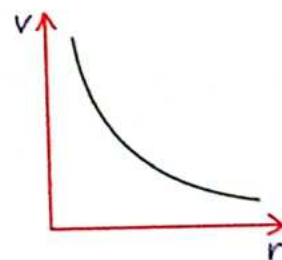
$$\bullet x_1 = \frac{r}{n+1} \quad \bullet x_2 = \frac{r}{n-1}$$

2 Potential due to

Point charge :-

$$\oplus V_P = \frac{KQ}{r}$$

$$\ominus V_P = - \frac{KQ}{r}$$

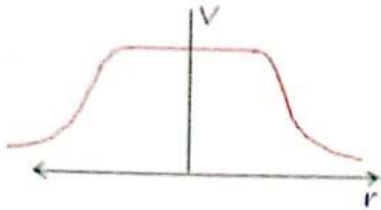


+ For n point charges $q_1, q_2, \dots, q_n$ located at distances $r_1, r_2, \dots, r_n$ from a point P, the electric potential V at P is given by:

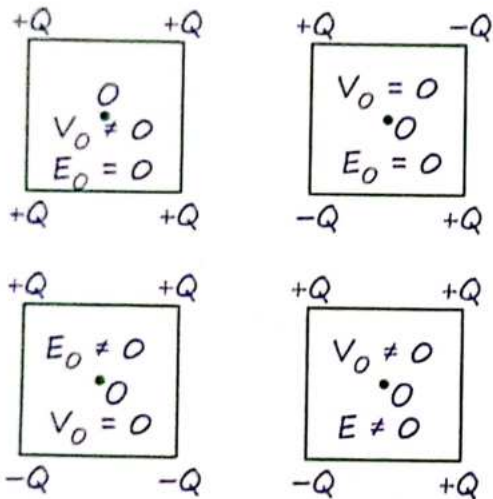
$$V = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i}$$

Ring :-

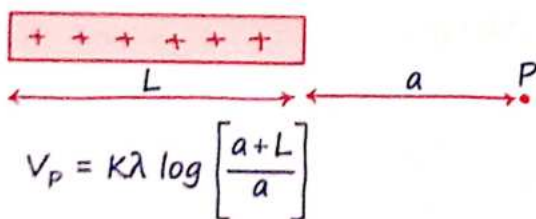
$$V_{\text{axis}} = \frac{KQ}{\sqrt{R^2 + X^2}}, V_C = \frac{KQ}{R}, V_{(X \gg R)} = \frac{KQ}{X}$$



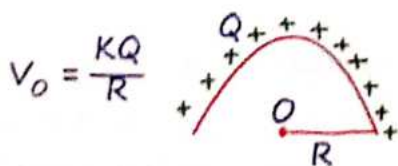
Charges at corner of square :-



Axis of line charge rod :-

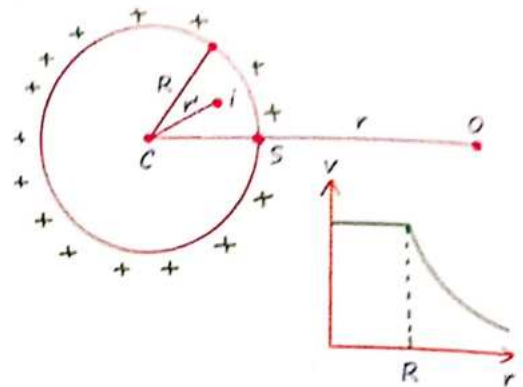


Half ring :-

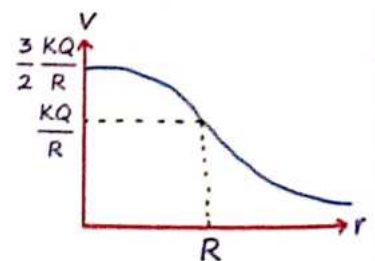
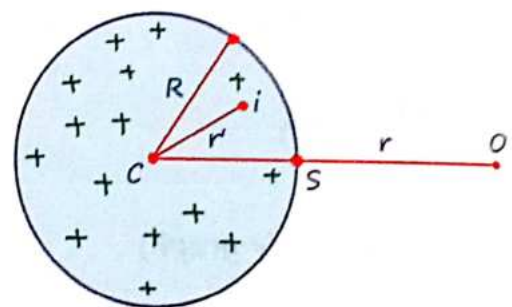


Hollow conducting / non-conducting or solid conducting sphere :-

$$\begin{aligned} + V_0 &= \frac{KQ}{r} & + V_s &= \frac{KQ}{R} \\ + V_i &= \frac{KQ}{R} & + V_c &= \frac{KQ}{R} \end{aligned}$$

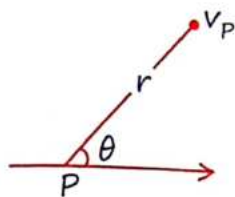


Solid non-conducting sphere :-



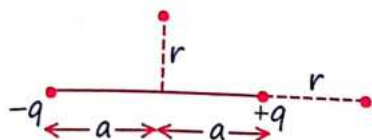
$$\begin{aligned} + V_0 &= \frac{KQ}{r} & + V_s &= \frac{KQ}{R} \\ + V_{in} &= \frac{KQ}{2R^3} [3R^2 - r^2] & + V_c &= \frac{3KQ}{2R} \end{aligned}$$

Electric Potential due to dipole at general point:-



$$+ V_P = \frac{K P \cos \theta}{r^2}$$

$$+ V_{\text{axial}} = \frac{K P}{(r^2 - a^2)} + V_{\text{equatorial}} = 0$$



+ If potential zero then electric field must be zero - False

+ If field is zero then potential must be zero - False

+ If potential constant then field is zero - True

+ If electric field constant then potential is zero - False

+ If we move perpendicular to field potential remains constant. - True

+ Potential increases if we move opposite to the direction of electric field. - True

Note:-

Potential difference does not depend upon reference point it's always the same between two points!

Potential Energy of System of Charged Particles :-

+ For a charge q in a potential V : $U = qV$

+ Using the electric field E : $U = q[-\int \vec{E} \cdot d\vec{r}]$

(i) Two charges system:

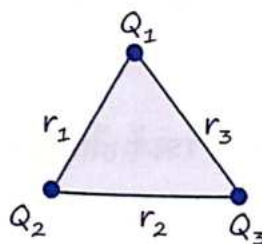
$$+ Q_1 \quad \text{---} r \quad \text{---} \quad + Q_2$$

$$U = \frac{K Q_1 Q_2}{r}$$

(ii) Three charges system:

+ For a system consisting of three point charges Q_1 , Q_2 , and Q_3 , arranged in a triangular configuration with distances r_1 , r_2 , and r_3 between them, the total potential energy U of the system is given by the sum of the potential energy contributions from each pair of charges.

$$U = \frac{1}{4\pi\epsilon_0} \left(\frac{Q_1 Q_2}{r_1} + \frac{Q_2 Q_3}{r_2} + \frac{Q_3 Q_1}{r_3} \right)$$

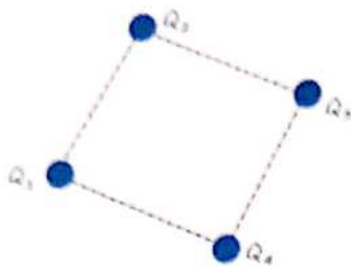


+ If $Q_1 = Q_2 = Q_3 = Q$, then

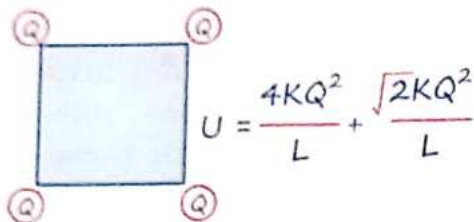
$$U_{\text{net}} = \frac{3KQ^2}{L}$$

+ For a system of four point charges Q_1 , Q_2 , Q_3 , and Q_4 , the electric potential energy U is

$$U = \frac{1}{4\pi\epsilon_0} \left(\frac{Q_4 Q_3}{r_{43}} + \frac{Q_4 Q_2}{r_{42}} + \frac{Q_4 Q_1}{r_{41}} + \frac{Q_3 Q_2}{r_{32}} + \frac{Q_3 Q_1}{r_{31}} + \frac{Q_2 Q_1}{r_{21}} \right)$$



* If $Q_1 = Q_2 = Q_3 = Q_4 = Q$



Vimp.

$$\text{Total no. of terms of PE} : - \frac{n(n-1)}{2}$$

$n = \text{no. of Charges}$

* $W_{\text{ext}} = + \Delta U = \Delta KE = q\Delta V$

Note:-

जो काम खुद हो रहा है जैसा होना चाहिए वैसा हो रहा है तो :-

$U \downarrow \quad \oplus \rightarrow \leftarrow \ominus \quad r \downarrow$

अगर बबरदस्ती किए :-

$U \uparrow \quad \oplus \rightarrow \leftarrow \oplus \quad r \downarrow$

* $\text{Pressure} :- \frac{\sigma^2}{2\epsilon_0}$

$$E = \frac{\sigma}{\epsilon_0}$$

3 Self Energy

Hollow conducting / non-conducting or solid conducting sphere :-

* $U = \frac{KQ^2}{2R}$

Solid non-conducting sphere :-

$$U_{\text{Total}} = \frac{0.6KQ^2}{R} = \underbrace{\frac{0.5KQ^2}{R}}_{\text{Surface to } \infty} + \underbrace{\frac{0.1KQ^2}{R}}_{\text{Centre to Surface}}$$

4 Equipotential Surface

* An equipotential surface is an imaginary surface where every point has the same electric potential. This means there is no potential difference between any two points on this surface.

Types of Equipotential Surfaces

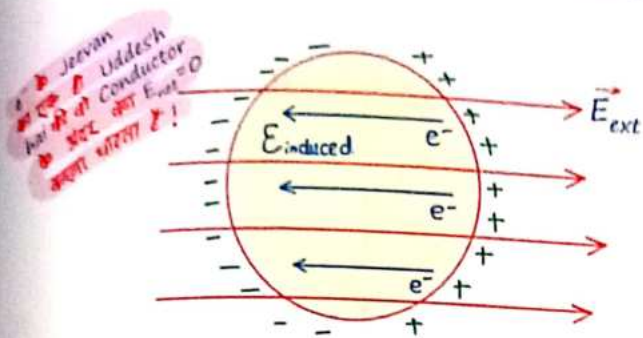
Charge Distribution	Shape of Equipotential Surface
Point Charge	Spherical
Infinite Line Charge	Cylindrical
Infinite Charged Plate	Plane
Large Distance from Ring/Charge Distribution	Spherical

Characteristics:-

- * Potential Difference (ΔV) is Zero.
- * Work (W) Done is Zero.
- * Equipotential surfaces are always at right angles (perpendicular) to electric field lines.

5 Electrostatics of Conductor

Electrostatic shielding



$$\therefore E_{net} = E_{ext} + E_{ind}$$

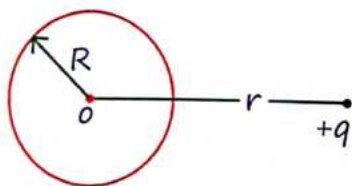
$$E_{net} = 0$$

$$E_{ext} = -E_0$$

$$E_{induced} = E_0$$

Jitna External E. Field hoga Utna opposite $\vec{E}_{induced}$ hoga.

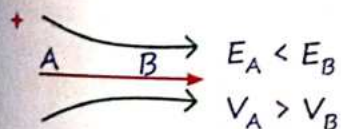
Q.1 Neutral conducting sphere, find electric field of center due to +q charge.



$$\text{Ans. } \vec{E}_q = \frac{Kq}{r^2} \quad E_{net} = 0$$

$$\vec{E}_{induced} = -\vec{E}_q$$

6 Important points



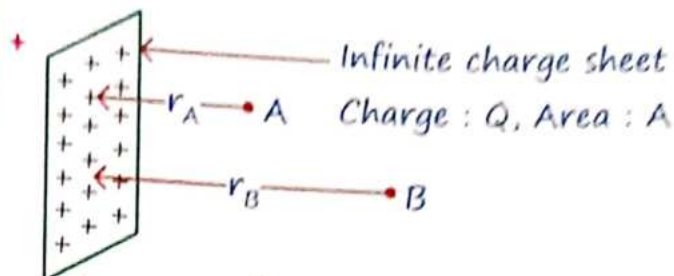
$$E_A < E_B$$

$$V_A > V_B$$



$$V_A - V_B = Er \cos \theta$$

Electric Potential

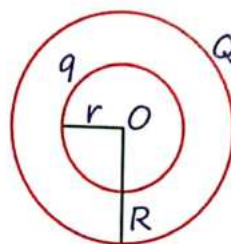


Infinite charge sheet

Charge : Q, Area : A

$$V_A - V_B = \frac{Q}{2AE_0} (r_B - r_A)$$

* Find potential difference b/w two concentric sphere



$$\Delta V = Kq \left[\frac{1}{r} - \frac{1}{R} \right]$$

* Potential difference does not depends on charge on outer sphere only on inner charge.

* When two concentric spherical charge connected with wire the potential difference becomes zero, all the charge of inner sphere will flow to outer sphere.

$$W_{ext} = \Delta U = q\Delta V$$

$$W_{electric\ field} = -\Delta U = -q\Delta V$$

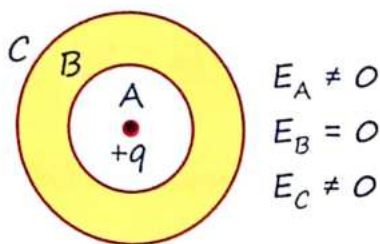
* If external force is absent, we can apply (C.O.M.E.) conservation of mechanical energy or $\Delta U_{loss} = K.E._{gain}$

$$K.E. = q\Delta V$$

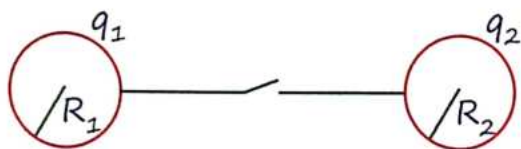
* Electric field inside conductor is zero
- False

- + Electric field inside isolated conductor where matter is present is always zero
- True

+ Thick conductor :-



- + When two conducting sphere is connected with wire (Not concentric)



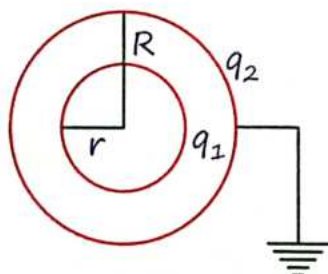
Final charge $q \propto \text{Radius (R)}$

Potential $V = \text{Same}$

Surface charge density $\sigma \propto \frac{1}{\text{Radius (R)}}$

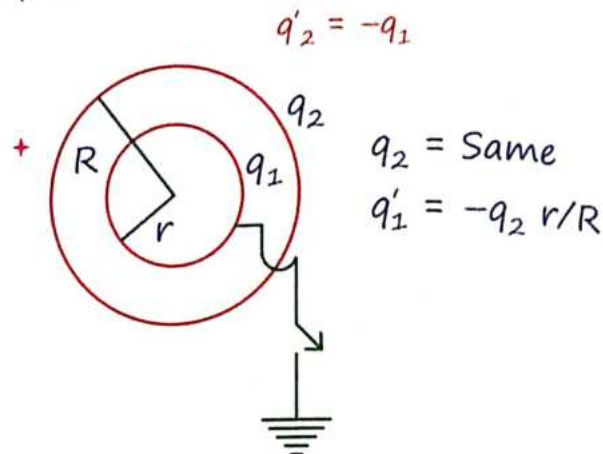
Electric field on surface $E \propto \frac{1}{\text{Radius (R)}}$

- + After earthing potential of conductor must be zero, charge may or may not be zero.
- + Final charge on each sphere when outer sphere is grounded.



Jisko ground nahi kiya hai uska charge same rahega. Jisko ground kiya hai, uska

potential zero hoga and final charge on outer sphere



- + From large distance then distance of closest approach

$q \rightarrow V_0$ (fixed $+Q$)
 $r = \frac{2KQq}{mV_0^2}$

Q.2 What amount of work is done in moving a charge of 4 coulombs from a point 220 volts to a point at 230 volts?

Sol. Potential difference between the two points

$$\Delta V = 230 - 220 \text{ volts}$$

$$\therefore \Delta V = 10 \text{ volts}$$

Amount of charge moving $q = 4$ coulombs

Thus work done $W = q\Delta V$

$$\therefore W = 4 \times 10 = 40 \text{ joules}$$

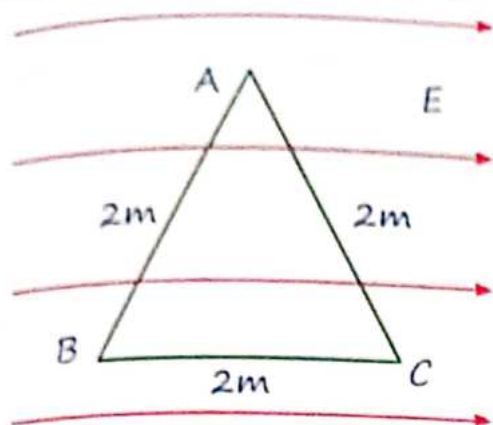
Q.3 Three point charge $-q$, $+q$ and $-q$ are placed along straight line at equal distance (say r meter). Electric potential energy of this system of charges will be if $+q$ charge is in the middle.

Sol. $-q \leftarrow r \rightarrow q \leftarrow r \rightarrow -q$

$$U = \frac{-Kq^2}{r} - \frac{Kq^2}{r} + \frac{Kq^2}{2r}$$

$$U = \frac{-Kq^2}{2r}$$

Q.4 In an uniform electric field $E = 10\text{N/C}$ as shown in figure, find $V_A - V_B$



Sol. We know that Electric field, distance and voltage are related by formula,
 $E \cdot d = V$

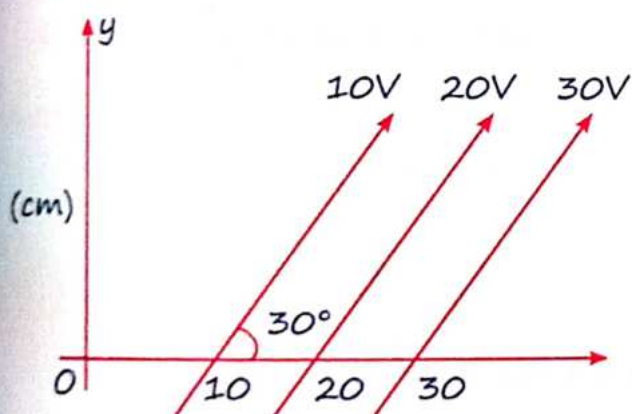
Substituting values,

$$V = 10 \times 2 \times \cos 120^\circ, \quad V = -10\text{V}$$

Q.5 The work done to move a charge on an equipotential surface is?

Sol. Workdone in equipotential surface is zero as $W = q(V_A - V_B)$ & $V_A = V_B$

Q.6 Equipotential surfaces are shown in fig. then the electric field strength will be



Sol. $\Delta V = -E \Delta r \cos \theta$

$$E = \frac{-\Delta V}{\Delta r \cos \theta}, \quad E = \frac{-(20-10)}{10 \times 10^{-2} \cos 120^\circ}$$

$$= \frac{-10}{10 \times 10^{-2} (-\sin 30^\circ)}$$

$$= \frac{-10^2}{-1/2} = 200 \text{ V/m}$$

Q.7 If potential at centre of Non conducting sphere is zero then find potential at surface of sphere?

Sol. $\Delta V = \text{Same}$, does not depends on reference

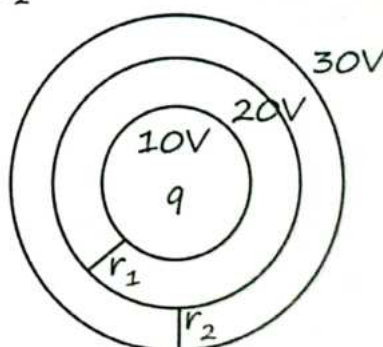
$$V_0 - V_S = \frac{3}{2} \frac{KQ}{R} - \frac{KQ}{R}$$

$$0 - V_S = \frac{KQ}{2R}$$

$$V_S = -\frac{KQ}{2R}$$

Q.8 Equipotential surface due to point charge then compare r_1 and r_2 .

Sol. Hint $E = -\frac{dv}{dr}$
 $r_1 < r_2$



MR*

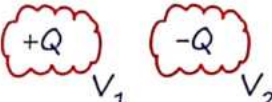
6 Aasma bhi jhukega tere aage yu hi
 junun ke had se guzarte raho pura
 jeevan ek sangarsh hai ladte raho or
 aage badhte raho. 9

1 Capacitor

- An electrical device which store electrostatic energy by storing charge.
- Generally it is combination of two conductor, having equal and opposite charge.

$$Q = CV \quad \frac{Q}{\text{Volt}} = \text{Farad}$$

* Depends on size, shape & Medium between them.

$$C = \frac{Q}{\Delta V} = \frac{Q}{V_1 - V_2}$$


→ Slope of Q/v graph

→ Scalar

→ unit q/v = farad

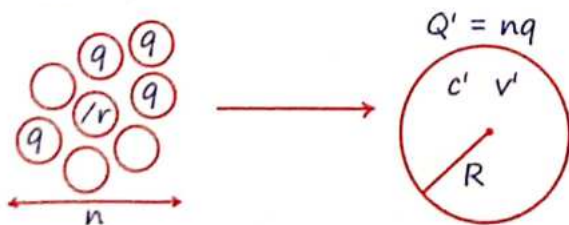
→ Dim → $[M^{-1}L^{-2}T^4A^2]$

2 Spherical Capacitor

$$C = \frac{Q \cdot R}{KQ} = 4\pi\epsilon_0 R$$

Special Case :-

- n small charged sphere combine to form a bigger sphere.



For bigger sphere,

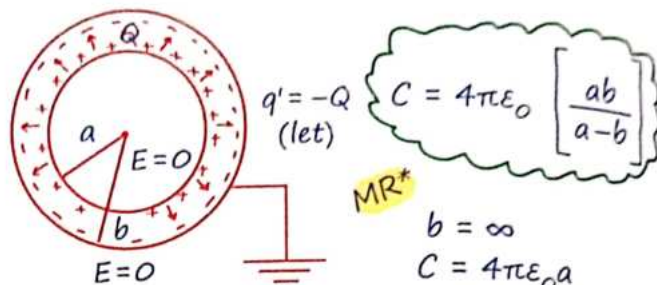
$$C' = n^{1/3} C$$

$$U' = n^{5/3} U_0$$

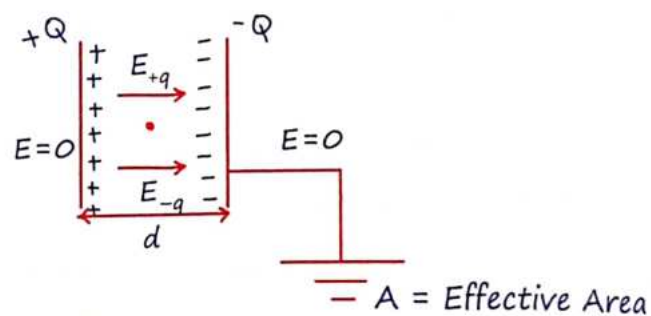
$$V' = n^{2/3} V$$

$$R = n^{1/3} r$$

Combination of Spherical Capacitor :-



Parallel Plate Capacitor :-



$$E \text{ (due to one plate)} = \frac{\sigma}{2\epsilon_0} = \frac{Q}{2A\epsilon_0}$$

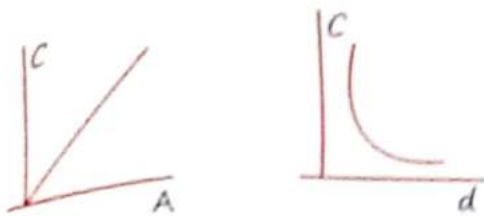
$$E_{\text{net}} = \frac{\sigma}{\epsilon_0} = \frac{Q}{A\epsilon_0}$$

$$\Delta V = \frac{\sigma d}{\epsilon_0} = \frac{Qd}{A\epsilon_0}$$

Capacitance in air

$$\text{(air)} C_0 = \frac{Q}{\Delta V} = \frac{A\epsilon_0}{d}$$

$$\text{(medium)} C' = \frac{A\epsilon_m}{d}, C' = k \frac{A\epsilon_0}{d}$$



• Dielectric introduce between plates :-

$$C = KC_0$$

Force between the Plates :-

$$F = \frac{Q^2}{2A\epsilon_0} = \frac{\sigma^2 A}{2\epsilon_0} \quad (\because F = qE)$$

Pressure on the plate :-

$$P = \frac{F}{A} = \frac{Q^2}{2A^2\epsilon_0} = \frac{\sigma^2}{2\epsilon_0}$$

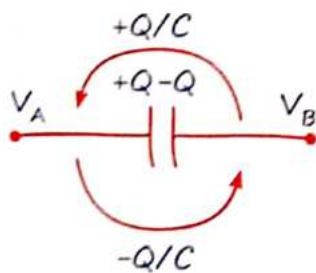
Energy density :-

$$U = \frac{\sigma^2}{2\epsilon_0} = \frac{\text{Energy}}{\text{Volume}} = \frac{Q^2}{2CA d}$$

• Between the plate net electric field only due to charge on Inner surface of plate

$$E_{\text{net}} = \frac{Q}{A\epsilon_0}, \quad Q = \text{charge of Inner surface.}$$

Note :-



A to B

$$V_A - Q/C = V_B$$

$$V_A - V_B = Q/C$$

B to A

$$V_B + Q/C = V_A$$

$$V_A - V_B = Q/C$$

Energy on Capacitor :-

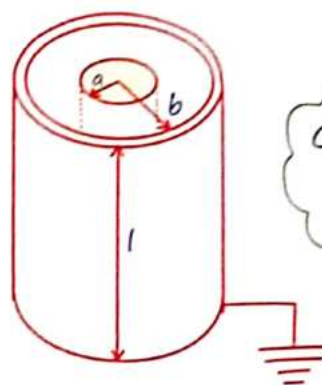
$$E = \frac{1}{2} CV^2 = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} QV$$

Self Energy :-

For Spherical Capacitor :-

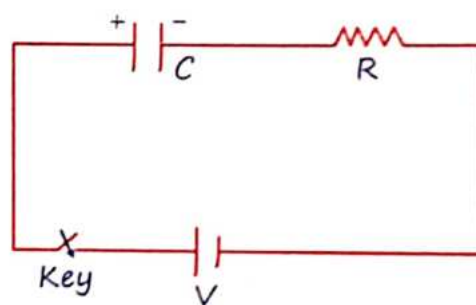
$$U = \frac{Q^2}{8\pi\epsilon_0 R}$$

Cylindrical Capacitor :-



$$C = \frac{2\pi\epsilon_0 l}{\log_e \left(\frac{b}{a} \right)}$$

Charging of Capacitor :-



• WD by battery in Charging Capacitor

$$= CV^2$$

• WD in Charging Capacitor = $\frac{1}{2} CV^2$

• $U_{\text{loss}} = \frac{1}{2} CV^2$ Heat

MR* for Calculating E_{loss}

$$U_i + W_{\text{battery}} = U_f + U_{\text{loss}}$$

$\frac{1}{2} CV_{\text{initial}}^2$ $Q \Delta V$ $\frac{1}{2} CV_{\text{final}}^2$
 charge transferred emf of battery

3 Connection of Two Capacitor

Same Terminal Connection :-

$$V_C = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2} = \frac{q_1 + q_2}{C_1 + C_2}$$

Loss in Energy :-

$$\Delta U = \frac{1}{2} \frac{C_1 C_2}{C_1 + C_2} (V_1 - V_2)^2$$

Final Charge :- $q'_1 = C_1 V_C$

$q'_2 = C_2 V_C$ Transfer :- $[q_1 - q'_1]$

Reverse Polarity Connection :-

$$V_C = \frac{C_1 V_1 - C_2 V_2}{C_1 + C_2}$$

$$\Delta U = \frac{1}{2} \frac{C_1 C_2}{C_1 + C_2} [V_1 + V_2]^2$$

- * An air filled capacitor charged with potential V connected with identical uncharged capacitor filled with k -dielectric then common potential and final charge on each.

$$V_C = \frac{CV + KC \times 0}{C + KC} = \frac{CV}{C(1+K)} = \frac{V}{1+K}$$

$$Q(\text{air}) = \frac{CV}{1+K} \quad Q(\text{dielectric}) = \frac{KCV}{1+K}$$

4 Combination of Capacitor

Series combination of capacitors :-

$Q = \text{Same}$

$V = V_1 + V_2 + \dots$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots$$

Law of Potential drop :-

$Q = CV = \text{same.}$

Jiska Capacitance Jyada hoga Usmein Voltage drop Kam hoga.

Parallel combination of capacitors :-

$V = \text{Same}$

$Q = q_1 + q_2 + \dots$

$C_{eq} = C_1 + C_2 + C_3 + \dots$

Law of Charge Distribution :-

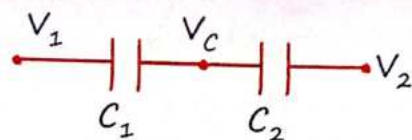
$q = CV$

MR*
Special

$q \propto C$

Jiska Capacitance jyada hoga woh jyada charge rakh lega!

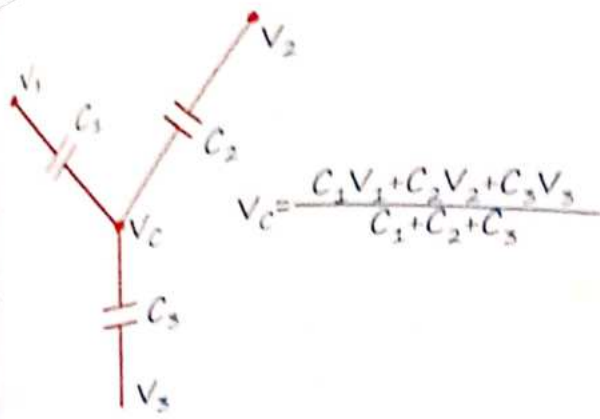
MR*



If $V_1 > V_2$

$$V_C = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$$

$$Q_1 = C_1 (V_1 - V_C) \\ Q_2 = C_2 (V_C - V_2)$$



Note:-

* If n-identical Connected in Series & Parallel :-

* Series :-

$$C_{eq} = \frac{C}{n}$$

C_{eq} is smaller than Smallest Capacitor

* Parallel :-

$$C_{eq} = nC.$$

C_{eq} is larger than largest Capacitor.

* Breakdown Voltage :-

V_{max} that can be applied across a Capacitor.

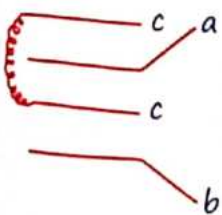
* Series :-

$$V_B = V_1 + V_2 + V_3 + \dots$$

* Parallel :-

$$V_B = V$$

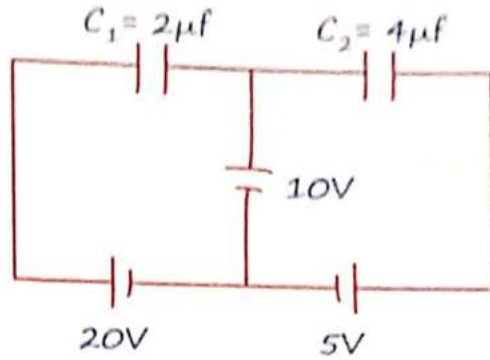
* Special Case :-



No. of gap = No. of Cap.

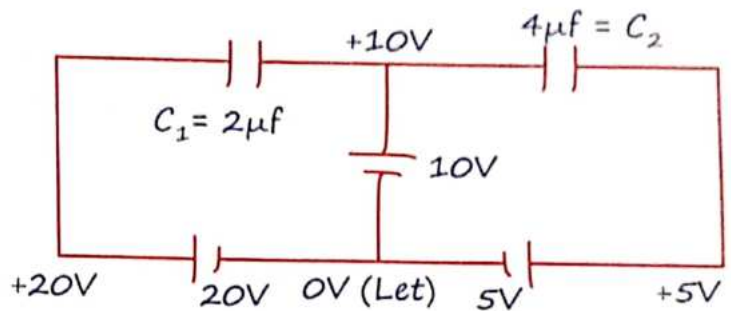
Capacitor

Q.1. Find charge and potential difference across each capacitor.



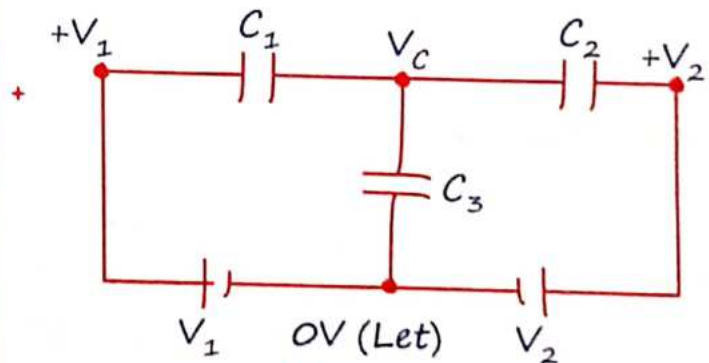
Sol.

MR* → You can assume potential as reference zero potential at any one point of circuit.



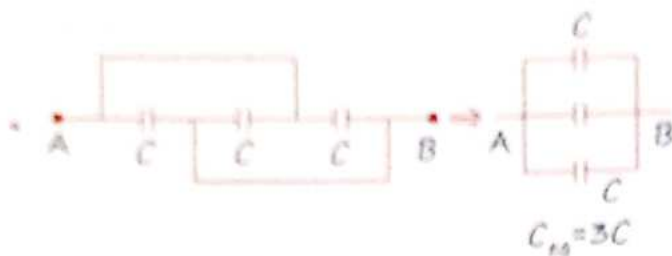
(Charge on C_1) $Q_1 = C_1 (\Delta V) = 2\mu f \times (10V) = 20\mu c$

(Charge on C_2) $Q_2 = C_2 (\Delta V) = 4\mu f \times (10-5) = 20\mu c$



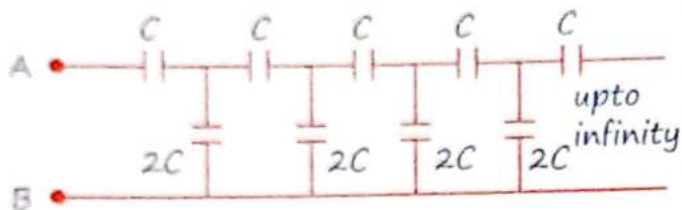
MR* Bade aaram से

$$V_c = \frac{C_1 V_1 + C_2 V_2 + C_3 \times 0}{C_1 + C_2 + C_3}$$

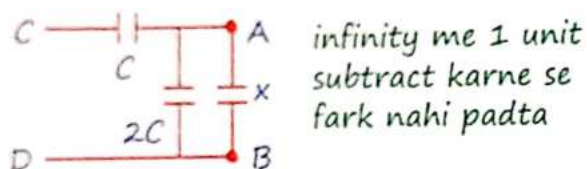


5 Ladder Network

Ladder Type - 1 :-



Sol.

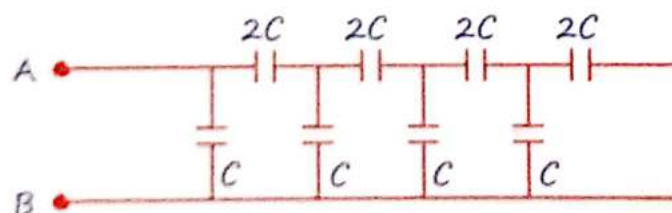


$X + 2C$ connect $\leftarrow X \& 2C$ parallel

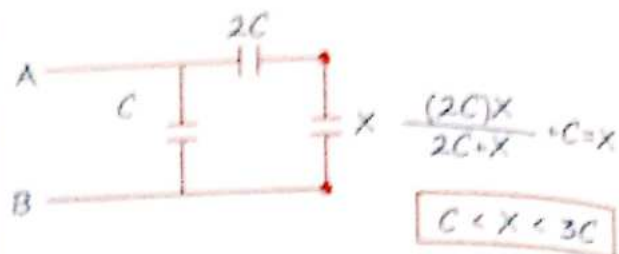
$$C_{eq} = \frac{(X+2C)C}{(X+2C)+C} \leftarrow \text{then it with } c \text{ in series to get } C_{eq}$$

MR* In series C_{eq} must be less than smallest
hence $C_{eq} < C$

Ladder Type - 2 :-



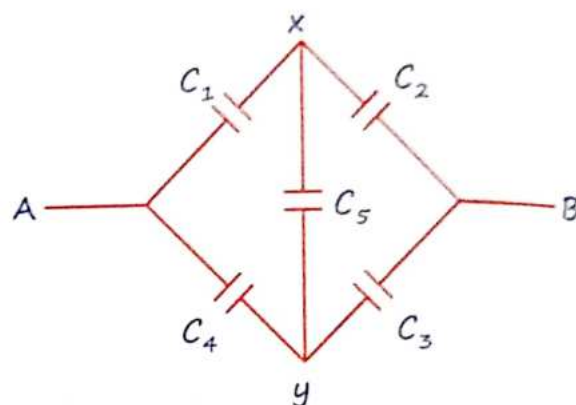
add $2C \& X$ in series then in parallel with C .



MR*

- Series main C_{eq} less than smallest and in parallel C_{eq} larger than largest

6 Wheat stone Bridge



- Balanced condition of a wheat stone bridge

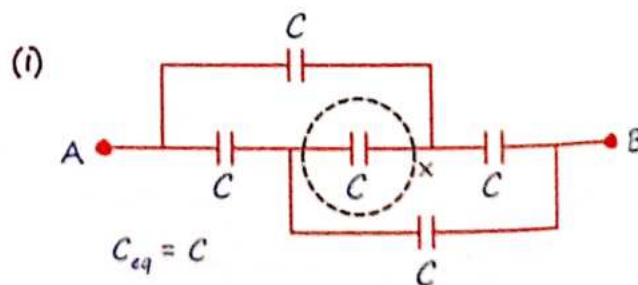
$$\frac{C_1}{C_4} = \frac{C_2}{C_3}$$

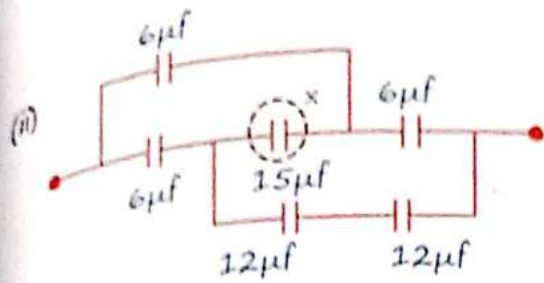
- $V_x = V_y$

($\therefore$ Potential difference between x and y is zero)

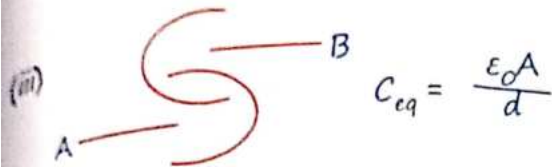
- Charge on $C_5 = 0$

Some example of Wheat Stone Bridge :-

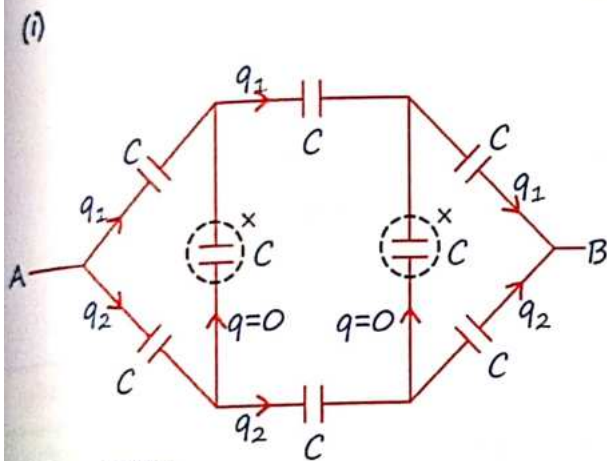




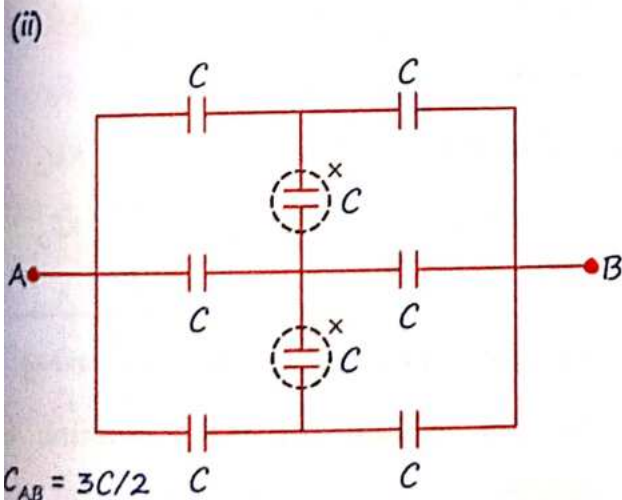
$$C_{eq} = 6\mu f$$



Some symmetric circuit :-

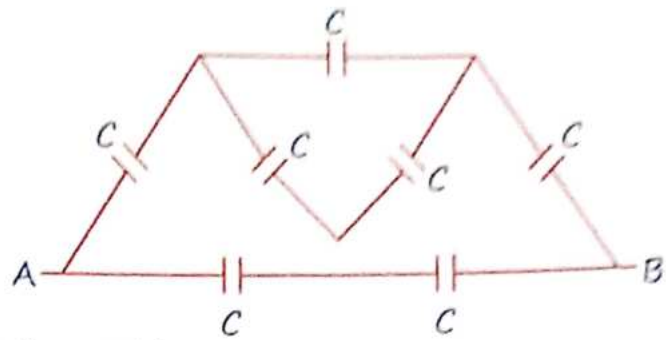


$$C_{eq} = 2C/3$$



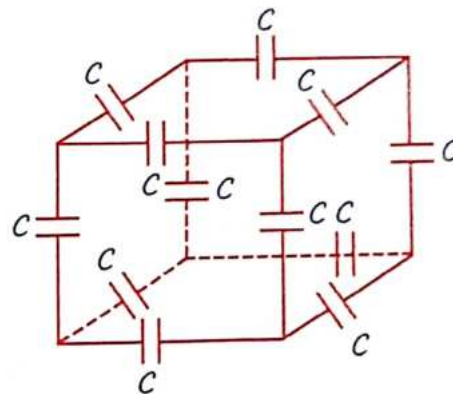
$$C_{AB} = 3C/2$$

(iii)



$$C_{AB} = 7C/8$$

+ 12-Identical capacitors or resistors in cubical form :-



MR*

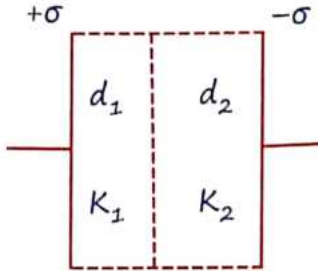
Along side	Along Face diag.	Along Body diag.
$C_{eq} = \frac{12}{7} C$	$C_{eq} = \frac{4}{3} C$	$C_{eq} = \frac{6}{5} C$
$R_{eq} = \frac{7R}{12} \Omega$	$R_{eq} = \frac{3R}{4} \Omega$	$R_{eq} = \frac{5R}{6} \Omega$

7 Dielectric Medium in an External E. Field

$E_{net} = \frac{E_0}{K}$ Jaise Coulomb F. dec. "K" time
Similarly E. Field dec. "K" times.

Series Combination of Dielectrics :-

Ek ke baad Ek daalte rehna!

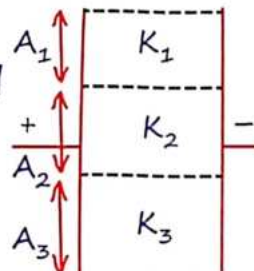
$$+ C = \frac{A\epsilon_0}{\left[\frac{d_1}{K_1} + \frac{d_2}{K_2}\right]} \quad +\sigma \quad -\sigma$$


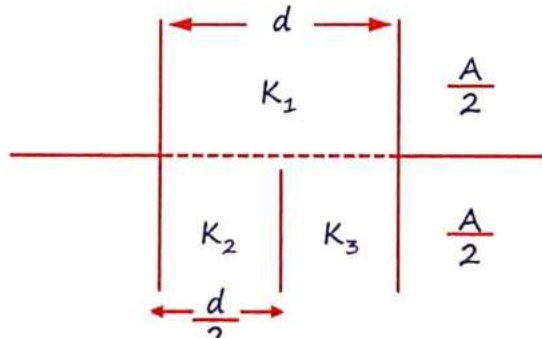
$$+ K_{eq} = \frac{d}{\frac{d_1}{K_1} + \frac{d_2}{K_2}}$$

Parallel Combination of Dielectrics :-

Saare dielectric
ka ek terminal
+ve और Dusra
-ve terminal

$$+ C_{eq} = \frac{\epsilon_0}{d} [K_1 A_1 + K_2 A_2 + \dots]$$

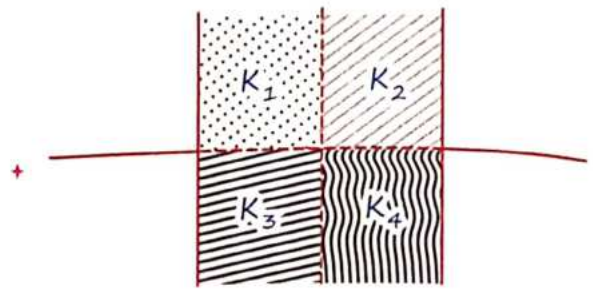
$$+ K_{eq} = \frac{K_1 A_1 + K_2 A_2 + \dots}{A_1 + A_2 + \dots}$$




$$C = \frac{\epsilon_0 A}{d \left(\frac{K_2 + K_3}{K_2 K_3} \right)} + \frac{\epsilon_0 A K_1}{2d}$$

$$C = \frac{\epsilon_0 A}{d} \left[\frac{K_2 K_3}{K_2 + K_3} + \frac{K_1}{2} \right]$$

MR* If $K_1 = K_2 = K_3$ then $C = \frac{\epsilon_0 A K}{d}$



$$C_{eq} = \frac{\epsilon_0 A}{d} \left(\frac{K_3 K_4}{K_3 + K_4} + \frac{K_1 K_2}{K_1 + K_2} \right)$$

MR* If $K_1 = K_2 = K_3 = K_4$ then $C_{eq} = \frac{\epsilon_0 A K}{d}$

MR* If $K=1$ $C' = C$ If $K=\infty$ Conductor $C' = \infty C$

Introduction of dielectric between plates of capacitor :-

+ Battery is Connected :-

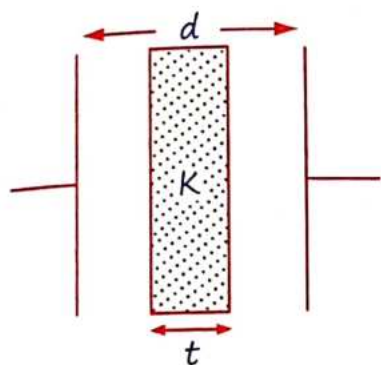
	Initially	Finally
Capacitance	C_0	KC_0
Potential	V_0	$*V_0$
Charge	$Q_0 = C_0 V_0$	KQ_0
E. Field	$E_0 = V_0/d$	E_0
U_0 (Energy)	$U_0 = Q_0^2/2C_0$	KU_0
F_0 (Force on Plate)	$F_0 = qE$	KF_0

+ Battery Ziddi hai potential same rakhegi!

+ Battery is removed then dielectric is placed :-

	Initially	Finally
Capacitance	C_0	KC_0
Potential	$V_0 = Q_0 / C_0$	$*V_0 / K$
Charge	$Q_0 = C_0 V_0$	$*Q_0$ <u>conserved</u>
E. Field	$E_0 = V_0 / d$	E_0 / K
U_0 (Energy)	$U_0 = Q_0^2 / 2C_0$	U_0 / K
F_0 (Force on Plate)	$F_0 = qE$	F_0 / K

plate isolated hai to charge conserved rahega.
Dielectric of width t placed between capacitor :-



$$C = \frac{\epsilon_0 A}{(d-t) + \frac{t}{K}}$$

if $t=0$

MR* $C = \frac{\epsilon_0 A}{d} \quad \left| \quad d=t \right. \quad C = \frac{\epsilon_0 KA}{d}$

Q.2. A parallel plate air capacitor has capacitance C . Half of space between the plate is filled with dielectric K as shown to Fig. Then new Capacitance C' is

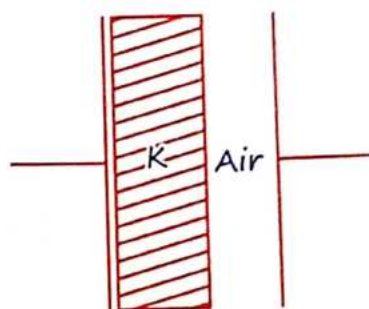
(a) $C' = C \left[\frac{K}{K+1} \right]$

(b) $C' = C \left[\frac{2K}{K+1} \right]$

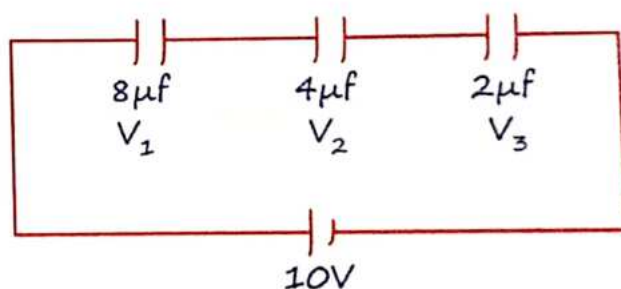
(c) $C' = \frac{2C}{K+1}$

(d) $C' = C \left[1 + \frac{K}{2} \right]$

Sol. (b) $C = \left[\frac{\epsilon_0 A}{d-t + \frac{t}{K}} \right] = \left[\frac{\epsilon_0 A}{d - \frac{d}{2} + \frac{d}{2K}} \right]$
 $= \left[\frac{2\epsilon_0 AK}{d(K+1)} \right] = 2C \left[\frac{K}{(K+1)} \right]$



Q.3. Find charge and potential drop across each capacitor.

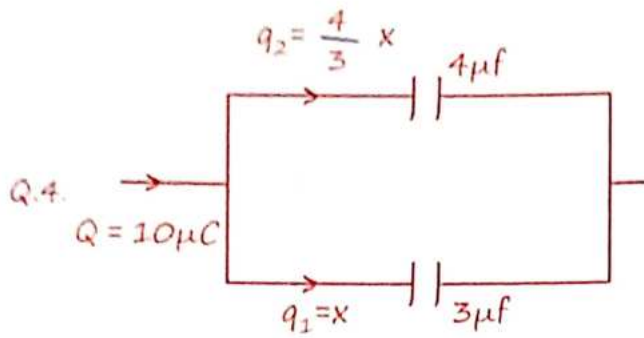


Sol. MR* $V = \frac{q}{C}$

$V_1 = x(\text{let}) \quad V_2 = 2x \quad V_3 = 4x$

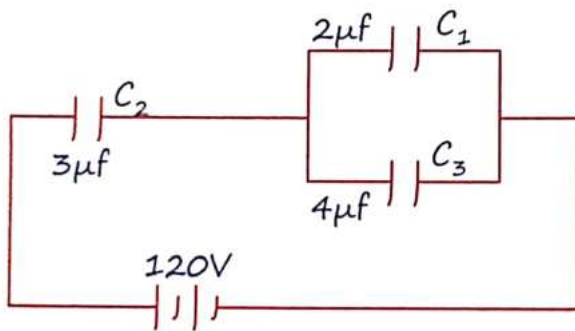
$x + 2x + 4x = 10V$

$x = \frac{10}{7} \text{ volt}$



Sol. $x + \frac{4}{3}x = 10$ **MR*** $Q = CV$
 $Q \propto C$
 $x = \frac{30}{7} \mu C$

Q.5. The charge on capacitors shown in the figure and the potential difference across each will be respectively.



Sol. The total capacitance of the circuit is,

$$C = 2 \mu f$$

$$Q = CV$$

$$\text{So, } Q = 240 \mu C$$

Voltage across $3 \mu f$ capacitor will be,

$$V_1 = \frac{240}{3} V = 80V$$

Voltage across $2 \mu f$ and $4 \mu f$ capacitor will be,

$$V_2 = (120 - \frac{240}{3}) V = 40V$$

Charge across the $2 \mu f$ capacitor will be,

$$Q_1 = (2 \times 40) \mu C = 80 \mu C$$

Charge across the $4 \mu f$ capacitor will be,

$$Q_2 = (4 \times 40) \mu C = 160 \mu C$$

1 Electric Current

The rate of directional flow of electric charge is called electric current.

Current

In time t

$$I_{\text{Avg.}} = \frac{\Delta Q}{\Delta t}$$

$$= \frac{\int I dt}{\int dt}$$

at time t

$$I_{\text{Inst.}} = \frac{dq}{dt}$$

= slope of charge-time graph

* Charge on circular path

$$Q_{\text{Avg.}} = \frac{Q}{T} = \frac{ne}{T}$$

$$= nef = \frac{nev}{2\pi r}$$

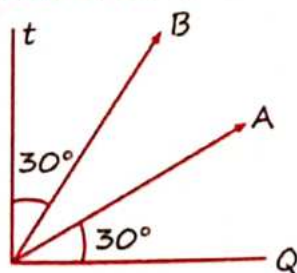
$\int dq = \int I dt$
 $\Delta Q = \text{Area of current time graph}$

scalar,
unit (Ampere),
dimension $[A^1]$

1. Through a given cross section n_1 electron per-sec are passing from left to right and n_2 proton are passing from right to left simultaneously then the electric current through this cross section.

Ans. $I = (n_1 + n_2)e$

2. Find ratio of current.



Ans. $\frac{I_A}{I_B} = \frac{\tan 60^\circ}{\tan 30^\circ} = \frac{3}{1}$

Isolated Conductor:-

- + Number Density of Electrons (n)
 $ne^- = 10^{28} e^-/m^3$
- + The average velocity of electrons in a conductor is
 $V = 10^3 \text{ m/s}$
- + In an isolated conductor, the electrons move in such a way that they cancel out any internal electric potential, resulting in a net average potential of zero.

$$V_{\text{avg}} = 0 \quad E_{\text{in}} = 0 \quad (\text{speed } V \propto \sqrt{T})$$

- + Kinetic Energy and Temperature

$$\frac{1}{2} mv^2 = \frac{3}{2} K_B T \quad K_B = 1.38 \times 10^{-23} \frac{\text{Js}}{\text{K}}$$

2 Battery Connected to Conductor

$$F_e = qE, \quad a = \frac{qE}{m_e} \quad E = \frac{V}{l}$$

→ acceleration
→ Force on electron

Drift velocity

$$V_D = \cancel{V} + aT \quad I = neAV_D$$

$$V_D = a_e T = \frac{eET}{m_e}$$

Where:

E = electric field

V_d = drift velocity

l = length of conductor

n = no. of electron per unit volume

τ = relaxation time

m_e = mass of electron

V = emf of battery

e = charge of electron

- The electric field in the conductor is due to the potential difference applied by the battery, distributed along the length of the conductor.
- Electrons gain velocity under the influence of the electric field, known as drift velocity. This is the average velocity attained by electrons in a conductor due to the applied electric field.

Conductivity(σ) and Resistivity(ρ):

- Conductivity is a measure of a material's ability to conduct electric current, while resistivity is the measure of how strongly a material opposes the flow of electric current. They are inversely related.

MR* feel

$$\sigma = \frac{ne^2\tau}{m} \quad \rho = \frac{1}{\sigma} \quad \rho = \frac{m}{ne^2\tau}$$

Microscopic form of Ohm's law!

$$V = \frac{m i l}{ne^2 \tau A} \quad V = iR \quad R = \frac{m l}{ne^2 \tau A} \quad R = \frac{\rho l}{A}$$

Mobility :-

- Property of charge carrier.
- Does not depend on drift velocity and electric field.

$$\mu = \frac{V_d}{E} = \frac{e\tau}{m}$$

- Relation between mobility and conductivity
 $\sigma = ne\mu$
- $\mu_e > \mu_p > \mu_{Deut.} = \mu_{\alpha}$

MR*

Q.3 If drift velocity is doubled then what about mobility?

Ans. Remains same

Current Density :- (Vector)

Current density is the amount of electric current flowing per unit area of the cross-section of a conductor.

$$J = \frac{I}{A} = neV_d$$

Where

I : Electric current

A : Cross sectional area of the conductor

n : Number of charge carriers per unit volume

e : Charge of an electron

V_d : Drift velocity of the charge carriers

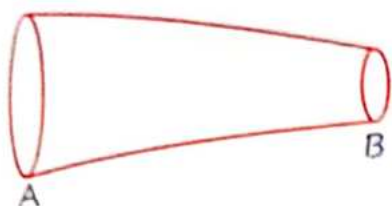
In vector form

$$\vec{I} = \vec{J} \cdot \vec{A} = JA \cos \theta$$
$$\vec{J} = ne\vec{V}_d$$

Vector form of Ohm's law!

$$\vec{J} = \sigma \vec{E}$$

The direction of the current density vector is the same as the direction of the flow of positive charge (or the direction of the conventional current).



Current : $I_A = I_B$
 Current density : $J_A < J_B$
 Drift velocity : $V_A < V_B$

Variation of Resistance :-

Material prop.
 $R = \frac{\rho L}{A}$

where R is the resistance, ρ is the resistivity of the material, L is the length of the conductor, and A is the cross-sectional area of the conductor.

(a) $l \rightarrow \text{Change}$ $A \rightarrow \text{Const}^n$ $R \propto l$	(b) $l \rightarrow \text{Change}$ $V \rightarrow \text{Const}^n$ $R \propto l^2$
(c) $A \rightarrow \text{Change}$ $l \rightarrow \text{Const}^n$ $R \propto 1/A$	(d) $A \rightarrow \text{Change}$ $V \rightarrow \text{Const}^n$ $R \propto 1/A^2$
(e) $R = \frac{\rho l^2}{M} \times \text{density}$ $R \propto \frac{l^2}{M}$	(f) $R = \frac{\rho M}{\text{density } A^2}$ $R \propto \frac{M}{A^2}$

If a wire of Resistance R stretched to double of its length the new resistance becomes $4R$.

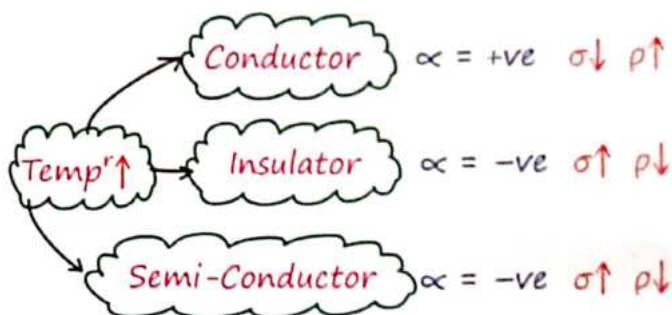
Q.4 The masses of the wire in the ratio of $1 : 3 : 5$ and their length are in ratio of $5 : 3 : 1$. The ratio of their resistance.

Ans. $R \propto \frac{l^2}{m}$ $R_1 : R_2 : R_3 = 125 : 15 : 1$

Ohm's Law :-

$\vec{J} = \sigma \vec{E}$, $V = \frac{m i l}{n e^2 \tau A}$ $V = i R$

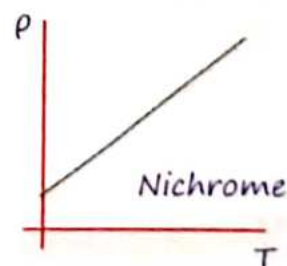
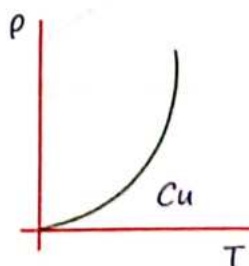
*Note :-



Temperature Dependence of Resistance & Resistivity :-

$\rho_T = \rho_0 [1 + \alpha \Delta T]$,

$\alpha = \frac{\Delta \rho}{\rho_0 \Delta T}$
 unit K^{-1}



* Two resistance R_1 and R_2 connected in series and their R_{eq} does not depends upon temperature then $R_1 \alpha_1 = -R_2 \alpha_2$

* $R_{t-c} = R_0 (1 + \alpha T)$ always valid

$$R_{t_2} = R_{t_1} [1 + \alpha (T_2 - T_1)]$$

Valid for small change in temperature

$\frac{R_{t_2}}{R_{t_1}} = \frac{1 + \alpha t_2}{1 + \alpha t_1}$ always valid	R_0 = resistance at 0°C R_{t_1} = Resistance at t_1 R_{t_2} = Resistance at t_2
---	---

Relation between Coefficient of :-

(a) linear expansion (α)

(b) Resistance (α_R)

(c) Resistivity (α_ρ)

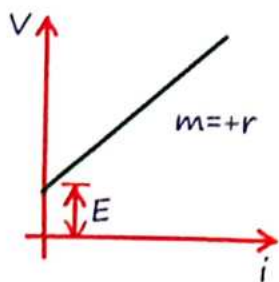
$$R = \frac{\rho l}{A}$$

$$\alpha_R + \alpha = \alpha_\rho$$

3 Battery

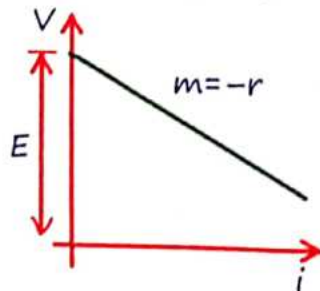
(a) Charging

$$\Delta V = E + ir$$



(b) Discharging

$$\Delta V = E - ir$$

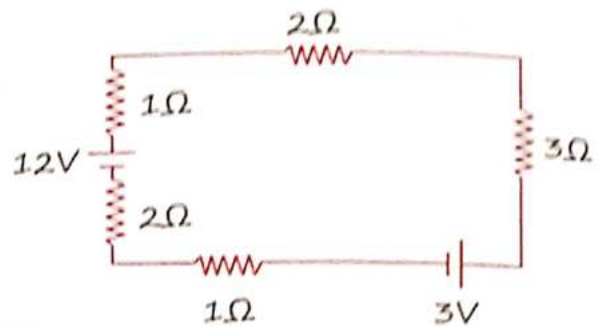


4 Kirchhoff's Law

Law (i) $\sum i = 0$ [Charge Conservⁿ]

Law (ii) Energy Conservⁿ.

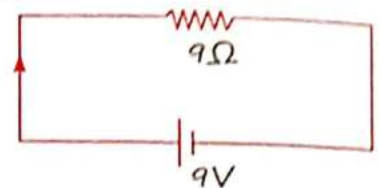
Note :- Household circuit maltaab parallel circuit.



MR*

Sare resistance ko series me ek sath add kar ke battery ko ek sath polarity ke sath add kare

$$I = \frac{9}{9} = 1A$$

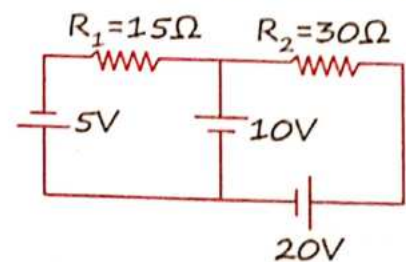


MR*

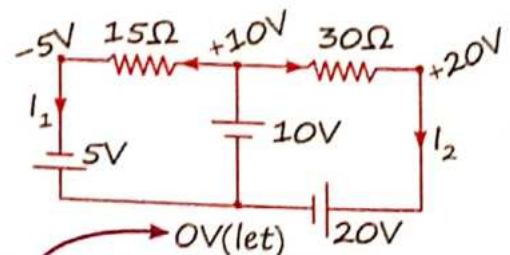
Point potential $\rightarrow$ current depends on potential difference not on potential and potential difference does not depends on reference, hence you can assume zero potential at any one point of circuit. (Sirf ek point pe hi zero man sakte hai)

Kishi bhi point ka Potential Apan zero Mann Sakte hai!

Q.5 Find current in 15Ω and 30Ω .



MR*



yaha ish liye kyki yha se sare point potential nikalna easy hai.

$$I_1 = \frac{10 - (-5)V}{15} = \frac{15}{15} = 1A$$

$$I_2 = \frac{(20-10)}{30} = \frac{10}{30} = \frac{1}{3} A$$

5 Combination of Resistance

Series :- i = Same potential different

$$+ V_{total} = V_1 + V_2 + V_3 + \dots$$

$$+ R_{eq} = R_1 + R_2 + R_3 + \dots$$

$$+ \text{If } n \text{ equal resistance, } R_{eq} = nR$$

$$+ R_{eq} \text{ will be larger than the largest Resistance}$$

$$V \propto R$$

$$+ \text{Jitna Jyada "R" Utina Jyada } V_{drop}.$$

Parallel :- [v = same] current different

$$+ I_{total} = I_1 + I_2 + I_3 + \dots$$

$$+ \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

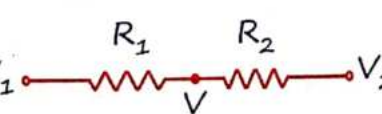
$$+ \text{If } n \text{ equal resistance, } R_{eq} = R/n$$

$$+ i \propto 1/R$$

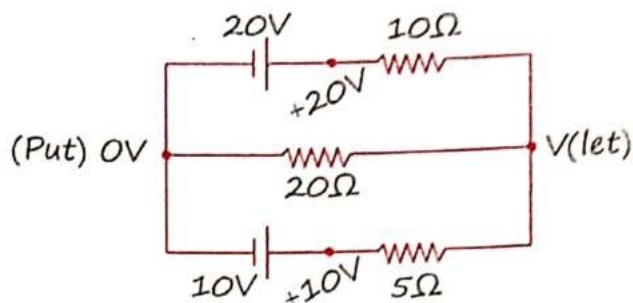
$$+ \text{Jitna Kam "R" Utina jyada "i"}$$

*Household circuit matlab parallel circuit.

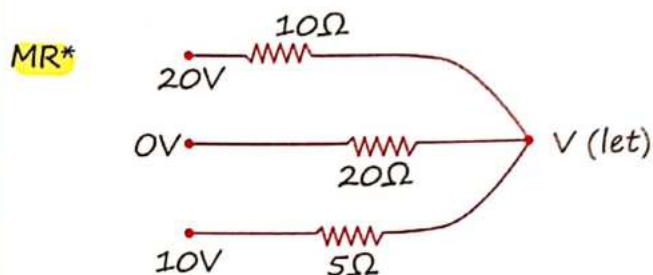
To Calculate Potential at Midpoint :-

$$V_{mid} = \frac{\frac{V_1}{R_1} + \frac{V_2}{R_2}}{\frac{1}{R_1} + \frac{1}{R_2}}$$


Q.6 Find current through each resistance?



Ans.

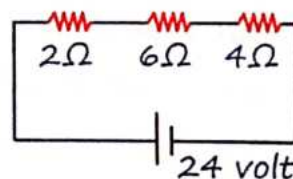


$$V = \frac{\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$

$$V = \frac{\frac{20}{10} + \frac{0}{20} + \frac{10}{5}}{\frac{1}{10} + \frac{1}{20} + \frac{1}{5}} = \frac{2 + 0 + 2}{\frac{2}{20} + \frac{1}{20} + \frac{4}{20}} = \frac{4}{\frac{7}{20}} = \frac{80}{7}$$

Now, we can calculate each current because we have $V = \frac{80}{7}$ hence we have potential difference.

Q.7 Find current and potential drop across each resistance



Ans. $V = IR$, $V \propto R$

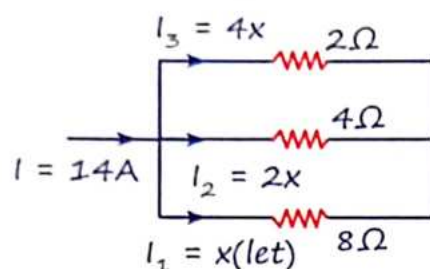
$$V_1 = x(\text{let}) \quad V_2 = 3x \quad V_3 = 2x$$

$$\text{Hence } x + 3x + 2x = 24V$$

$$V_1 = x = 4 \text{ volt}, V_2 = 12 \text{ volt}, V_3 = 8 \text{ volt}$$

$$I = \frac{V_{\text{net}}}{R_2} = \frac{24}{12} = 2 \text{ Amp}$$

Q.8 Find current through each resistance



Ans. $V = IR$

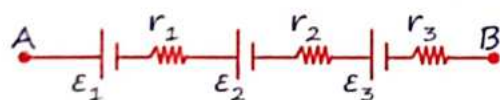
$$I \propto \frac{1}{R}$$

$$x + 2x + 4x = 14$$

$$x = 2 \text{ Amp}$$

6 Combination of Battery

Series :-



$$\epsilon_{\text{net}} = \epsilon_1 + \epsilon_2 + \epsilon_3$$

$$r_{\text{net}} = r_1 + r_2 + r_3$$

- If there are "n" identical battery is connected in series :

$$\epsilon_{\text{net}} = n\epsilon$$

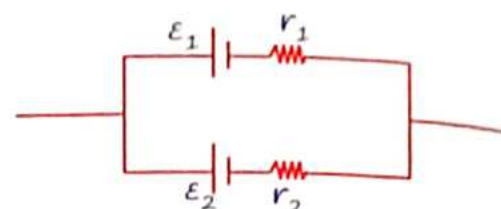
$$r_{\text{net}} = nr$$

- If "n" identical battery is connected in series out of which "m" reversed :

$$\epsilon_{\text{net}} = [n - 2m]E$$

$$r_{\text{net}} = nr \text{ [Maximum]}$$

Parallel :-



$$V = \frac{\frac{\epsilon_1}{r_1} + \frac{\epsilon_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}}, \text{ \& } \frac{1}{r_{\text{eq}}} = \frac{1}{r_1} + \frac{1}{r_2}$$

- If there are n-identical cell in parallel then,

$$\Sigma_{\text{net}} = E(\text{emf})$$

$$r_{\text{eq}} = r/n \text{ [Minimum]}$$

Mixed Grouping :-

- n-Identical battery (E,r) connected in series then this series combination connected m-times parallel with external resistance R.

$$n - \text{series} \Rightarrow nE, nr$$

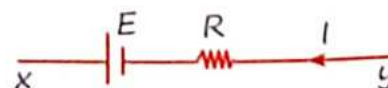
$$\text{then } m\text{-times parallel } (nE) \left(\frac{nr}{m} \right)$$

$$I = \frac{nE}{R + \frac{nr}{m}}$$

This I will be maximum when

$$R = \left(\frac{nr}{m} \right) \text{ (Internal resistance)}$$

- Circuit Mai chalna Important Hai :

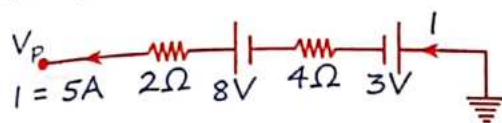


$$\rightarrow V_y - IR + E = V_x$$

$$\rightarrow V_x - E + IR = V_y$$

- Current ki direction me resistance ko cross karne par potential drop hoga $(-IR)$.
- Current ke opposite potential increase hoga $(+IR)$.
- Battery ko lower to higher cross karne pe potential increase hoga $(+E)$.
- Higher to lower cross karne pe potential decrease $(-E)$.
- Current ki direction se fark nahi pedega.

Q.9 Find V_P ?



Ans. Move from 'P' to ground

$$V_P + 5 \times 2 - 8 + 4 \times 5 + 3 = 0$$

$$V_P = -33 + 8 = -25 \text{ volt}$$

7 Power

$$P = iV, P = i^2R, P = \frac{V^2}{R}$$

Series: $- P \propto R$

Parallel: $- P \propto 1/R$

8 Joule's Law of heating

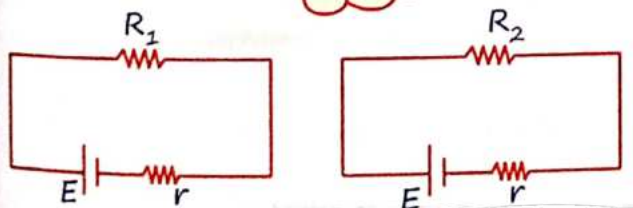
Heat produced (H) in a resistor when current flows through it;

$$H = i^2Rt = iVt = ms\Delta\theta$$

If, $i = \text{Variable}$: $- H = R \int i^2 \cdot dt.$

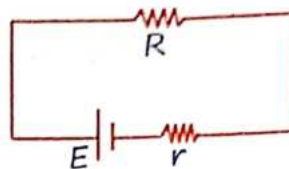
Special Case

$$r = \sqrt{R_1 R_2}$$



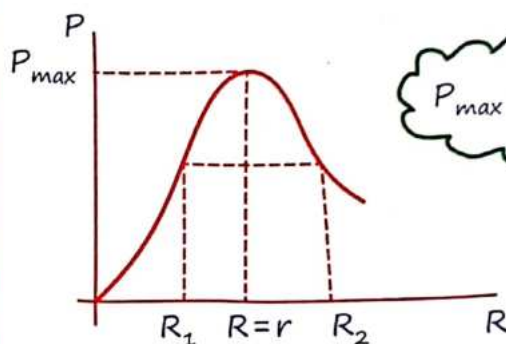
agar ye internal resistance between two circuit iss pattern mein raha toh P_{drop} Same hoga woh doh circuit mein!

Power drop in ext. Circuit with maximum power theorem :-



$$\text{Power drop across } R, P = I^2 R = \left[\frac{E}{R+r} \right]^2 R$$

Power drop will be maximum when $r=R$



$$P_{\text{max}} = \frac{E^2}{4R}$$

9 Bulb (Pure Resistance $\frac{W}{V}$)

+ Rated power and rated voltage given to calculate resistance of bulb.

$$R_{\text{bulb}} = \frac{V_R^2}{P_R}$$

$$R_{\text{Bulb}} \propto \frac{1}{P_{\text{Rated}}}$$

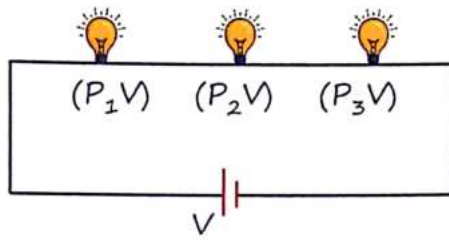
$$P_{\text{Cons.}} = \left[\frac{V_S}{V_R} \right]^2 P_{\text{Rated}}$$

Q.10 If two bulb of power (60W, 110V) and (100W, 110V) are connected in series with supply of 220V then?

Ans. Potential drop across 60W bulb will be greater than 110V hence it will fuse.

Series Combination :-

Bakwas Combination



$$\frac{1}{P_{\text{cons.}}} = \frac{1}{P_1} + \frac{1}{P_2} + \frac{1}{P_3}$$

If all are identical bulb then,

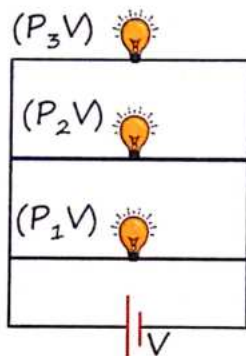
$$P_{\text{cons.}} = \frac{P}{n} \quad P_{\text{cons.}} \text{ is smaller than the smallest } P_{\text{rated}} \text{ bulb.}$$

$$P_{\text{consumed}} \propto R_{\text{bulb}} \propto \frac{1}{P_{\text{rated}}}$$

$$P_{\text{consumed}} = i^2 R_{\text{bulb}}$$

Joh Kam P_{rated} Ka hoga woh Jyada Chamkega!

Parallel Combination :-



$$P_{\text{cons.}} = P_1 + P_2 + P_3$$

If all are identical bulb then,

$$P_{\text{cons.}} = nP \quad P_{\text{consumed}} \propto \frac{1}{R_{\text{bulb}}} \propto P_{\text{rated}}$$

$$P_{\text{consumed}} = \frac{V^2}{R_{\text{bulb}}}$$

Joh Jyada " P_{rated} " Ka hoga woh Jyada Chamkega!

$$[1 \text{ kWh} = 36 \times 10^5 \text{ J}]$$

Time Taken by Heater Coils :-

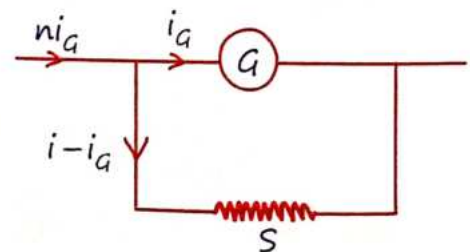
$$+ \text{ Series :- } t = t_1 + t_2$$

$$+ \text{ Parallel :- } t = \frac{t_1 t_2}{t_1 + t_2}$$

10 Electrical Instruments

Galvanometer :-

- + An instrument use to detect or measure small current.
- + Very sensitive, produce large error.
- + I_G = Maximum current that can flow.
- + G = resistance of galvanometer
- + **As Ammeter** :- Connected in series in circuit.
 - o Small resistance shunt connected in parallel with galvanometer.
 - o Ideal $R = 0$; Behave as simple wire.
 - o % Error = $\frac{i_T - i_M}{i_T} \times 100$



Where i_T = True current

i_M = Measured current

$$R_{\text{ammeter}} = \frac{GS}{G+S}$$

$$S = \frac{G}{n-1} \quad R \downarrow \quad \% \text{ Error } \downarrow$$

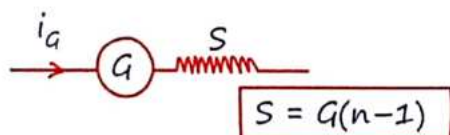
$$n = \frac{\text{i we want to Measure}}{\text{i jitna Galvano-meter se jayega.}}$$

$$= \frac{\text{Total current}}{\text{Current through galvanometer}}$$

As Voltmeter :- Connected parallel in circuit

o large resistance connected in series with galvanometer

o Ideal $R = \infty$ infinite (Behave as open wire)



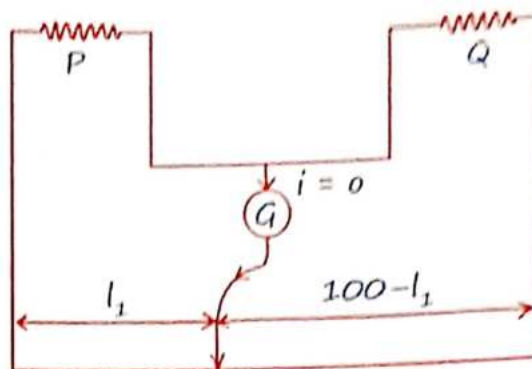
$$\% \text{ Error} = \frac{V_T - V_M}{V_T} \times 100$$

Potentiometer Bridge :-

Use to find value of Resistance.

Working based on wheat-stone bridge.

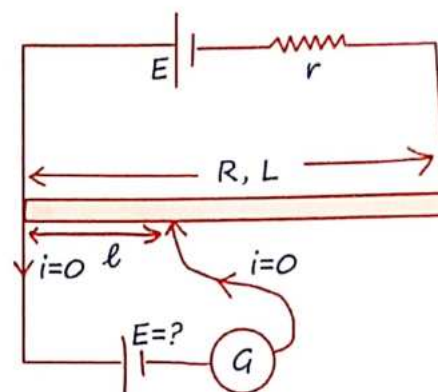
$$\frac{P}{l_1} = \frac{Q}{100-l_1}$$



Potentiometer wire :-

Working based on potential gradient.

+ To Find EMF :-



$$\text{*Step-i :- } i = \frac{E}{R+r}$$

$$\text{*Step-ii :- } v = iR$$

$$\text{*Step-iii :- } K = \frac{V}{L}$$

Potential gradient

Potential drop per unit length in wire.

$$E = kl$$

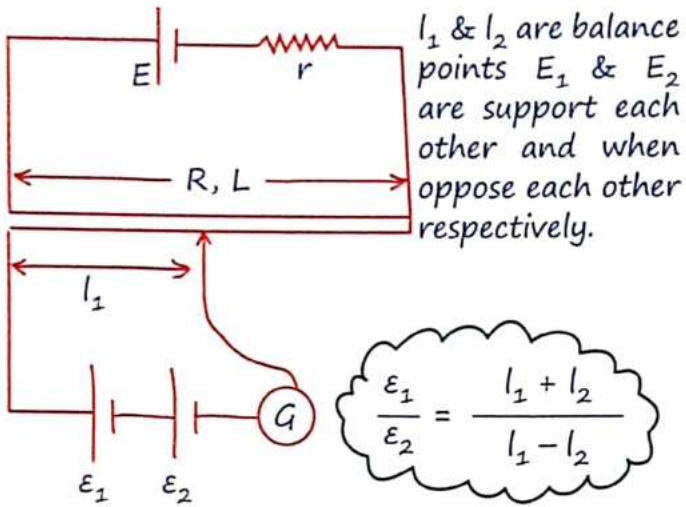
l = Balancing length where current through galvanometer is zero.

Caution !

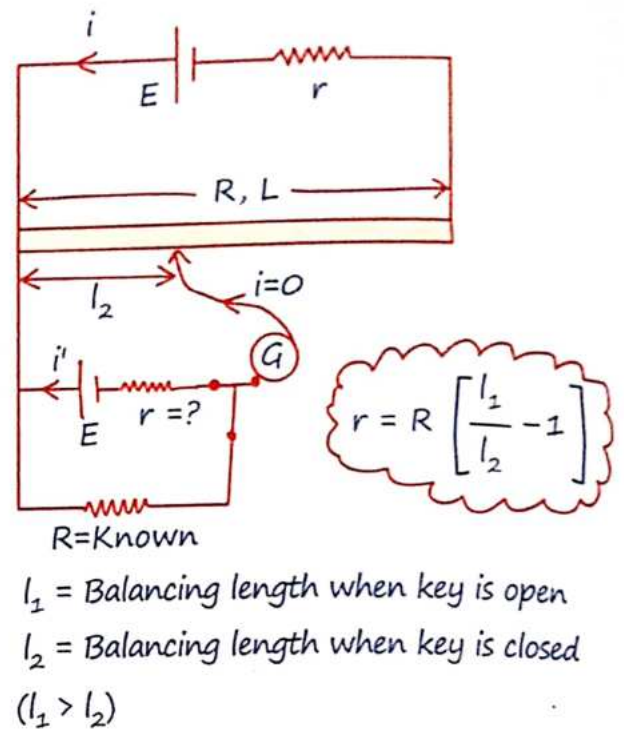
*EMF of the battery must be less or equal to the Potential drop in wire.

*Upar Ke battery Ki polarity aur niche ke battery Ki polarity supportive honi chahiye nai toh balance point asambhav hai... !

• To Compare EMF :-



• To Find internal Resistance :-



MR*

‘खुद की समझदारी भी अहमियत रखती है,
 वरना दुर्योधन और अर्जुन दोनों
 के गुरु एक ही थे।’

1 Oersted Experiment

Electric field outside current carrying wire is zero

Electric field inside current carrying wire may or may not be zero.

But moving charge near to current carrying wire experience force hence there must be a field that is magnetic field.

$$F_{\text{rest } Q} = 0 \rightarrow F_{\text{moving } Q} \neq 0$$

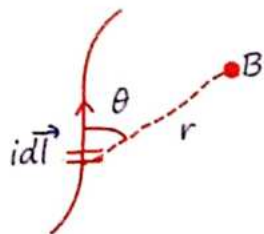
Note:- *Current carrying wire produces MF around it.

2 BIOT-SAVART'S LAW

The magnetic field (dB) due to a small element of current carrying wire (idL) at a point is given by:

$$dB = \frac{\mu_0}{4\pi} \frac{idl \sin\theta}{r^2}$$

(Scalar form)



Where:

μ_0 = permeability of free space ($4\pi \times 10^{-7}$ Tm/A)

i = current through the wire.

dl = length of the small element of the wire.

θ = angle between the element dl and the line joining the element to the point where the field is being calculated.

r = distance from the element to the point.

Unit conversion:

$$*1T = 10^4 G = 1 \frac{wb}{m^2}$$

Vector Form :-

$$\vec{dB} = \frac{\mu_0}{4\pi} \frac{id\vec{l} \times \hat{r}}{r^2} = \frac{\mu_0}{4\pi} \frac{(id\vec{l} \times \vec{r})}{r^3}$$

MR*

To find direction of magnetic field

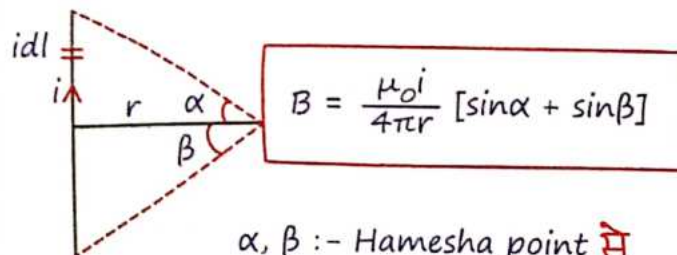
$$\vec{dB} = id\vec{l} \times \vec{r}$$

(Result) (1^{st} vector) (2^{nd} vector)

Place your four-finger (palm) of right hand along 1^{st} vector ($id\vec{l}$) slap 2nd vector ($\vec{r}$), thumb will represent $\vec{B}$.

3 Magnetic Field due to Straight Wire

* The magnetic field (B) at a point due to a straight current carrying wire is given by:



α, β :- Hamesha point lena hai !

Where

- r is the perpendicular distance from the wire to the point.
- α and β are the angle subtended by the wire at the point of interest.

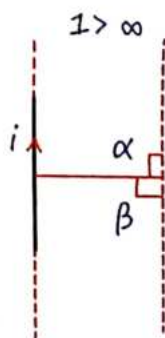
MR* feel

Dimensional Format

$$B = \frac{\mu_0 i}{\text{dist}^n}$$

For infinitely long wire :-

- For infinitely long wire :-

$$B = \frac{\mu_0 i}{4\pi r} \times 2$$


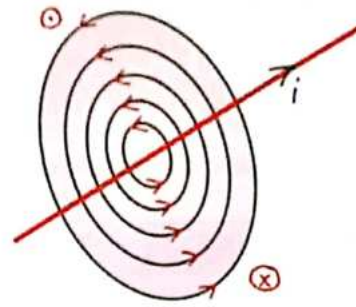
- Semi- ∞ wire :-

$$B = \frac{\mu_0 i}{4\pi R}$$


Direction of Magnetic Field :-

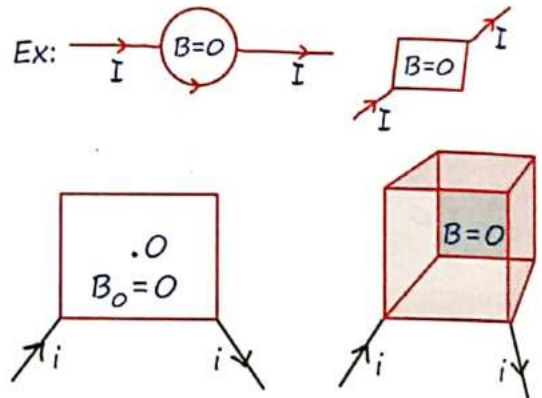
Right hand Rule:-

- Place thumb in the direction of current then curling finger will represent direction of magnetic field

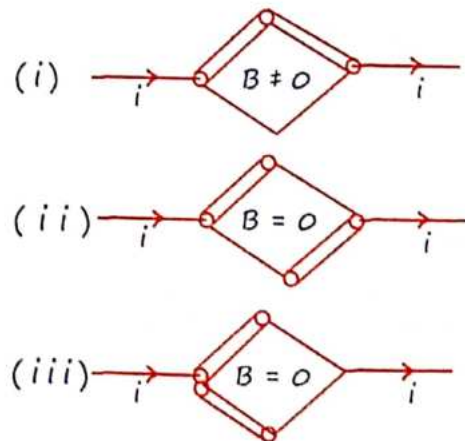


Symmetrical Object :-

Symmetrical objects with uniform current distribution often result in zero magnetic fields inside the object.



Combination of two thick and two thin wire :-



Parallel wires with different currents:

- Consider two parallel wires separated by a distance r , with one wire carrying current $I_1 = I$ and the other carrying current $nI = I_2$